

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398397
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Carrillo Rincón; Andrés Felipe et al.

Inducible promoters

Abstract

Provided herein are nucleic acid constructs that comprise an inducible promoter. Dual expression systems are provided comprising two nucleic acid constructs or a single nucleic acid construct with two inducible promoters. Also provided are methods of expressing transcripts by transforming a nucleic acid construct described herein into a prokaryotic cell and contacting the prokaryotic cell with an inducer. Methods of producing a carotenoid are also disclosed herein.

Inventors:	Carrillo Rincón; Andrés Felipe (Worcester, MA), Farny; Natalie G. (Worcester, MA)
Applicant:	Worcester Polytechnic Institute (Worcester, MA)
Family ID:	1000008778007
Assignee:	Worcester Polytechnic Institute (Worcester, MA)
Appl. No.:	18/136660
Filed:	April 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240018529 A1	Jan. 18, 2024

Related U.S. Application Data

us-provisional-application US 63332507 20220419

Publication Classification

Int. Cl.: C12N15/63 (20060101); C12N15/11 (20060101)

U.S. Cl.:

CPC C12N15/635 (20130101); C12N15/111 (20130101); C12N2830/36 (20130101); C12N2840/002 (20130101)

Field of Classification Search

CPC: C12N (15/635); C12P (23/00)

USPC: 435/320.1; 435/91.1

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4551433	12/1984	DeBoer	435/69.3	C12P 21/02
6117651	12/1999	Schultz et al.	N/A	N/A
7495092	12/2008	Barrangou	435/243	C12N 15/746
8846374	12/2013	Sharpe	435/254.2	C12P 7/6432
2014/0140959	12/2013	Szalay et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2019123324	12/2018	WO	N/A

OTHER PUBLICATIONS

Aoyama, T. et al. “Essential structure of *E.coli* promoter: effect of spacer length between the two consensus sequences on promoLer function” pp. 5855-5864. Nucleic Acids Research. vol. 11, No. 17. 1983; Abstract; DOI:

10.1093/nar/11.17.5855. cited by applicant

Spencer, J. et al.. “A TATA Binding Protein Mutant with Increased Affinity for DNA Directs Transcription from a Reversed TATA Sequence In Vivo” pp. 8744-8755. Molecular and Cellular Biology. vol. 22, No. 24. Dec. 2002; Abstract; p. 8748; DOI: 10.1128/MCB.22.24.8744-8755.2002. cited by applicant

International Search Report and Written Opinion in International Application No. PCT/US2023/065954 mailed Jan. 31, 2024. cited by applicant

Harley, C. et al., “Analysis of *E. coli* promoter sequences,” Nucleic Acids Research, Jan. 1987, vol. 15, No. 5., pp. 2343-2361. cited by applicant

Lanzer, M. et al., “Promoters largely determine the efficiency of repressor action,” Proceedings of the National Academy of Sciences, Dec. 1988, vol. 85., pp. 8973-8977. cited by applicant

Rincón, A. et al., “Unlocking the strength of inducible promoters in Gram-negative bacteria,” Microbial Biotechnology, May 2023, vol. 16., pp. 961-976. cited by applicant

Primary Examiner: Fox; Allison M

Assistant Examiner: Abuzeineh; Hanan Isam

Attorney, Agent or Firm: Greenberg Traurig, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/332,507 filed Apr. 19, 2022, the entirety of which is incorporated herein by reference.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML copy, created on Aug. 4, 2023, is named 110697-015901_US_SL.xml and is 133,847 bytes in size.

BACKGROUND

(2) Inducible promoters are ubiquitous biotechnology tools for manufacturing proteins, providing molecular models of biosynthesis pathways, and as synthetic switches for a variety of environmental, physiological, and cellular tools. Inducible promoters have a consistent architecture including two key elements: the operator region recognized by transcriptional regulatory proteins and consensus sequences that recruit the sigma (σ) subunits of RNA polymerase to initiate transcription of the inducible gene. Despite their widespread use, leaky transcription in the “OFF” state remains a challenge for inducible promoters. Therefore, improved inducible promoters, cellular systems, and methods of generating proteins are needed to enable advances in protein production and various biotechnology applications.

BRIEF SUMMARY

(3) Provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TXTXTGT sequence at position −10, relative to a transcriptional start site of the promoter; (b) a TXGXCX-sequence at position −35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence, wherein X is any nucleobase. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAAT sequence at position −10, relative to a transcriptional start site of the promoter; (b) a TTGACA sequence at position −35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. In some embodiments, the nucleic acid construct further comprises a transgene. In some embodiments, the modified inducible

promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a prokaryotic cell in the presence of an inducer; and wherein: (i) the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and (ii) the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. In some embodiments, the nucleic acid construct further comprises a transgene. In some embodiments, the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a prokaryotic cell in the presence of an inducer; and wherein: (i) the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and (ii) the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. In some embodiments, a strong promoter increases the amount of expression of a transgene provided herein relative to a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TXTXXTGT, or TXGXCX. In some embodiments, the strong promoter increases the amount of expression of a transgene provided herein by at least 10% relative to the expression of a transgene expressed by a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TXTXXTGT, or TXGXCX.

(4) Provided herein are methods of expressing a transgene in a cell, the methods comprising: (a) transforming a nucleic acid construct into the cell, wherein the nucleic acid construct comprises a modified inducible promoter, wherein the modified inducible promoter comprises: (i) a TATAAT sequence or a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; (iii) a nucleic acid sequence encoding a bacterial ribosome binding sequence; and (iv) a transgene; and (b) contacting the cell with an inducer, thereby expressing the transgene. In some embodiments, when cloned into a comparable nucleic acid construct that comprises a strong promoter, the transgene inhibits growth of the cell prior to the contacting step (b) with the inducer, thereby preventing expression of the transgene via the comparable nucleic acid. In some embodiments, the cell is a prokaryotic cell. In some embodiments, the cell is a eukaryotic cell.

(5) Provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: (a) a first modified inducible promoter sequence, wherein the first modified inducible promoter comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the first modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the first modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a first bacterial ribosome binding sequence; and (b) a second modified inducible promoter sequence, wherein the second modified inducible promoter sequence comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a second bacterial ribosome binding sequence. In some embodiments, the nucleic acid constructs further comprise a transgene. In some embodiments, wherein the first modified inducible promoter sequence, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is in the presence of an inducer.

(6) A composition comprising two or more of a nucleic acid construct provided herein.

(7) Provided herein are isolated prokaryotic cells, wherein the isolated prokaryotic cells comprise a nucleic acid construct provided herein. Provided herein are isolated eukaryotic cells, wherein the isolated eukaryotic cells comprise a nucleic acid construct provided herein. Provided herein are cell-free systems, wherein the cell-free systems comprise a nucleic acid construct provided herein.

(8) Provided herein are non-naturally occurring organisms, wherein the non-naturally occurring organisms comprise: a nucleic acid construct comprising: (a) a first inducible promoter sequence comprising: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence; (b) one or more of a biosynthesis pathway transgene; and (c) a second modified inducible promoter sequence comprising: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. In some embodiments, the non-naturally occurring organisms comprise prokaryotic organisms. In some embodiments, the prokaryotic organisms comprise a population of bacteria. In some embodiments, the biosynthesis pathway transgene comprises a carotenoid synthesis gene.

(9) Provided herein are compositions comprising a non-naturally occurring organism provided herein and an inducer.

(10) Further provided herein are methods of expressing a transgene in a cell that does not comprise a T7 RNA polymerase, wherein the methods comprise: (a) transforming a nucleic acid construct into the cell, wherein the nucleic

acid construct comprises a modified inducible promoter, wherein the modified inducible promoter comprises: (i) a TATAAT sequence or a TATAATGT-sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA-sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a bacterial ribosome binding sequence; and a transgene: and (b) contacting the cell with an inducer, thereby expressing a protein encoded by the transgene. In some embodiments, the expression of the protein in the absence of the inducer is less than an amount of expression of the protein in the absence of the inducer of the transgene operatively coupled to a T7 promoter in a comparable nucleic acid construct. In some embodiments, the expression of the protein encoded by the transgene in the presence of the inducer is greater than an amount of expression of the protein in the presence of the inducer of the transgene operatively coupled to the T7 promoter in the comparable nucleic acid construct. In some embodiments, the transgene is toxic to a prokaryotic cell that does not express a modified inducible promoter. In some embodiments, the cell is a prokaryotic cell. In some embodiments, the cell is a eukaryotic cell.

(11) Further provided herein are methods of expressing a transgene in a cell-free system, wherein the methods comprise: (a) transforming a nucleic acid construct into the cell-free system, wherein the nucleic acid construct comprises a modified inducible promoter, wherein the modified inducible promoter comprises: (i) a TATAAT sequence or a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a bacterial ribosome binding sequence; and a transgene: and (b) contacting the cell-free system with an inducer, thereby expressing a protein encoded by the transgene. In some embodiments, the cell-free system further comprises: an RNA polymerase, ribonucleotides, and/or a buffer.

(12) Provided herein are methods for producing a protein, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the protein. Further provided herein are methods for producing a carotenoid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the protein. Provided herein are methods for producing an organic compound, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the organic compound. Further provided herein are methods for producing a carotenoid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the carotenoid. In some embodiments, the organic compound or the protein comprises a polyketide, a terpene, a non-ribosomal peptide, or an enzyme protein. Provided herein are methods for producing benzoic acid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the benzoic acid. Further provided herein is a composition comprising benzoic acid made by a method provided herein. Further provided herein is a composition comprising a carotenoid made by a method provided herein.

(13) Provided herein are kits, wherein the kits comprise a nucleic acid construct provided herein, a cell provided herein, a cell-free system provided herein, or a non-naturally occurring organism provided herein, packaging and materials, therefore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Novel features of exemplary embodiments are set forth with particularity in the appended claims. A better understanding of the features and advantages will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosed systems and methods are utilized, and the accompanying drawings of which:

(2) FIGS. 1A-1B show schematic diagrams of exemplary plasmids produced in accordance with the teachings of the present disclosure. FIG. 1A shows a schematic of a constitutive version of the synthetic lac and tet expression systems. FIG. 1B shows a schematic of inducible versions of the synthetic lac and tet expression systems in pColE1 backbone. Plasmids contain the Kanamycin selection marker and the attP sites specific fox Bxb1 integrase. # indicates the version of the promoter (e.g. V1Tc, V2lac, etc).

(3) FIGS. 2A-2C show the architecture of exemplary synthetic lac and tet promoters. The sequences of the operators lacO and tetO are overlined, in bold text. The -35 and -10 sequences of the promoters are underlined. Boxes represent the native -35 and -10 sequences, and consensus $\sigma 70$ -35 and -10 sequences. Conserved regions of the original lac promoter and the original tacI promoter are shown. FIG. 2A shows the original lac (i) and synthetic lac (ii-vii) promoters. SEQ ID NOS: 3-9 are shown. FIG. 2B shows the original tet (i) and synthetic tet (ii-v) promoters. SEQ ID NOS: 10-14 are shown. FIG. 2C shows schematic representation of the function of synthetic lac and tet promoters. Each transcriptional unit is insulated by terminators. LacI/TetR regulators bind to the operators lacO/tetO. Addition of the inducers (yellow ovals) IPTG or anhydrotetracycline (aTc) remove the repressor allowing transcription from the promoters.

(4) FIGS. 3A-3D show a graphical comparison of sfGFP and mCardinal emissions in Gram-negative bacteria. FIG. 3A shows absolute values of fluorescence measured in arbitrary units with far red wavelengths (excitation 604, emission 659, left panel), and green wavelengths (excitation 485, emission 510) *E. coli* DH10B, *P. putida* and *V. natriegens*. Note that these strains are wild type and do not contain any fluorescent protein genes. N=3. Error bars +/-SD. FIGS. 3B-3D show graphs of the fluorescence signal-to-background ratio of recombinant strains expressing sfGFP and mCardinal from the

constitutive *tacI* promoter (these represent “signal”) vs. wild type strains of (FIG. 3B) *E. coli* (FIG. 3C) *P. putida* & (FIG. 3D) *V. natriegens* (these wild type strains represent “background”). Signal-to-background measurement ratios are then plotted over time (left panels) The fluorescence values (in arbitrary units, AU) measured to create the signal-to-background ratios are shown in the middle (sfGFP) and right (mCardinal) panels. N=3. Error bars +/-SD.

(5) FIGS. 4A-4C show graphs of the time course of mCardinal production in *E. coli* DH10B with synthetic lac and tet promoters. FIG. 4A shows the synthetic lac promoters, and FIG. 4B shows the synthetic tet promoters, with and without the transcriptional regulators tetR and lacI and induced after 3 hours with aTc or IPTG, respectively. FIG. 4C shows direct comparison of each promoter under evaluation against the strong constitutive promoter *tacI*. The recombinant DH10B carrying *tacI*-mCardinal was normalized to 100% mCardinal production and the wild type DH10B strain normalized to 0% mCardinal production. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=4. Error bars +/-SD.

(6) FIGS. 5A-5B show an SDS-Page gel and graph of the expression profile of CocE in *E. coli*. FIG. 5A shows gel electrophoresis of total crude extracts (top), the insoluble fraction analysis (middle panel), and the soluble fraction (bottom). The expected size of CocE is 63 kDa. *E. coli* expressing the pET and V2TcR expression systems, uninduced and induced (induced samples indicated with an asterisk). FIG. 5B shows a graph of benzoic acid production by CocE present in the soluble fraction, using cocaine as substrate. N=3. Error bars +/-SD. Key for X axis Groups: I: DH10B pσ_{sup.70} V2TcR-CocE; II: DH10B pσ_{sup.70} V2TcR-CocE aTc; III: DH10B pσ_{sup.70} V2TcR19-CocE; IV: DH10B pσ_{sup.70} V2TcR19-CocE aTc; V: BL21 pET21-cocE; VI: BL21 pET21-cocE IPTG.

(7) FIGS. 6A-6C show graphs of the time course of mCardinal production in *P. putida*. Synthetic lac (FIG. 6A) and tet (FIG. 6B) promoters in their constitutive, repressed and induced states. All constructs were integrated in a single copy into the same genomic locus. FIG. 6C shows direct comparison of each promoter under evaluation against the constitutive promoter *tacI* in *P. putida*. *P. putida* with a single-copy integration of *tacI*-mCardinal was normalized to 100% mCardinal production and the wild type *P. putida* strain normalized to 0% mCardinal production. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=4. Error bars +/-SD.

(8) FIGS. 7A-7C show graphs of the time courses of mCardinal production in *V. natriegens*. Synthetic lac (FIG. 7A) and tet (FIG. 7B) promoters in their constitutive, repressed and induced states are represented. FIG. 7C shows direct comparison of each promoter under evaluation against the constitutive promoter *tacI* in *V. natriegens*. *V. natriegens* carrying *tacI*-mCardinal was normalized to 100% mCardinal production and the wild type *V. natriegens* strain normalized to 0% mCardinal production. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=4. Error bars +/-SD.

(9) FIG. 8 shows a graphical comparison of the pET21a expression system in *E. coli* BL21 versus the synthetic V2Lac/lacI and V2Tc/tetR promoters in *E. coli* DH10B. Cultures were induced after 3 hours of cultivation and mCardinal mean was normalized by the cell density (OD_{sub.600}). N=4. Error bars +/-SD.

(10) FIGS. 9A-9F shows a schematic representation of an exemplary dual expression system. FIG. 9A shows the design of the V2TcR-V2(3)LacI system. FIG. 9B shows a dual expression system controlling expression of mCardinal. FIG. 9C shows a dual expression system controlling expression of sfGFP and mCardinal. FIG. 9D shows a V2TcR expression system controlling the LYC operon. FIG. 9E shows a V2TcR expression system controlling expression of crtEBIY. FIG. 9F shows a dual expression system controlling expression of crtEBI and crtY. Exemplary sequences tested include SEQ ID NOS: 69-73 and 76-79.

(11) FIGS. 10A-10C show graphs of the time course of mCardinal production in *E. coli* (FIG. 10A), *P. putida* (FIG. 10B) & *V. natriegens* (FIG. 10C). Left graphs show the recombinant strains containing the V2TcR-mCardinal (SEQ ID NO: 56) and V2(3)LacI-mCardinal expression systems and the right graphs show the recombinant strains containing the dual V2TcR-mCardinal/V2(3)LacI-mCardinal expression system. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=3. Error bars +/-SD.

(12) FIGS. 11A-11C shows graphs of the time course of the reporter systems mCardinal and sfGFP in *E. coli* (FIG. 11A), *P. putida* (FIG. 11B), and *V. natriegens* (FIG. 11C) containing the dual expression system V2TcR-sfGFP/V2(3)LacI-mCardinal. Left graphs measure mCardinal production and right graphs measure sfGFP. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) and sfGFP signal (excitation 485, emission 510) was normalized by the cell density (OD600). N=3. Error bars +/-SD.

(13) FIGS. 12A-12B shows graphs of the production of lycopene and β-carotene with the V2TcR expression system in *E. coli*, *P. putida*, and *V. natriegens*. Cultures were induced with aTc at OD600. 0.7 and cultured for 4 hours. FIG. 12A shows Lycopene levels that were extracted and measured by UHPLC. FIG. 12B shows B-carotene levels that were extracted and measured by UHPLC. N=3. Error bars +/-SD.

(14) FIGS. 13A-13B shows graphs of the production of lycopene and beta (β)-carotene by the dual expression system in *E. coli*, *P. putida*, and *V. natriegens*. Cultures were induced with aTc, IPTG or both inducers at OD600. 0.7 and cultured for 4 hours. FIG. 13A shows lycopene levels that were extracted and measured by UHPLC. FIG. 13B shows β-carotene levels that were extracted and measured by UHPLC. N=3. Error bars +/-SD.

(15) FIG. 14 shows a schematic of an exemplary dual expression system. Exemplary sequences include SEQ ID NO: 54 and SEQ ID NO: 55.

DETAILED DESCRIPTION

(16) Overview

(17) Disclosed herein are nucleic acid constructs that comprise a modified inducible promoter. Such constructs can be utilized to express a transgene operatively coupled to the promoter upon transformation into a cell or a cell-free system. As described herein, the modified inducible promoter does not substantially express the transgene in the absence of an inducer and in the presence of a specific transcriptional regulator. In some cases, a transgene is toxic to the cell when expressed utilizing the modified inducible promoter described herein. In some cases, a transgene is not toxic to the cell when expressed utilizing the modified inducible promoter described herein. Expression of toxic genes is one of many advantages to the modified inducible promoter systems provided herein. In addition, non-toxic gene product yields (e.g., proteins or organic compounds) are also higher as compared to inducible promoters that are not modified.

(18) In contrast, many inducible promoters produce significant expression of the transgene even in the absence of an inducer. Such promoters, known as “leaky” promoters, cannot be used to express transgenes that are toxic to a cell, as the leaky expression of the transgene inhibits growth of the cell culture and thus prevents overexpression or avoids formation of inclusion bodies.

(19) Further, nucleic acid constructs provided herein comprising a modified inducible promoter that generates the expression of a protein encoded by a transgene in the presence of an inducer that is comparable or to a greater extent than the expression of a protein encoded by the transgene when expressed using a different promoter (e.g., a T7 promoter such as a pET vector). In some embodiments, expression of the transgene can be carried out in various prokaryotic cells, eukaryotic cells, or cell-free systems without the need for T7 lysogenization.

(20) Definitions

(21) The terminology used herein is for the purpose of describing particular cases only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising”.

(22) The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, such as plus or minus 10%. Where ranges and/or subranges of values are provided, the ranges and/or subranges include the endpoints of the ranges and/or subranges.

(23) The term “substantially” as used herein refers to a value approaching 100% of a given value. For example, an expression system described herein that does not “substantially” express a transgene in the absence of an inducer can indicate that less than 10% of the transgene (e.g. less than 5%, less than 1%, less than 0.1%, or less than 0.01%) is expressed, relative to an amount of transgene expressed in the presence of the inducer.

(24) As used herein, the term “operably linked” indicates that a promoter is in a correct functional location and/or orientation in relation to a nucleic acid sequence it regulates to control transcriptional initiation and/or expression of that sequence.

(25) Modified Inducible Promoters

(26) Inducible promoters often have a consistent architecture including two elements: (1) the operator region recognized by the transcriptional regulator proteins (e.g., lacI and tetR, and the -10) and -35 consensus sequences that recruits the sigma (σ) subunits of RNA polymerase to initiate transcription. Improvements to the lac promoter can be made to increase its strength and improve its regulation. For example, the -10 and -35 boxes of the original lac promoter (FIG. 1Ai) can be updated to generate improved variants such as the lacUV5 (FIG. 1Aii) and tacI (FIG. 1Aiii) promoters. In the lacUV5 promoter, the -10 box of the original lac promoter can be replaced by the Pribnow box (e.g., TATAAT) or any one of TATAATGT, TTXXT, TATXGT, TATXGT, TATAATGT or TATAATGT. The lacUV5 and trp promoters shown in the working examples can be combined to create the tacI promoter, which increased transcription 11-fold compared to its predecessor with the incorporation of the -35 consensus box TTGACA. The tet promoter (FIG. 1Bi) shares the highly conserved -35 hexamer with tac promoter, but does not contain the Pribnow box.

(27) Despite their ubiquitous use in biology, there remain problems with the current lac-based inducible expression systems. Leaky transcription in the OFF state remains a consistent challenge. Tight transcriptional control is indispensable to produce high yields of challenging recombinant proteins, including toxic genes and proteins that are impossible for a heterologous host to process or fold. For the pET system, target proteins are driven indirectly by controlling expression of the T7 polymerase under the lacUV5 promoter, and then driving the target gene transcription by the T7 promoter. Still, low level T7 transcription in the uninduced state leads to leaky transcription of the target gene. To combat this problem further, *E. coli* strains such as pLysS and pLysE with integration of the T7 lysozyme that inhibits low level T7 activity are used to obtain tighter transcriptional control. However, a host strain with T7 RNA polymerase under control of the lacUV5 promoter, and integration of T7 lysozyme, is required to tightly regulate target gene expression. Expression with the tet system has historically provided tighter transcriptional control than lac derived promoters, does not required specialized strains (such as the BL21), and fully induction is achieved by anhydrotetracycline (aTc) at concentrations that do not cause growth defect due the high affinity of aTc to TetR, and its imperceptible antibiotic activity. Remarkably, at the uninduced state just one mRNA molecule per three cells is produced, and up to 5000-fold induction has been reported, however, the yields of recombinant protein obtained by the tet expression system remains low compared to the pET expression system when using *E. coli* as heterologous host.

(28) The repertoire of organisms used both in academic and industrial settings is rapidly expanding. To address challenges

related to complex protein expression in *E. coli*, other chassis organisms such as *Pseudomonas putida* and *Vibrio natriegens* have been employed to produce challenging proteins (e.g., carotenoids) in a variety of biotechnological processes. Both lac and tet expressions systems can be adapted to *P. putida* and *V. natriegens*, though in some instances with lower total protein yield than achieved in pET. However, previous systems for inducible expression were not configured for use across various cell types (e.g., prokaryotic cells). Accordingly, disclosed herein are universal expression systems that can be directly ported between different gram-negative species, yield high quantities of recombinant protein comparable to expression via a pET in *E. coli*, and maintain tight transcriptional repression in the uninduced state. The expression system of the current disclosure is a significant improvement over the pET system in providing tight OFF state control while achieving similar yields of recombinant protein, and with the advantage of direct portability to alternative host species. Further, additional modifications can be performed to enable the expression of the transgene to be turned OFF after induction, thus allowing for reversible control of expression.

(29) A modified, inducible promoter of the current disclosure comprises modified architecture of the lac and tet expression systems to improve their strength, control, and portability. In some embodiments, the genetic architecture of the lac and tet expression systems were modified in three ways: (1) addition of the consensus -10 and -35 sequence boxes to be strongly targeted by σ .sup.70, (2) incorporation of a nucleic acid sequence that encodes a strong ribosome binding site recognized by a broad spectrum of gram-negative bacteria, and (3) independent control of the transcriptional regulators by appropriately-tuned constitutive promoters.

(30) The nucleic acid constructs provided herein can comprise DNA, RNA, an artificial nucleic acid analog, or any combination thereof. In some embodiments, the nucleic acid constructs provided herein comprise, a nucleic acid modification (e.g., a chemical modification), a nucleobase substitution, or a nucleotide substitution. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises one or more nucleobase substitutions. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TXTXXT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TXTXXTGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATXXTGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATXXTGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATAATGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATAATGT, where X is any nucleobase.

(31) In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises one or more nucleobase substitutions. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TXGXCX, where X is any nucleobase. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TTGXCX, where X is any nucleobase. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TTGACX, where X is any nucleobase. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TTGACA, where X is any nucleobase.

(32) In some embodiments, a nucleic acid construct provided herein comprises a nucleotide analogue. Nucleotide analogues include nucleotides having modifications in the chemical structure of the base, sugar and/or phosphate, including, but not limited to, 5-position pyrimidine modifications, 8-position purine modifications, modifications at cytosine exocyclic amines, substitution of 5-bromo-uracil, and the like; and 2'-position sugar modifications, including but not limited to, sugar-modified ribonucleotides in which the 2'-OH is replaced by a group selected from H, OR, R, halo, SH, SR, NH.sub.2, NHR, NR.sub.2, or CN. shRNAs also can comprise non-natural elements such as non-natural bases, e.g., inosine and xanthine, sugars, e.g., 2'-methoxy ribose, or non-natural phosphodiester linkages, e.g., methylphosphonates, phosphorothioates and peptides.

(33) The nucleic acid construct provided herein can comprise a sequence encoding a ribosome binding site. Ribosome binding sites (RBSs) are nucleic acid sequences that promote efficient and accurate translation of mRNAs for protein synthesis, and are also provided for use in the inducible promoters provided herein to permit modulation of the efficiency and rates of synthesis of the proteins encoded by the system. An RBS affects the translation rate of an open reading frame in two main ways—i) the rate at which ribosomes are recruited to the mRNA and initiate translation is dependent on the sequence of the RBS, and ii) the RBS can also affect the stability of the mRNA, thereby affecting the number of proteins made over the lifetime of the mRNA. Accordingly, one or more nucleic acid sequence encoding a ribosome binding site (RBS) or an RBS mRNA can be added to the nucleic acid constructs described herein to control expression of proteins.

(34) In some embodiments, a nucleic acid construct provided herein further comprises a terminator sequence. Terminators are sequences that usually occur at the end of a gene or operon and cause transcription to stop, and are also provided for use in the modules and engineered systems described herein to regulate transcription and prevent transcription from occurring in an unregulated fashion, i.e., a terminator sequence prevents activation of downstream modules by upstream promoters. A terminator or termination signal can include the DNA sequences involved in specific termination of an RNA transcript by an RNA polymerase. Thus, in certain embodiments a terminator that ends the production of an RNA transcript is contemplated. A terminator can be necessary for use in vivo to achieve desirable message levels.

(35) The most commonly used type of forward terminator is a forward terminator. When placed downstream of a nucleic acid sequence that is usually transcribed, a forward transcriptional terminator will cause transcription to abort. In some embodiments, bidirectional transcriptional terminators are provided. Such terminators will usually cause transcription to terminate on both the forward and reverse strand. Finally, in some embodiments, reverse transcriptional terminators can be used to terminate transcription on the reverse strand only.

(36) In some embodiments, a nucleic acid construct provided herein comprises additional regulatory elements that increase or decrease transgene expression in a cell or cell-free system depending on the absence or presence of a particular inducer or set of inducers. In some embodiments, the regulatory element is an enhancer. Additional non-limiting examples of regulatory elements include: lasR activator (e.g., from *P. aeruginosa*), cinR activator, toxicity-gene activator (e.g., ToxR, from *Vibrio cholerae*), lacI (e.g., a wild-type, derivative, or variant thereof), lacI repressor, tetracycline repressor (e.g., TetR from transposon Tn10), mnt repressor, TP901, heat shock proteins, and any derivative or variant thereof.

(37) Thus, in some embodiments, a nucleic acid construct provided herein can be part of a synthetic gene network. Synthetic gene networks can include an engineered composition that comprises at least one nucleic acid construct provided herein and can perform a function including, but not limited to, sensing the presence or absence of an analyte or inducer, a logic function, or a regulatory function. In some embodiments of a synthetic gene network comprising at least two nucleic acid constructs, the nucleic acid constructs can interact with each other directly or indirectly. A synthetic gene network can comprise a nucleic acid encoding a transgene operably linked to a modified inducible promoter provided herein.

(38) The nucleic acid constructs provided herein can be used to visualize chemical, analyte, or protein production in the presence and absence of an inducer provided herein. In some embodiments, a nucleic acid construct provided herein further comprises a reporter gene. In some embodiments, the reporter gene is mCardinal or a green fluorescent protein (e.g., superfolder GFP or sfGFP). A reporter gene encoding any fluorescent protein can be applicable to the nucleic acid constructs and methods of use provided herein. Additional examples of genes encoding fluorescent proteins that can be used in accordance with the compositions and methods described herein include, without limitation, enhance yellow fluorescent protein (EYFP), engineered cyan fluorescent protein (ECFP), mOrange, mCherry, Venus YFP, Cerulean, mBanana, orange fluorescent protein (OFP), derivatives, or variants thereof.

(39) In some embodiments, the reporter gene encodes for a colorimetric protein enzyme. Colorimetric enzymes can cleave a substrate (e.g., a chemical) to yield a color-changing product. In some embodiments, the protein tag is chitinase (which cleaves colorless 4-Nitrophenyl N,N'-diacetyl-beta-D-chitobioside substrate to yield a yellow p-nitrophenol product). In some embodiments, the reporter gene is LacZ (which encodes beta-galactosidase) or a fragment thereof. When LacZ is expressed, the enzyme cleaves the yellow chlorophenol Red-β-D-galactopyranoside (CPRG) substrate to produce the purple chlorophenol red product.

(40) In some embodiments, the reporter gene comprises a catalytic nucleic acid. Examples of catalytic nucleic acids include, but are not limited to, a ribozyme, an RNA-cleaving deoxyribozyme, a group I ribozyme, RNase P, a Hepatitis delta ribozyme, and DNA-zymes.

(41) In some embodiments, the reporter gene comprises an antigen for which a specific antibody or antibody fragment is available. In some embodiments, a reporter gene comprises an antibody, which when expressed, binds to a complementary antigen.

(42) Methods of Expression

(43) Provided herein are methods for expressing a transgene or a protein encoded by a transgene sequence using a nucleic acid construct provided herein having a modified inducible promoter as described herein. In some embodiments, a transgene can be expressed in a host cell that is toxic to the host cell or a non-naturally occurring organism (e.g., a bacterium). Such transgenes can be difficult to express utilizing a pET or other vector containing a leaky promoter, because the leaky expression upon transformation can inhibit growth of the host cell and thereby either prevent expression or express the transgene as an inclusion body. Indeed, tight OFF state control can allow the host cell to reach mid-log phase growth prior to induction, thus allowing for improved expression of the toxic transgene in the prokaryotic cell relative to expression via a vector having a leaky promoter.

(44) The nucleic acid constructs provided herein can comprise a transgene encoding a protein or a fragment thereof. In some embodiments, the transgene or a protein encoded by the transgene is not toxic to the host cell. A protein and/or peptide or fragment thereof can be any protein of interest, for example, but not limited to: biological proteins; mutated proteins; therapeutic proteins; truncated proteins, and the like. Proteins can also be selected from a group comprising: mutated proteins, genetically engineered proteins, peptides, synthetic peptides, recombinant proteins, chimeric proteins, antibodies, midibodies, tribodies, humanized proteins, humanized antibodies, chimeric antibodies, modified proteins and fragments thereof. A transgene provided herein can comprise a gene that is, for example, part of a biosynthesis pathway for the production of a protein, an organic compound, or a molecule of interest. The cells, systems, and organisms provided herein provide for a facile method of manufacturing such proteins.

(45) The modified inducible promoters provided herein can be introduced to any cell type or any system that can transcribe and translate the protein encoded by the transgene downstream of the promoter. In some embodiments, a cell provided herein is a prokaryotic cell or a eukaryotic cell. In some embodiments, the system is a cell-free system that can be used to produce the protein or organic compound of interest. A cell-free system is a composition comprising a set of

reagents capable of providing or supporting a biosynthetic reaction (e.g., transcription reaction, translation reaction, or both) in vitro in the absence of cells. For example, to provide for a transcription reaction, a cell-free system comprises promoter-containing DNA, RNA polymerase, ribonucleotides, and a buffer system. Cell-free systems can be prepared using enzymes, coenzymes, and other subcellular components either isolated or purified from eukaryotic or prokaryotic cells, including recombinant cells, or prepared as extracts or fractions of such cells. A cell-free system can be derived from a variety of sources, including, but not limited to, eukaryotic and prokaryotic cells, such as bacteria including, but not limited to, *E. coli*, *P. putida*, *V. natriegens*, thermophilic bacteria and the like, wheat germ, rabbit reticulocytes, mouse L cells, Ehrlich's ascitic cancer cells, HeLa cells, CHO cells, or budding yeast. In some embodiments, the cell-free system comprises an RNA polymerase. In some embodiments, the cell-free system comprises components sufficient for the translation reaction. In some embodiments, the cell-free system comprise ribosomes, aminoacyl transfer RNAs, translation factors, and a buffer system. The components can also comprise amino acids or amino acids and aminoacyl tRNA synthetases. Components of translation factors are disclosed, for example, in Shimizu and Ueda, "Pure Technology," Cell-Free Protein Production: Methods and Protocols, Methods in Molecular Biology, Endo et al. (Eds), Humana 2010, the contents of which is incorporated herein by reference in its entirety. Exemplary translation factors include, but are not limited to, factors responsible for protein biosynthesis are initiation factors (IF1, IF2, and IF3), elongation factors (EF-G, EF-Tu, and EF-Ts), and release factors (RF1, RF2, and RF3), as well as RRF for termination.

(46) Prior to induction, the modified inducible promoter provided herein does not substantially express the transgene in the absence of an inducer (e.g. isopropyl β -d-1-thiogalactopyranoside (IPTG), anhydrotetracycline, and the like). In some embodiments, the amount of expressed transgene produced in the absence of inducer is less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, less than 1%, less than 0.9%, less than 0.8%, less than 0.7%, less than 0.6%, less than 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, less than 0.1%, less than 0.09%, less than 0.08%, less than 0.07%, less than 0.06%, less than 0.05%, less than 0.04%, less than 0.03%, less than 0.02%, or less than 0.01% of an amount of expressed transgene produced in the presence of the inducer. In some embodiments, the amount of expressed transgene produced in the absence of inducer is less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, less than 1%, less than 0.9%, less than 0.8%, less than 0.7%, less than 0.6%, less than 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, less than 0.1%, less than 0.09%, less than 0.08%, less than 0.07%, less than 0.06%, less than 0.05%, less than 0.04%, less than 0.03%, less than 0.02%, or less than 0.01% of an amount of expressed transgene produced by a comparable nucleic acid having a weak promoter (e.g. lacI) promoter in the absence of the inducer.

(47) Expression of the transgene is performed by contacting the host cell or cell-free system provided herein with an inducer (e.g. isopropyl β -d-1-thiogalactopyranoside (IPTG), anhydrotetracycline, and the like). Non-limiting examples of inducers include: a chemical, a compound, and organic compound, a protein, an analyte, tetracycline and derivatives thereof, metallothionine, ecdysone, cocaine, hormones, steroids, and antibiotics (e.g., rapamycin, kanamycin). Exemplary environmental inducers include exposure to heat (i.e., thermal pulses or constant heat exposure), light (e.g., photoirradiation within the defined range of wavelengths), various steroidal compounds, divalent cations (including Cu.sup.2+ and Zn.sup.2+), galactose, tetracycline, IPTG (isopropyl- β -D thiogalactoside), as well as other naturally occurring and synthetic inducing agents and gratuitous inducers.

(48) In some embodiments, the amount of expressed transgene produced in the presence of inducer is at least equal to the amount of expressed transgene produced by a comparable nucleic acid having a lacUV5 promoter and a T7 polymerase in the presence of the inducer. In some embodiments, the amount of protein expressed in the presence of the inducer is at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, at least 100%, at least 105%, at least 110%, at least 115%, at least 120%, at least 125%, at least 130%, at least 135%, at least 140%, at least 145%, at least 150%, at least 155%, at least 160%, at least 165%, at least 170%, at least 175%, at least 180%, at least 185%, at least 190%, at least 195%, at least 200%, at least 300%, at least 400%, at least 500%, at least 600%, at least 700%, at least 800%, at least 900%, or at least 1000% an amount of expressed transgene produced by a comparable nucleic acid having a strong (e.g., T7) promoter in the presence of the inducer.

(49) Such expression can be carried in a variety of prokaryotic host cells. Indeed, while most expression is carried out in T7 lysogenized *E. coli* DE3 strains, the nucleic acid constructs of the present disclosure having a modified inducible promoter can be utilized in prokaryotic cells in the absence of T7 lysogenization.

(50) The host cell for expression of a nucleic acid provided herein (e.g., a transgene) can comprise a prokaryotic cell or a eukaryotic cell. In some embodiments, the prokaryotic cell is a bacterial cell. In some embodiments, a prokaryotic cell can include any Gram-positive strain bacterial cell. In some embodiments, a prokaryotic cell can include any Gram-negative strain bacterial cell. Examples of bacterium that can be used include but are not limited to: non-T7 lysogenized *E. coli* strains such as DH5 α , DH10 β , or W3110, *P. putida*, *P. aeruginosa*, *H. influenzae*, *C. trachomatis*, *P. mirabilis*, *P. vulgaris*, *C. pneumoniae*, *K. pneumoniae*, *N. gonorrhoeae*, *H. pylori*, *A. cholera*, *S. aureus*, *S. enterica*, *C. jejuni*, *B. fragilis*, *L. pneumophila*, *V. parahaemolyticus*, and *V. natriegens*.

(51) The host cell for expression can be a eukaryotic cells, e.g., a mammalian cell, an insect cell, a yeast cell, a fungal cell, and the like. A nucleic acid construct provided herein can be regulated in a cell-specific or tissue-specific manner

such that it is only active in describing the associated coding region of a given transgene in a specific tissue type(s).

(52) Exemplary Biosynthesis Pathways for Producing Gene Products

(53) Provided herein are methods of producing a protein, an organic compound, or a molecule using the inducible promoters provided herein, a cell expressing an inducible promoter provided herein, a non-naturally occurring organism provided herein, or kits provided herein. In some embodiments, the non-naturally occurring organism provided herein comprises a nucleic acid construct comprising one or more inducible promoters provided herein and one or more sequence encoding a transgene. In some embodiments, the one or more transgene comprises a sequence encoding a protein in a biosynthesis pathway. In some embodiments, the nucleic acid construct provided herein comprise a biosynthesis pathway transgene. In some embodiments, the biosynthesis pathway is a polyketide synthesis pathway, a terpene synthesis pathway, a non-ribosomal peptide biosynthesis pathway, or a carotenoid biosynthesis pathway. In some embodiments, the cell expressing an inducible promoter provided herein or the a non-naturally occurring organism provided herein are cultured for a period of time. In some embodiments, the period of time is for at least 4 hours, 6 hours, 10 hours, 12 hours, or more. In some embodiments, the period of time is for at least 24 hours. In some embodiments, the period of time is for at least 96 hours.

(54) In some embodiments, the biosynthesis pathway is a carotenoid synthesis pathway. Industrially useful carotenoids are generally produced by chemical synthesis processes for which possibility of undesired actions such as contamination of synthesis auxiliary materials is a major concern for the quality of the product. In addition, tastes of consumers tend to lean toward naturally-occurring carotenoids. However, there is a limit to extraction of carotenoids from plants and natural products, and an effective industrial process is not entirely established. As a production method of naturally-occurring carotenoids, microbial fermentation methods have been used. However, none of such cases enable production of carotenoids in an amount which is enough for economical industrial production. In many cases, through classical mutation and breeding, wild-type of carotenoid producing microorganisms do not generate enough carotenoid product for large-scale manufacturing. carotenoid biosynthesis pathway is made up of various enzymes, and genes encoding such enzymes have been analyzed by many researches. In a typical pathway, for example, carotenoid is synthesized in its early stage by an isoprenoid biosynthesis pathway which is shared by steroid and terpenoid, starting from mevalonic acid which is a basic metabolite. Farnesyl pyrophosphate having 15 carbons (C15) generating through the isoprenoid basic synthesis system is condensed with isopentenyl diphosphate (IPP) (C5), to give geranylgeranyl diphosphate (GGPP) (C20). Then through condensation of two molecules of GGPP, colorless phytoene which is the first carotenoid is synthesized. The phytoene is then converted into lycopene through a series of unsaturation reactions, and then the lycopene is converted into β -carotene through a cyclization reaction. Then, a hydroxyl group and a keto group are introduced into the (3-carotene, which leads synthesis of various xanthophylls represented by astaxanthin.

(55) Provided herein are methods for producing a carotenoid, the method comprises culturing the non-naturally occurring organism provided herein or a plurality of non-naturally occurring organisms under conditions and for a sufficient period of time and contacting the non-naturally occurring organism with an inducer, thereby producing the carotenoid. In some embodiments, the cells or non-naturally occurring organisms provided herein are cultured in a cell culture medium (e.g., a lysogeny broth, also called LB broth or Luria Broth). In some embodiments, the cell or non-naturally occurring organisms are cultured in a bioreactor, a spinning flask, or a vessel suitable for cell growth and survival.

(56) In some embodiments, the transgene encodes a polypeptide having such an enzymatic activity that converts a methylene group at 4 position in β -ionone ring into a keto group. In some embodiments the transgene comprises a crtW gene. In some embodiments, the transgene encodes a polypeptide having such an enzymatic activity that adds one hydroxyl group to a carbon at 3-position of 4-keto- β -ionone ring and/or at 3-position of β -ionone ring. In some embodiments, the transgene comprises a crtZ gene sequence. In some embodiments, the transgene encodes for a polypeptide having such an enzymatic activity that converts lycopene into β -carotene. In some embodiments, the transgene comprises a crtY gene sequence. In some embodiments, the transgene encodes for a polypeptide having such an enzymatic activity that converts phytoene into lycopene. In some embodiments, the transgene comprises a crtI gene sequence. In some embodiments, the transgene encodes for a polypeptide having prephytoene synthase activity. In some embodiments, the transgene comprises a crtB gene sequence. In some embodiments, the transgene encodes for a polypeptide having geranylgeranyl diphosphate synthase activity. In some embodiments, the transgene comprises a crtE gene sequence. In some embodiments, the transgene encodes for a polypeptide having a lycopene elongase/hydratase activity. In some embodiments, the transgene comprises a crtEB gene sequence. In some embodiments, the transgene comprises a crtEBI gene sequence. In some embodiments, the transgene comprises a crtEBIY gene sequence. In some embodiments, the transgene comprises a crtEBIYZ gene sequence. In some embodiments, the transgene comprises a crtEBI-YZW gene sequence. In some embodiments, the transgene comprises an ABA1 gene sequence. In some embodiments, the transgene comprises an ABA2 gene sequence. In some embodiments, the transgene comprises a sequence that is at least 85% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the comprises a sequence that is at least 90% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the comprises a sequence that is at least 95% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the comprises a sequence that is at least 99% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the comprises any one of SEQ ID NOS: 69-73, 76-79.

(57) Further provided herein are methods of producing benzoic acid. In some embodiments, the biosynthetic pathway gene is a CocE gene. In some embodiments, the transgene comprises a sequence that is at least 85% identical to SEQ ID

NO: 74. In some embodiments, the transgene comprises a sequence that is at least 90% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises a sequence that is at least 95% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises a sequence that is at least 99% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises SEQ ID NO: 74.

(58) Exemplary Embodiments:

(59) Provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TXTXXTGT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TXGXCX sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence, wherein X is any nucleobase. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAAT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprises a transgene. Further provided herein are nucleic acid constructs, wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a prokaryotic cell in the presence of an inducer; and wherein: (i) the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and (ii) the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAATGT-sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA-sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a transgene. Further provided herein are nucleic acid constructs, wherein the transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, an ABA1 gene, an ABA2 gene, and a CocE gene. Further provided herein are nucleic acid constructs, wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a cell in the presence of an inducer. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein a strong promoter increases the amount of expression of a transgene provided herein relative to a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TXTXXTGT, or TXGXCX. Further provided herein are nucleic acid constructs, wherein the strong promoter increases the amount of expression of a transgene provided herein by at least 10% relative to the expression of a transgene expressed by a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TXTXXTGT, or TXGXCX. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a sequence encoding a reporter gene. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise one or more regulatory element. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a terminator sequence.

(60) Provided herein are nucleic acid constructs comprising: a sequence that is at least 85% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence that is at least 90% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence that is at least 95% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence that is at least 99% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73.

(61) Provided herein are nucleic acid constructs comprising: (a) a first modified inducible promoter sequence provided herein; and (b) a second modified inducible promoter sequence, wherein the second modified inducible promoter sequence comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a second bacterial ribosome binding sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid construct further comprises a transgene. Further provided herein are nucleic acid constructs, wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is in the presence of an inducer. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the

absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a terminator sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise one or more regulatory elements. Further provided herein are nucleic acid constructs, wherein the transgene comprises a carotenoid biosynthesis pathway gene. Further provided herein are nucleic acid constructs, wherein the transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, an ABA1 gene, an ABA2 gene, and a CocE gene.

(62) Provided herein are compositions, wherein the compositions comprise two or more nucleic acid constructs provided herein.

(63) Provided herein are cells comprising a nucleic acid construct or a composition provided herein. Further provided herein are cells, wherein the cells are prokaryotic cells. Further provided herein are cells, wherein the cells are bacterial cells. Further provided herein are cells, wherein the cells are Gram-negative bacterial cells. Further provided herein are cells, wherein the cells are *P. putida* bacterial cells or *V. natriegens* bacterial cells. Further provided herein are cells, wherein the cells are eukaryotic cells. Further provided herein are cells, wherein the cells are mammalian cells.

(64) Provided herein are cell-free systems comprising a nucleic acid construct provided herein, a composition provided herein and/or an RNA polymerase. Further provided herein are cell-free systems, wherein the cell-free systems further comprise ribosomes, aminoacyl transfer RNAs, translation factors, and a buffer.

(65) Provided herein are methods of expressing a protein encoded by a transgene in a cell, the methods comprising: (a) transforming a cell with a nucleic acid construct, wherein the nucleic acid construct comprises a modified inducible promoter that comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; (iii) a nucleic acid that encodes a bacterial ribosome binding sequence; and a transgene; and then (b) contacting the cell with an inducer, thereby expressing the protein encoded by the transgene. Further provided herein are methods, wherein the transgene when cloned into a comparable nucleic acid construct that comprises a strong promoter inhibits growth of the cell prior to the contacting with the inducer, thereby preventing expression of the transgene via the comparable nucleic acid. Further provided herein are methods, wherein the cell is a prokaryotic cell. Further provided herein are methods, wherein the prokaryotic cell is a bacterium. Further provided herein are methods, wherein the cell does not comprise a T7 promoter or a T7 polymerase. Further provided herein are methods, wherein when in the absence of the inducer, the expression of the protein encoded by the transgene is lower relative to the expression of the protein in a comparable cell comprising the transgene operably linked to a T7 promoter. Further provided herein are methods, wherein when in the presence of the inducer the expression of the protein encoded by the transgene is greater relative to the expression of the protein in a comparable cell comprising the transgene operably linked to a T7 promoter.

(66) Provided herein are non-naturally occurring organisms comprising a nucleic acid construct comprising: (a) a first inducible promoter sequence comprising: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence; (b) one or more of a biosynthesis pathway transgene; and (c) a second modified inducible promoter sequence comprising: (i) a TATAATGT consensus sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. Further provided herein are non-naturally occurring organisms wherein the one or more biosynthesis pathway transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, and a CocE gene. Further provided herein are non-naturally occurring organisms, wherein the non-naturally occurring organism comprises a bacterium. Further provided herein are non-naturally occurring organisms wherein the bacterium is a Gram-negative bacterium. Further provided herein are non-naturally occurring organisms, wherein the Gram-negative bacterium is a *P. putida* bacterium or a *V. natriegens* bacterium. Further provided herein are non-naturally occurring organisms, wherein the bacterium is a Gram-positive bacterium. Further provided herein are non-naturally occurring organisms, wherein the nucleic acid construct comprises a terminator sequence. Further provided herein are non-naturally occurring organisms, wherein the nucleic acid construct comprises one or more regulatory element. Further provided herein are non-naturally occurring organisms, wherein the nucleic acid construct comprises a reporter gene.

(67) Provided herein are compositions, wherein the compositions comprise: the non-naturally occurring organism provided herein; and an inducer. Further provided herein are compositions, wherein the inducer comprises: anhydrotetracycline (aTc), isopropyl β -d-1-thiogalactopyranoside, cocaine, metallothionine, ecdysone, an antibiotic agent, galactose, a steroid, or a divalent cation.

(68) Provided herein are methods for producing a carotenoid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the carotenoid. Further provided herein are methods, wherein the non-naturally occurring organism comprises a

nucleic acid comprising a gene selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, and a crtEBI-YZW gene.

(69) Provided herein is a carotenoid produced by the methods provided herein, wherein the carotenoid is lycopene, beta-carotene, zeaxanthin, canthaxanthin, or astaxanthine.

(70) Provided herein are methods for producing benzoic acid, the methods comprising: culturing the cell provided herein; or the non-naturally occurring organism of provided herein; and contacting the cell or the non-naturally occurring organism with an inducer, thereby producing benzoic acid. Further provided herein are methods, wherein the cell or the non-naturally occurring organism comprises a nucleic acid comprising a CocE gene.

(71) Provided herein are compositions comprising benzoic acid, wherein the benzoic acid is produced by the methods provided herein.

(72) Provided herein are kits comprising the nucleic acid construct provided herein, packaging, buffers, and materials therefor.

(73) Provided herein are kits comprising the cell provided herein, packaging, culture medium, buffers, and materials therefor.

(74) Provided herein are kits comprising the non-naturally occurring organism provided herein, packaging, culture medium, buffers, and materials therefor.

(75) Provided herein are nucleic acid constructs comprising a modified inducible promoter, wherein the modified inducible promoter comprises: a TATAAT consensus sequence at position -10, relative to a transcriptional start site of the promoter; a TTGACA sequence at the -35 position, relative to a transcriptional start site of the promoter; a bacterial ribosome binding sequence; wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a cell in the presence of an inducer; and wherein: the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct.

(76) Provided herein are isolated prokaryotic cells comprising a nucleic acid construct provided herein.

(77) Provided herein are methods of expressing a transgene toxic to a prokaryotic cell, the methods comprising: transforming a nucleic acid construct into the prokaryotic cell, wherein the nucleic acid construct comprises a modified inducible promoter comprises: a TATAAT consensus sequence at position -10, relative to a transcriptional start site of the promoter; a TTGACA sequence at the -35 position, relative to a transcriptional start site of the promoter; a bacterial ribosome binding sequence; and contacting the prokaryotic cell with an inducer, thereby expressing the transgene; wherein the transgene when cloned into a comparable nucleic acid construct that comprises a strong promoter inhibits growth of the prokaryotic cell prior to the contacting with the inducer, thereby preventing expression of the transgene via the comparable nucleic acid.

(78) Provided herein are methods of expressing a transgene in a prokaryotic cell that does not comprise a T7 RNA polymerase, the methods comprising: transforming a nucleic acid construct into the prokaryotic cell, wherein the nucleic acid construct comprises a modified inducible promoter comprises: a TATAAT consensus sequence at position -10, relative to a transcriptional start site of the promoter; a TTGACA sequence at the -35 position, relative to a transcriptional start site of the promoter; a bacterial ribosome binding sequence; and contacting the prokaryotic cell with an inducer, thereby expressing the transgene; wherein the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a T7 promoter in a comparable nucleic acid construct; and the expression of the transgene in the presence of the inducer is greater than an amount of expression in the presence of the inducer of the transgene operatively coupled to the T7 promoter in the comparable nucleic acid construct.

(79) Provided herein are nucleic acids, nucleic acid constructs, compositions, cells, non-naturally occurring organisms, cell-free systems, kit, or method provided herein.

(80) For a better understanding of the present disclosure and of its many advantages, the following examples are given by way of illustration and without limiting the scope of this disclosure.

EXAMPLES

Example 1: Inducible Promotor Architecture

(81) In exemplary embodiments, the -10 and -35 boxes of the original lac promoter (FIG. 1Ai), which are targeted by sigma 70 (σ_{70}), were modified to generate improved variants such as the lacUV5 (FIG. 1Aii) and tacI (FIG. 1Aiii) promoters. In the lacUV5 promoter, the -10 box of the original lac promoter was replaced by the Pribnow box (TATAAT). Later the lacUV5 and trp promoters were combined to create the tacI promoter, which increased transcription 11-fold compared to its predecessor with the incorporation of the -35 consensus box TTGACA. The tet promoter (FIG. 1Bi) shares the highly conserved -35 hexamer with tac promoter, however it does not contain the Pribnow box.

(82) Provided herein are exemplary lac and tet expression systems with improved strength, control, and portability.

(83) The genetic architecture of the lac and tet expression systems were modified in three ways: (1) addition of the consensus -10 and -35 sequence boxes to be strongly targeted by σ_{70} ; (2) incorporation of a strong ribosome binding site recognized by a broad spectrum of Gram-negative bacteria; and (3) independent control of the transcriptional regulators by appropriately-tuned constitutive promoters.

(84) The results were validated with the reporter protein mCardinal, which significantly improves the dynamic range of promoter measurements over more commonly used green fluorescent proteins. The improvement seen in the mCardinal dynamic range is due to intrinsic fluorescence of many bacterial species that interferes with measurements in the green wavelengths, and the bacteria provided herein have reduced autofluorescence in the far red spectrum. Additionally, the inducible promoters provided herein were compared with the pET system with the production of the cocaine esterase CocE, a thermosensitive enzyme capable of metabolizing cocaine into benzoic acid. CocE, which is prone to form inclusion bodies in leaky *E. coli* expression systems, is expressed as an inducible and soluble protein using the promoters provided herein. The results and assays provided in Example 2 further support that the expression system provided herein is a significant improvement over available expression systems in providing tight OFF state control while achieving high yields of recombinant protein, and with the advantage of direct portability to alternative host species.

Example 2: Generation of Inducible Promoters in Gram Negative Bacteria

(85) Bacterial Strains

(86) *E. coli* DH10B, *P. putida* JE90 derivative of KT2440 with BxB1int-attB, and *V. natriegens* Vmax X2 (Codex DNA, Inc.) were used to evaluate the synthetic lac and tet promoters. The plasmid pET21a-mCardinal was evaluated in *E. coli* BL21 strain. Selective markers kanamycin (50 µg/mL for *E. coli* and *P. putida*, and 400 µg/mL for *V. natriegens*), ampicillin (100 µg/mL for *E. coli*) and spectinomycin (60 µg/mL for *E. coli* and *P. putida*, and 250 µg/mL for *V. natriegens*) were supplemented to LB medium when required. *E. coli* strains were transformed by electroporation and *V. natriegens* by chemical transformation. Integration of plasmids into *P. putida* chromosome was performed by electroporation following the protocol as described, for example, in Elmore et al. *Metab Eng Commun*, 5:1-8 (2017), the contents of which is incorporated herein by reference in its entirety.

(87) Plasmid information is listed in Table 1 below. Any plasmid containing any version of the σ .sup.70-based promoters as shown in FIG. 2A and FIG. 2B has been labeled “ σ .sup.70” followed by information about the construct and the specific promoter. “Lac” and “Tc” constructs are constitutive and do not express a repressor protein; “LacI” and “TcR” constructs are inducible and include expression of LacI or TetR, respectively. Numbers in parentheses indicate a vector backbone other than pJH0204 ((13) for pJH0228 with the pCloDF13 origin of replication, and (19) for pUC19).

(88) TABLE-US-00001 TABLE 1 List of Plasmids and Vectors. Origin of Name Backbone replication Selective marker Construct pJH0204 pJH0204 pColE1 Kanamycin pJH0228 pJH0228 pCloDF13 Spectinomycin/Streptomycin pJH0204F pJH0204 pColE1 Kanamycin pFctacI-GFP pJH0204 pColE1 Kanamycin tacI-sfGFP pFctacI-cardi pJH0204 pColE1 Kanamycin tacI-mCardinal σ .sup.70 V1Lac-cardi pJH0204 pColE1 Kanamycin V1lac-mCardinal σ .sup.70 V2Lac-cardi13 pJH0228 pCloDF13 Spectinomycin/Streptomycin V2lac-mCardinal σ .sup.70 V3Lac-cardi pJH0204 pColE1 Kanamycin V3lac-mCardinal σ .sup.70 V4Lac-cardi pJH0204 pColE1 Kanamycin V4lac-mCardinal σ .sup.70 V1LacI-cardi pJH0204 pColE1 Kanamycin V1lac/lacI-mCardinal σ .sup.70 V2LacI-cardi pJH0204 pColE1 Kanamycin V2lac/lacI-mCardinal σ .sup.70 V2Lac-cardi13 pJH0204 pCloDF13 Spectinomycin/Streptomycin V2lac/lacI-mCardinal σ .sup.70 V3LacI-cardi pJH0204 pColE1 Kanamycin V3lac/lacI-mCardinal σ .sup.70 V4LacI-cardi pJH0204 pColE1 Kanamycin V4lac/lacI-mCardinal σ .sup.70 V1Tc-cardi pJH0204 pColE1 Kanamycin V1Tc-mCardinal σ .sup.70 V2Tc-cardi pJH0204 pColE1 Kanamycin V2Tc-mCardinal σ .sup.70 V3Tc-cardi pJH0204 pColE1 Kanamycin V3Tc-mCardinal σ .sup.70 V4Tc-cardi pJH0204 pColE1 Kanamycin V4Tc-mCardinal σ .sup.70 V1TcR-cardi pJH0204 pColE1 Kanamycin V1Tc/tetR-mCardinal σ .sup.70 V2TcR-cardi pJH0204 pColE1 Kanamycin V2Tc/tetR-mCardinal σ .sup.70 V3TcR-cardi pJH0204 pColE1 Kanamycin V3Tc/tetR-mCardinal σ .sup.70 V4TcR-cardi pJH0204 pColE1 Kanamycin V4Tc/tetR-mCardinal σ .sup.70 V1TcR-cardi19 pUC19 pMB1 Ampicillin V2Tc/tetR-mCardinal pET21a-cardi pET21a pET Ampicillin pET

Molecular Cloning:

(89) A schematic diagram of plasmid construction are shown in FIGS. 1A-1B. Polymerase chain reactions (PCR) were carried out with Phusion DNA polymerase (ThermoFisher Scientific, USA). Digestion of DNA was performed with fast digest restriction enzymes and DNA fragments were join by T4 DNA ligase or Gibson assembly kit (ThermoFisher Scientific, USA). Oligonucleotides and DNA synthesis were ordered to IDT (IDT, USA). DNA sequencing was performed (QuintaraBio, USA). Shuttle vectors pJH0204 and pJH0228 were used. The synthetic Vlac and VTc promoters were synthesized as gBlocks and incorporated into the MCS of pJH0204 using BamHI and XhoI. Further, sfGFP and mCardinal ORF were codon optimized for *P. putida* and inserted downstream the synthetic promoters using NdeI and HindIII. The transcriptional regulator tetR gene was synthesized with the PJ23119 promoter in a single gBlock and tetR was codon optimized for *P. putida* using the IDT DNA codon optimization tool (available on the world wide web at <https://www.idtdna.com/pages/tools/codon-optimization-tool>) and inserted after mCardinal with EcoRI and XhoI. The lacI gene was PCR amplified from pET21a vector and placed under the control of PJ23119 using NcoI and XhoI, but this combination resulted toxic for *E. coli*, therefore lacI was PCR amplified together with its native promoter and ligated after mCardinal using EcoRI and XhoI. pJH0228-V2lac-mCardinal and pJH0228-V2lac/lacI-mCardinal vectors were produced using the same cloning strategy. mCardinal was inserted into pET21a using NdeI and XhoI. The transcriptional unit V2Tc/tetR-mCardinal was inserted into the pUC19 vector by Gibson removing the NdeI RE nucleotide sequence in the plasmid and incorporating terminators insulating the transcriptional unit. DNA sequence of CocE was codon optimized for *P. putida* using the IDT DNA codon optimization tool and cloned into the pET21a and V2Tc-tetR expression system using NdeI and HindIII. A complete list of plasmids generated in this study is shown in Table 1. Primers used in this study are listed in Table 2.

(90) TABLE-US-00002 TABLE 2 List of Primers name SEQ ID NO: Sequence 5' to 3' pUC19
 15 attgcCTCGAGACTGATTTTTTAAGGCGACTGATGAGTCGCCTTTTTTTTGTGTC modified FW
 TAGCTAACTCACATTAATTGCGTTGCGCT XhoI pUC19 16
 gcaatGGATCCagtcaaaagcctccgaccggaggcttttgactAGTACAATC modified FW TGCTCTGATGCCGCATAGTT XhoI
 204F1_fwd 17 Ggcgcgtactccaaaaggatctaggtgaagatc 204F1_rev 18 gagtcttctgattagacaaaaaaaggcgac 204F2_fwd 19
 ttttttgtctaatacagaagaactcgtaagaagg 204F2_rev 20 gcccgcgaggcgaggatctcgctgacgcatg 204F3_fwd 21
 cacgacgagatcctcgccgtcgggcatccgc 204F3_rev 22 gtctagactgcaggaattcaagcttcatatgggatccggacaaaacgAAA 204F4_fwd 23
 aagcttgaattcctgcagcttagaccatggctcgaggacgaacaataaggc 204F4_rev 24 Cctagatccttttgagtagcgccccgggga placI upstream
 25 TACTGGTTTCACATTCACCAC lacI downstream 26 GTGGTGAATGTGAAACCAGTAA lacI upstream 27
 TTTCCAGTCGGGAAACCT lacI 28 AGGTTTCCCGACTGGAAA downstream2 tetR upstream 29
 CCAATACAACGTGGGTGCT FwplacI 30 AACCGGAATTCccccgtggccggg LXhoI RV 31
 AACCGCTCGAGTCACTGCCCCGCTTTCC Fw PstI-lacI 32 attgcCTGCAGTCACTGCCCGCTTTCCAGTCG Rv
 XbaI-lacI 33 gcaatTCTAGAagtcaaaagcctccgaccggaggcttttgactCCCCGTGGC CGGGGGACTG XbaI-V3Lac- 34
 gcctccggtcgaggcttttgact NcoI fwd CTAGAGAGCTGTTGACACTTTATG XbaI-V2Lac- 35
 taatgagctcctacccttactcacCATGGGTACATTCCTCTCTTACTGC NcoI rev XbaI-V3Lac- 36 taatgagctcctacccttactcac
 NcoI_rev 37 CATGGGATTCTGTTTCCTGTGTG Rv cardi XhoI 38
 TCAGTCTCGAGTTACTTGTCTCATCATCCTTAT

(91) Primers were used as sequencing primers (e.g., placI upstream, tetR upstream). 204F primers were used to incorporate BamHI, NdeI, HindIII, KpnI, EcorI, PstI, XbaI, NcoI, XhoI RE into the MCS of pJH0204 and remove NcoI from kanamycin cassette. The pUC19 modified FW XhoI primers were used to incorporate terminators into pUC19 vector and BamHI/XhoI sites.

(92) Sequences of cocE, tetR, and mCardinal, codon optimized for *P. putida*, and the PJ23119 promoter sequence, are provided below.

(93) DNA sequence of mCardinal codon optimized for *P. putida*:

(94) TABLE-US-00003 (SEQ ID NO: 39)

ATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAACATGCACATGAAGC
 TGTATATGGAGGGCACCGTAAACAACCACCACTTCAAGTGTAACACCGA
 GGGTGAAGGTAAACCCTACGAGGGGACGCAGACCCAACGCATCAAGGTC
 GTGGAGGGCGGCCCCGCTGCCTTTTCGATTTCGACATTCTGGCGACCTGTT
 TTATGTACGGCTCGAAGACCTTCATCAACCACACCCAAGGCATCCCGGA
 CTTCTTCAAGCAGAGCTTCCCTGAGGGCTTCACCTGGGAGCGCGTCACC
 ACGTATGAAGACGGTGGGGTGCTCACCGTGACCCAGGACACGAGCTTGC
 AGGATGGCTGCTTGATTTACAACGTCAAGCTGCGCGGGGTGAACTTCCC
 TAGCAACGGGCCAGTGATGCAGAAAAAGACGCTGGGTGTTGGGAGGCCACC
 ACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATA
 TGGCCCTGAAATTGGTCGGCGGCGGTCAATTTGCACTGCAATCTCAAGAC
 CACGTACCGCTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTGGTGTT
 TATTTTGTGCGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGA
 CGTACGTGGAACAGCACGAAGTGGCCGTGGCTCGTTATTGCGATCTGCC
 GTCGAAGCTGGGTCACAAACCTGAACGGCATGGATGAGCTGTACAAAGAT
 TATAAGGATGATGACGACAAGTAA

(95) DNA sequence of sfGFP codon optimized for *P. putida*:

(96) TABLE-US-00004 (SEQ ID NO: 40)

ATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCCATCTGTTGG
 AGCTGGATGGCGACGTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGG
 TGAGGGCGACGCCACGAACGGTAAGCTGACGCTGAAATTCATTTGCACC
 ACCGGCAAATTGCCTGTACCCTGGCCCCACCCTGGTGACCACGCTCACCT
 ACGGCGTACAGTGCTTCAGCCGTTACCCGGACCACATGAAGCGTCACGA
 CTTCTTCAAAAGCGCCATGCCGGAGGGTTACGTGCAGGAGCGTACGATT
 AGTTTCAAGGACGACGGCACCTATAAGACCCGTGCCGAAGTGAAGTTTCG
 AAGGCGATACGTTGGTGAACCGTATCGAGTTGAAGGGTATCGACTTTAA
 GGAAGACGGCAACATCCTGGGCCATAAGCTGGAGTACAATTTCAACAGC
 CATAACGTTTACATCACCGCCGATAAACAGAAGAACGGCATTAAAGCCA
 ACTTTAAGATCCGCCACAACGTCTGAAGACGGCTCGGTGCAGCTGGCCGA
 CCATTATCAGCAAAACACCCCCATCGGTGATGGGCCCGTGCTGCTGCCG
 GATAACCATTATCTGAGCACGCAGTCGGTGCTCAGCAAGGACCCTAACG
 AAAAGCGCGATCACATGGTGCTGCTGGAGTTCGTCACGGCGGCGGGGAT
 CACCCATGGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATGAC AAGTAAA

(97) DNA sequence of tetR codon optimized for *P. putida*:

(98) TABLE-US-00005 (SEQ ID NO: 41)

ATGTCCCGCCTGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGC

TGAATGAAGTCGGTATCGAGGGGGCTGACGACCCGTA AATTGGGCACAAAA
GTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGG
GCATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATACCC
ATTTCTGTCCACTGGAGGGCGAGTCCTGGCAGGACTTTCTCCGCAACAA
CGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAG
GTGCACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGGA
ATCAATTGGCGTTTTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTCT
CTATGCGCTCTCGGCTGTGCGGGCACTTTACGTTGGGGTGCGTGTTGGAG
GACCAGGAGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACCACGG
ACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATCATCA
GGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGT
TTGGA AAAACA ACTCAAGTGTGAAAGCGGGTCCTAA

(99) DNA sequence of cocE codon optimized for *P. putida*:

(100) TABLE-US-00006 (SEQ ID NO: 42)

CATATGGTGGACGGTAATTATTCGGTAGCGTCCAACGTTATGGTGCCGATGCGCGACGGGGTGCGCTTGGCTGTA
GATCTGTACCGCCCGGACGACGAGATGGCCCTGTACCGGTCTGCTGGTCCGCAACCCCTACGACAAATTTCGACGTG
TTCGCTTGGAGTACGCAGAGCACGAACTGGCTGGAATTTGTGCGCGATGGGTACGCCGTCGTCATCCAAGACACC
CGGGGCCTCTTTGCATCCGAAGGTGAGTTTCGTTCCACATGTTGATGACGAGGCGGATGCGGAAGACACGCTGAGC
TGGATCTTGGAAACAAGCATGGTGGCAGGGCAATGTGGGTATGTTTCGGTGTAAGCTACCTGGGCGTTACGCAGTGG
CAAGCTGCTGTTAGCGGTGTGGGTGGTTTGAAGGCAATCGCCCCGAGCATGGCGAGCGCGGATCTGTACCGTGCC
CCCTGGTACGGTCTTGGCGGCGCCCTGAGCGTGGAAGCACTCCTGGGCTGGAGCGCATTGATCGGTACGGGCCTG
ATTACCAGCCGTAGCGATGCCCCGCCGGAAGACGCAGCCGACTTCGTACAGCTGGCAGCCATCCTGAACGATGTG
GCCGGTGCCGCAAGCGTGACCCCTCTGGCCGAACAGCCCTTGTGGGCCGCTGATCCCTTGGGTGATCGACCAG
GTGGTGGACCATCCAGACAACGACGAGTCGTGGCAGAGCATCTCGCTCTTTGAACGTTTGGGTGGGCTCGCTACC
CCGGCCTTGATTACCGCCGGGTGGTACGATGGCTTCGTGGGCGAGAGCCTCCGTACCTTCGTAGCTGTGAAGGAC
AACCGCGGATGCGCGTCTGGTGGTGGGGCCGTGGAGCCACAGCAATCTGACCGGCCGTAATGCCGACCGTAAGTTT
GGGATCGCCGCGACCTACCCCATCCAGGAGGCGACGACCATGCACAAGGCTTTTTTCGACCGGCACCTCCGTGGC
GAGACCGATGCCCTGGCAGGGGTGCCCAAGGTGCGCCTCTTCGTAATGGGTATCGATGAGTGGCGCGACGAGACC
GACTGGCCATTGCCAGATACCGCTTACACGCCTTTTTACCTCGGGGGCTCCGGTGCGGCCAACACGAGCACGGGT
GGTGGGACCCTGTGACCTCGATCAGCGGCACGGAGTCGGCGGACACCTACCTGTATGATCCTGCCGACCCCGTG
CCAAGTCTGGGCGGCACCCTCCTCTTCCATAATGGGGACAACGGTCCAGCTGACCAGCGCCCGATTACGATCGC
GACGACGTGCTGTGCTACTCCACCGAGGTGTTGACCGACCCCGTGGAAGTAACGGGGACGGTTTTCGGCTCGCCTG
TTCGTGTCCTCGTCGGCCGTGGATACCGATTTTACCGCCAAGTTGGTTCGACGTGTTCCCCGATGGTCGGGCAATC
GCTCTCTGCGACGGCATCGTGCGTATGCGCTACCGGGAGACCTTGGTAAATCCTACGCTCATTGAGGCCGGTGAG
ATTTACGAGGTGGCTATTGATATGCTGGCCACCAGCAACGTGTTTTTGCCGGGCCACCGCATCATGGTGCAAGTT
AGCAGCTCGAACTTCCCGAAGTACGACCGCAACTCCAACACCGGCGGCGTCATCGCTCGCGAGCAACTGGAGGAA
ATGTGCACCGCCGTAAACCGCATTACCGCGGCCCCCGAACACCCGTCCCATATCGTGCTGCCGATCATTAAGCGC
GACTATAAGGACGACGACGATAAGTGAAAGCTT

(101) DNA sequence of pJ23119 promoter:

(102) TABLE-US-00007 (SEQ ID NO: 43)

TTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGG AATGTACAC

SDS-PAGE

(103) SDS-PAGE was carried out on a 4-12% Bis-tris Midi Protein Gel in an XCell4 SureLock Midi system (Invitrogen, USA). *E. coli*, *P. putida* and *V. natriegens* cell extracts were obtained from diluting an overnight culture 1:50 in 10 mL of fresh LB media and grown at 37° C. for *E. coli* and 30° C. for *P. putida* and *V. natriegens* at 220 rpm until the culture reached an optical density (OD) at 600 nm (OD.sub.600) of 0.7 measured spectrophotometrically, then induced with 0.2 mM IPTG or 0.1 µg/mL aTc accordingly. Induced cultures were grown for 5 hours, and 2 mL culture were spun down at 14,000 rpm and 4° C. for 10 minutes and frozen at -20° C. for further analysis. Cells were lysed with Complete Bacterial Protein Extraction Reagent (ThermoFisher Scientific), and crude extracts were analyzed by SDS-PAGE. Total protein concentration was estimated with Thermo Scientific NanoDrop one and ~10 mg total protein of each cell extract was loaded into the SDS-gel.

(104) Expression of CocE and Measurement of Benzoic Acid Production

(105) Recombinant *E. coli* BL21 strain carrying the pET21a-cocE and DH10B containing either pσ_{sup}.70 V2TcR-cocE or pσ_{sup}.70 V2TcR-cocE19 plasmids were cultivated overnight at 30° C. at 220 rpm and fresh cultures were started with 5% of the ON culture and induced at ~OD.sub.600 0.7 with 0.2 mM IPTG or 0.1 µg/mL aTc accordingly. 1 mL samples were collected each hour for 6 hours by centrifugation at 4° C. and 14000 rpm and stored at -80° C. Cell pellets were disrupted using 150 µL of B-PE Complete Bacterial Protein Extraction Reagent (ThermoFisher Scientific) for 25 minutes at room temperature and soluble fractions were collected after centrifugation for 25 minutes at 4° C. and 14000 rpm. ~3 mg/mL of soluble fractions were incubated with 0.015 mg of cocaine for 20 minutes at 28° C. and benzoic acid production was estimated using the benzoic acid detection kit for feed (Attogene, EZ2013-03).

Example 3: Design of Synthetic lac and tet Promoters

(106) A schematic diagram of plasmid construction is shown in FIGS. 1A-1B. All plasmids used in the assay are listed in Table 1. The original lac promoter (FIG. 2Ai) lacks the -35 conserved box recognized by the σ .sup.70 subunit of RNA polymerase. Original lac and tet (FIG. 2Bi) promoters lack the Pribnow (-10) box. Three synthetic variants of each promoter were constructed containing the highly conserved σ .sup.70 consensus hexamers at positions -10 and -35. Version 1 (V1lac, FIG. 2Aiv, and V1Tc, FIG. 2Bii) of each promoter resembles the original lac and tet promoters, with the incorporation of a strong RBS from *P. putida*. The version 2 (V2lac, FIG. 2Av, and V2Tc, FIG. 2Biii) replaces the -35 and -10 boxes of Version 1 for the highly conserved σ .sup.70 consensus hexamers TTGACA and TATAAT, respectively. Version 3 (V3lac, FIG. 2Avi, and V3Tc, FIG. 2Biv) incorporates an additional lacO operator and the RBS of the tacI promoter. Version 4 (V4lac, FIG. 2Avii, and V4Tc, FIG. 2Bv) displaces the location of the -10 and -35 boxes (V4lac) and incorporates an additional -35 box in V4Tc. The synthetic promoters were cloned into the MCS of the shuttle vector pJH0204 containing the origin of replication pColE1 with 25-30 copies per cell in *E. coli*. The design allowed the incorporation of the transcriptional regulators lacI and tetR after a terminator L3S2P21 and under the control of the constitutive promoter PJ23119 (FIG. 2C).

(107) No aberrant phenotypes in the *E. coli* strains carrying the PJ23119-tetR construct were observed. However, *E. coli* failed to maintain the PJ23119-lacI construct. The plasmids containing the lacI repressor gene in this configuration produced slow-growing colonies and yielded plasmids with aberrant restriction patterns, indicating gross plasmid rearrangements. Therefore, the strong PJ23119 promoter was replaced for the native lacI promoter and observed no further toxicity.

(108) The reporter protein mCardinal was incorporated downstream of the synthetic promoters and the plasmids were transformed into *E. coli* DH10B, *P. putida* JE90 derivative KT2440, and *V. natriegens* Vmax X2 strains. The V2lac-mCardinal constructs could not be maintained in *E. coli*, likely due to the high strength of the V2lac promoter. To avoid this abnormal phenotype, the genetic circuit V2lac-mCardinal was produced in the pJH0228 vector containing the CloDF13 origin of replication at about 10 copies per cell, allowing stable maintenance of the V2lac promoter.

Example 4: Selection of the Fluorescent Reporter System

(109) GFP and other green fluorescent protein derivatives are widely used as reporters in bacterial expression systems. However, the intrinsic green fluorescence produced by endogenous molecules, such as proteins containing aromatic amino acids, negatively impact the interpretation of the exogenous green fluorescence generated by the reporter system. The autofluorescence is exacerbated in *P. putida*, a member of the fluorescent *Pseudomonas* species. Under iron limited conditions *P. putida* secretes the siderophore pyoverdine, a soluble fluorescent yellow green pigment. Far-red fluorescent proteins were explored, specifically mCardinal, a monomeric far red shifted derivative of mKate, as wavelengths between 600 and 900 nm are not absorbed by cells and organic molecules, thus reducing the noise of endogenous autofluorescence.

(110) To quantify the impact of autofluorescence on the selected Gram-negatives strains, the endogenous fluorescence of these strains cultivated in LB over time were measured. All three species emit fluorescence in the green spectrum, as expected (FIG. 3A, right panel). This fluorescence is more or less constant over time with *E. coli*, however *P. putida* and *V. natriegens* show variable production of molecules that absorb green light at different cell densities. In contrast, when measured in the far-red spectrum specific for mCardinal, there was approximately 400 times less detection of autofluorescence, in arbitrary units on the same instrument, in all strains as compared to the measurements in the green spectrum (FIG. 3A, left panel). Again, *E. coli* displayed rather consistent far red autofluorescence over time whereas *P. putida* and *V. natriegens* autofluorescence varies with cell density.

(111) To further validate the benefit of using mCardinal instead of sfGFP, the inherent noise of each reporter system was quantified by measuring the fluorescence signal-to-background ratio of both reporter systems expressed under the control of the constitutive tacI promoter in the three species. After 16 hours of growth, the recombinant *E. coli*, *P. putida* and *V. natriegens* expressing sfGFP emitted 64, 11 and 5 times more green fluorescence than their respective wild type strains (not expressing a fluorescent protein). Meanwhile, mCardinal-expressing strains displayed 294, 34 and 13-fold higher red-light emission in *E. coli*, *P. putida* and *V. natriegens* compared to their wild type controls (FIGS. 3B-3D). Overall, the use of mCardinal rather than sfGFP significantly improves the dynamic range and facilitates measurement of promoter strength, as there is little endogenous autofluorescence in the far-red spectrum.

Example 5: Strength and Regulation of Synthetic lac Promoters in *Escherichia coli*

(112) The lac promoter, and its derivatives, are constitutively active in the absence of its transcriptional regulator lacI. The strength of the synthetic lac promoters were evaluated without lacI and compared these against the strong tacI promoter, which drives high levels of transcription and can result in recombinant protein production of up to 30% of total protein. The V1Lac promoter was 10-fold weaker than the tacI promoter (FIG. 4A), consistent with previous data observing an 11-fold difference between these two promoters. The V3lac promoter matched the tacI promoter strength, while the V4lac promoter showed 5 times less mCardinal than tacI (FIG. 4A). The V2lac promoter had to be tested in the low copy plasmid pCloDF13 because the medium copy pColE1 derived plasmid could not be maintained in *E. coli*. Even in the low copy configuration, the V2Lac promoter surpassed by ~1.4-fold the tacI promoter and was therefore the strongest constitutive promoter despite its expression in a low copy plasmid. The result suggests that maintenance of a promoter of this strength within a medium copy plasmid exceeds the sustainable metabolic burden of *E. coli* (FIG. 4A). These results indicate that the incorporation of σ .sup.70 consensus sequences at positions -35 and -10 efficiently increase the strength of the lac promoter.

(113) Next, the transcriptional regulator lacI was incorporated into these circuits to quantify the efficiency of the OFF state. The V1Lac and V4lac constructs containing the lacI repressor produced a red fluorescence signal comparable to wild type *E. coli* (FIG. 4A), indicating tight transcriptional repression. The V3lac promoter could not be totally repressed by lacI, showing a slight tendency to leak, even with the presence of two lacO operators. The V2lac promoter with lacI was stably maintained by *E. coli* in both pColE1 and CloDF13-derived vectors. However, the repressed state of the V2lac promoter in the low and medium copy plasmids showed rather a constitutive behavior. LacI repressed ~9-fold the V2Lac promoter in the CloDF13 derived plasmid as compared to its constitutive construct (FIG. 4A), while the medium copy version of the V2lac-lacI system (which was not sustainable as a constitutive circuit) exhibited a strong constitutive expression ~1.3-fold higher than the tacI promoter, thus showing that one lacO operator region is not sufficient to block transcription of V2Lac containing both $\sigma_{sup.70}$ consensus sequences by LacI, but incorporation of a second lacO reduces the leakage as observed in the V3Lac promoter (FIG. 4A).

(114) To verify the inducibility of the synthetic lac promoters, *E. coli* was exposed to IPTG at the beginning of exponential phase. The V1lac and V4lac promoters showed minimal induction of mCardinal (FIG. 4A). The V3lac promoter, which mimics the V2lac promoter with an additional lacO operator, demonstrated a dynamic range of ~17-fold versus the uninduced state, but could only produce 24% of its full constitutive potential (FIG. 4A). The V2lac promoter in the low copy plasmid produced ~2-fold more mCardinal upon induction; similar behavior was observed in the medium copy version, where production of mCardinal reached the maximum values of all promoters tested at ~2.6-fold stronger than the tacI promoter (FIG. 4A). Overall, the results demonstrate that incorporation of $\sigma_{sup.70}$ consensus sequences to the lac promoter improve its strength (V2Lac), producing more mCardinal than the strong tacI promoter in both, the constitutive LacI-less version in the low copy plasmid and the repressed and induced states in the medium copy plasmid. However, LacI could not repress the transcription of the synthetic V2Lac promoter. Thus, additional lacO operators are necessary to turn the lac promoter OFF containing the perfect $\sigma_{sup.70}$ boxes, as observed in V3Lac, though full induction then becomes impossible in this architecture.

Example 6: Strength and Regulation of Synthetic tet Promoters in *Escherichia coli*

(115) The tet promoter also drives transcription constitutively without TetR. The expression profile of the constitutive synthetic tet promoters were evaluated and compared against the tacI promoter. As observed for the lac promoter, replacement of the -10 and -35 sequences with the $\sigma_{sup.70}$ consensus boxes in the tet promoter also increased the constitutive efficiency of the V2Tc promoter over the original V1Tc, in this case by 20%, and reached the yields obtained with tacI promoter (FIG. 4B). The V4Tc promoter showed similar results to the V2Tc, while the V3Tc lost any capability to initiate transcription (FIG. 4B). Further, incorporation of the tetR repressor to each of the four circuits under the control of the strong constitutive PJ23119 promoter completely silenced mCardinal production from all the synthetic tet promoters, even in the strong V2Tc promoter which harbors the optimal combination of consensus sequences recognized by the $\sigma_{sup.70}$ (FIG. 4B).

(116) To confirm the induction of each promoter and determine its functional dynamic range, each circuit was induced with anhydrotetracycline (aTc). The original tet promoter (V1Tc) only achieved ~37% of its full constitutive potential, reaching only 30% of mCardinal compared to tacI promoter confirming the middle level strength of the tet promoter (FIG. 4B). The V2Tc promoter achieved maximal induction reaching the full potential dynamic range and produced similar yields as the strong tacI promoter (FIG. 4B). Interestingly, the V4Tc promoter could not be de-repressed by aTc and showed a weak induction of only 5%, and as expected, the V3Tc promoter showed no activity (FIG. 4B). Together these results confirm that in *E. coli* the incorporation of the consensus boxes targeted by the $\sigma_{sup.70}$ is key to boost the expression profile of promoters recognized by this transcription initiation factor and achieve the maximal transcriptional capacity of these promoters upon induction.

Example 7: Direct Comparison of Synthetic lac and tet Expression Systems to pET in *Escherichia coli*

(117) In *E. coli*, the pET series of expression plasmids are the most popular systems for recombinant protein production. The efficiency of the synthetic lac and tet promoters were compared directly against pET21a. The pET expression system in the BL21 strain can produce more than 50% of the target gene as the total protein per cell, mCardinal production was measured in BL21. As expected, the pET system induced by IPTG produced ~1.5-fold more mCardinal than the strong constitutive tacI promoter, which is expected to accumulate up to 30% of total cell protein. Despite the massive production of mCardinal by the pET system, its OFF or uninduced state can be considered as a medium high constitutive expression system, yielding 30% of the constitutive tacI promoter as evidenced by mCardinal production (FIG. 4B, FIG. 4C). This transcriptional leakage in the pET system is well known in the scientific community.

(118) Among the synthetic lac and tet promoters, all tested in *E. coli* DH10B, only the V2Lac promoter surpassed the pET system in the medium copy plasmid pColE1. The IPTG induced V2Lac expression system produced ~1.8 times more recombinant protein than the pET system, however, the leakage of V2Lac promoter equals the induced state of the pET system (FIG. 4A and FIG. 8), therefore cancelling the advantages of an inducible expression system. The V3Lac expression system showed a tighter control of the OFF state, therefore, to increase its strength the V3Lac-lacI construct containing mCardinal was inserted into the pUC19 vector, thus increasing the copy number from 30 to 250 copies per cell. *E. coli* failed to maintain the V3lac-lacI construct in the pUC19 vector, likely due to the toxicity of LacI. The V2Tc-tetR expression system was evaluated in the pUC19 vector. *E. coli* stably maintained the V2Tc-tetR construct and no phenotypic or genotypic abnormalities were observed as in pUC19-V3Lac-lacI. mCardinal production by pUC19-V2Tc-tetR matched the pET21a expression system, with one exceptional difference, the pUC19-V2Tc-tetR maintained complete

repression in the uninduced after 12 hours (FIG. 4B and FIG. 8). Thus, pUC19-V2Tc-tetR shows significant improvement in transcriptional control and dynamic range over pET21a (FIG. 4B, C). Further, equivalent protein production and tighter transcriptional control were achieved without the obligatory use of the BL21(DE3) strain, suggesting the system provided herein is readily portable to other strains and species.

Example 8: Expression of the Cocaine Esterase CocE and Production of Benzoic Acid

(119) To further validate the advantage of the expression system provided herein over the pET expression system, the production of a functional protein product, the cocaine esterase CocE, was assayed. This enzyme hydrolyses cocaine into benzoic acid and could expand the use of the narcotic compound as raw source in the production of the carboxylic acid widely used as precursor and preservative in the food and pharmaceutical industries. Expression of CocE has previously been shown to be a difficult and laborious task with the pET expression system, because CocE forms inclusion bodies; consequently, long incubation methods at low temperatures are required to isolate sufficient yields of functional CocE. *E. coli* could not support CocE expression in the repressor-less variants of the synthetic promoters, thus confirming this is a toxic protein for *E. coli*. Next, CocE in the V2Tc-tetR expression system was assessed in both the medium and high copy plasmid backbones, pJH0204 and pUC19 respectively (p σ 70 V2TcR-cocE and p σ 70 V2TcR-cocE, see Table 1), and compared against the pET expression system for the production of benzoic acid. Benzoic acid production is an indication that CocE was correctly folded and properly hydrolyzing cocaine.

(120) The soluble fraction recovered from the recombinant *E. coli* strain containing the pET21a-cocE expression system showed equivalent benzoic acid production in the presence of cocaine in both the uninduced and induced states (FIG. 5B), with a clear tendency to form inclusion bodies, as evidenced by the accumulation of protein in the insoluble fraction upon activation by IPTG (FIG. 5A). Thus, while induction of CocE by IPTG in the pET system does result in the additional production of protein, the additional protein made is insoluble and non-functional protein. Functionally speaking, the expression of CocE in pET21a is not inducible, but instead behaves constitutively, as there is no difference in cocaine hydrolysis in the uninduced and induced states (FIG. 5B).

(121) The soluble fractions from the recombinant *E. coli* V2Tc-tetR-cocE expression system, in both the pJH0204 and pUC19 plasmids, showed no benzoic acid production in the absence of the inducer. The addition of aTc triggered production of CocE with protein production and function roughly equivalent to the soluble pET21a-cocE plasmid (FIG. 5A, FIG. 5B). In both plasmids, benzoic acid production was observed after only one hour of aTc induction, and reached maximal production by 3 hours (FIG. 5B). These results confirmed the advantage of the incorporation of the σ .sup.70 boxes to the tetracycline promoter, keeping the toxic enzyme CocE OFF in the uninduced state and producing sufficient amounts of mature CocE after 3 hours of induction. This is significantly shorter than induction of 16 hours previously recorded (data not shown). Further, the expression system provided herein produced soluble and functional protein without evidence of massive toxicity or inclusion body formation. This method facilitates the scalability of this bioprocess which has potential as an alternative, environmentally friendly method to obtain benzoic by replacing petroleum-based starting materials.

Example 9: Strength and Regulation of Synthetic lac and tet Promoters in *Pseudomonas putida*

(122) Genetic control in the soil-dwelling species *P. putida* is of great interest for biotechnological applications due to the ability of this microorganism to synthesize complex natural products and metabolize a variety of xenobiotic compounds. The constitutive strength, repression, and inducibility of synthetic promoters in *P. putida* were assayed. The lac-based promoter systems (which include the tacI promoter) are reported to have poor dynamic range and instability in high-copy plasmids when harbored in *P. putida* as replicative plasmids. To address this problem, the genetic circuits were integrated into the *P. putida* chromosome. The constitutive strength of V1lac, V2lac, V3lac and V4lac promoters were measured using mCardinal, and found out that tacI is stronger than all the synthetic lac variants by ~3, 1.5, 10 and 33-fold respectively (FIG. 6A, FIG. 6C). The incorporation of the transcriptional regulator efficiently repressed the V1lac, V3lac and V4lac promoters, but as seen in *E. coli*, the V2lac showed a tendency to leak, though not as egregiously as in *E. coli* (FIG. 6A, FIG. 6C). Induction by IPTG was barely detected in the V1lac, V3lac and V4lac promoters, thus indicating the weakness of the original lac promoter in *P. putida* (FIG. 6A, FIG. 6C). Interestingly and analogously to *E. coli*, the V2lac promoter showed ~27-fold increase in mCardinal production against the uninduced state and surpassed its theoretical dynamic range as compared to the constitutive version, ultimately achieving similar yields as the strong tacI promoter (FIG. 6A, FIG. 6C). These results confirm that the incorporation of the σ 70 consensus sequences to the lac promoter (V2Lac) significantly improve the strength of this promoter in the *P. putida* host.

(123) The tet promoters in *P. putida* were also evaluated. The original tet promoter with a strong RBS (V1Tc) was the strongest among all synthetic tet promoters in the absence of tetR, followed by V4Tc, V2Tc and V3Tc (FIG. 6B, FIG. 6C). Incorporation of the tetR repressor under the control of the PJ23119 promoter silenced all the synthetic tet promoters, as in *E. coli* (FIG. 6B, 6C). Addition of aTc activated the four tet promoters and remarkably, the induced V2Tc promoter showed the highest mCardinal induction exceeding its constitutive expression by ~10-fold and matching the tacI promoter. The V2Tc promoter was revealed to be ~2 and 3-fold more efficient than the V1Tc and V4Tc promoters after induction respectively, while the V3Tc proved to be inefficient in *P. putida* (FIG. 6B, FIG. 6C). These results highlight the importance of the consensus sequences targeted by σ 70 to amplify the strength of inducible promoters also in *P. putida*.

Example 10: Strength and Regulation of Synthetic lac and tet Promoters in *Vibrio natriegens*

(124) The circuits were next evaluated in the marine bacterium *V. natriegens* which has gained popularity for routine molecular biology applications due its ability to double in <10 min. In this host, previous studies indicated that the tacI

promoter produced GFP upon activation with IPTG, while induction of the tet promoter resulted in low GFP yields. The constitutive and inducible versions of the synthetic lac and tet promoters were evaluated in the pColE1 derived plasmid with ~300 copies per cell, except for the constitutive V2Lac, which was evaluated in pCldDF13 derived vector with ~64 copies per cell in this strain. The constitutive V2Lac and V4Tc promoters outperformed tacI by ~16 and ~2-fold respectively, while V3Lac, V1Tc and V2Tc produced similar levels as tacI (FIGS. 7A-7C). V1Lac and V3Lac underperformed tacI by ~2.7-fold, and V3Tc showed no activity (FIGS. 7A-7C). Incorporation of the transcriptional regulators lacI and tetR turned OFF all the synthetic promoters except for V2Lac, which continued to show leakage, as in *E. coli* and *P. putida* (FIG. 7A). Induction of the synthetic lac promoters by IPTG revealed that only the V2Lac promoter was fully activated, surpassing its constitutive version by ~1.3-fold and showed a potential dynamic range of ~25-fold (FIG. 7A). The V3Lac reached 30% of its full potential, and the V1Lac and V4Lac showed no induction (FIG. 7A). While addition of aTc activated all four synthetic tet promoters, the V3Tc and V4Tc promoters rather showed weak mCardinal production, and V1Tc produced 2-fold more mCardinal than its constitutive variant (FIG. 7B). The V2Tc promoter, which harbors the 2 consensus $\sigma_{sup.70}$ boxes, demonstrated strong inducibility producing 11-fold more mCardinal than the tacI promoter and displaying full potential of its dynamic range producing 100-fold more mCardinal than its OFF version (FIG. 7B, C). These results confirmed that the adaptation of the two consensus boxes recognized by the $\sigma_{sup.70}$ in the V2lac and V2Tc promoters improved the performance of the inducible lac and tet promoters in *V. natriegens*.

Example 11: Additional Considerations for Inducible Promoter Design

(125) Production of recombinant proteins is and will continue to be one of the main tools scientists use to understand biological processes and transfer academic results to industrial applications. The pET system is well known to induce the formation of inclusion bodies, a major drawback in the production of soluble proteins. pET requires specific strains that carry the T7 RNA polymerase, it lacks real tuneability, and fails to keep the target gene OFF. Leakiness in the OFF state negates the main advantage of an inducible system, which is intended to permit time- or context-specific control of gene expression.

(126) In contrast to pET, the results provided in Examples 2-9 show that inclusion of the $\sigma_{sup.70}$ consensus sequences into the lac and tet inducible promoters improve both repression and induction. The tight transcriptional control does not require any particular strain background, and permits rapid expression of soluble proteins, including toxic proteins such as CocE. The inducible promoters provided herein can be easily optimized to be used in a variety of Gram-negative hosts, including *P. putida* and *V. natriegens*, thus widening the applicability of these tools to a broad spectrum of bacteria. The inducible promoters provided herein offer a significant advance to the biotechnology industry in offering additional platforms for exogenous protein expression and purification.

(127) The V2lac promoter expressed very highly across three Gram-negative species. Indeed, in *V. natriegens* the V2lac promoter is the strongest promoter yet described, as it surpasses the widely used tacI promoter by 16-20-fold (FIG. 7). Additional factors can affect promoter performance, for instance, the length and sequence of the spacer between the -10 and -35 elements can have dramatic effects on promoter strength.

(128) The results in FIG. 5 suggest that complete OFF state control is key to avoiding common problems with protein expression of challenging proteins, such as the formation of inclusion bodies. Future promoter configurations could be aided by the combination of additional mechanisms to achieve complete OFF state control, such as the additional of small RNA transcriptional and translational regulators. These approaches have increased appeal over the pET system with pLysE, for example, as they permit facile portability among strains and species. OFF state control of the lac promoters was less of an issue in *P. putida* and *V. natriegens*, where better repression of the strong V2lac promoter was achieved (FIGS. 6A-7C). Without being bound by a particular theory, it is likely that species-specific differences in RNA polymerase binding and promoter clearance may affect the behavior of inducible promoters.

(129) The challenges of leakiness were entirely eliminated by the use of tet-based promoters in the system provided herein. An often-cited drawback of tet promoters for protein expression was the limitation of protein production as compared to lac-based systems. Incorporation of the $\sigma_{sup.70}$ consensus sequence into the tet promoter (V2Tc) significantly increased expression above that of the original tet promoter (V1Tc) to the point where the promoter output equaled (and in the case of *V. natriegens*, exceeded) that of the tacI promoter. As a complementary approach to inducible promoter manufacturing and design, the sigma factor can be replaced with $\sigma_{sup.70}$ sequences to assess promoter performance. Promoter performance was quantified herein by using far-red reporters, which significantly decreased background fluorescence and increased dynamic range in the bacterial species assayed.

(130) Different yields of recombinant protein can be achieved based on promoter and host selection. Overall, the incorporation of the consensus -35 sequence and Pribnow (-10) box unlocks the strength of the lac and tet promoters in Gram-negative bacteria, facilitating the production of any given target gene in different host with the same set of plasmids.

(131) The V2Tc promoter provided herein offers an improvement over the pET-based protein expression systems that remain very widely used. V2Tc is tightly regulated, robustly inducible, and drives protein production comparable to or better than the tacI promoter in all three species that were examined. No specific strain backgrounds were necessary, as it does not rely on the presence of the T7 polymerase.

Example 12: Development of the Dual Expression System V2TcR/V2(3)LacI

(132) The $\sigma_{sup.70}$ adapted expression systems V2TcR and V3LacI have tight regulation and strong induction in *E. coli* in the presence of anhydrotetracycline (aTc) and IPTG respectively, while in *P. putida* and *V. natriegens* the V2TcR and

V2LacI showed the best performance as inducible promoters in these host bacteria. A dual expression system was developed using as backbone the pJH0204 vector containing the origin of replication pColE1 and the attB sites specific for bxb1 recombinase. Each transcriptional unit of the dual expression system was insulated by terminators and the transcriptional regulators tetR and lacI were located in-between and in opposite direction of the inducible promoters V2Tc and V2/3Lac to block undesired transcription of the open reading frames (ORF). Further, 2 different multicloning sites were located after the inducible promoters V2Tc and V2/3Lac to facilitate the incorporation of target genes (FIGS. 9A-9F, and FIG. 14).

(133) The dual expression systems were tested in *E. coli*, *P. putida* and *V. natriegens* to evaluate i) synergy ii) expression of different recombinant proteins and iii) production of the biosynthetic gene cluster (BGC) leading to the biosynthesis of lycopene and B-carotene. Synergy was evaluated with the reporter system mCardinal which was accommodated in both inducible promoters of the dual expression system (FIG. 9B). Expression of different recombinant proteins was performed by adapting the sfGFP under the control of V2TcR and mCardinal controlled by V2/3LacI (FIG. 9C). Finally, production of the terpenoids lycopene and B-carotene by the dual expression system was achieved by the assembly of the crtEBIY genes from *Pantoea ananatis*. The crtEBIY genes were synthesized with the codon usage for *E. coli*. The crtEBI operon was assembled in the pUC19 vector via Gibson adapting the ribosome binding site (RBS) of J23119 9 to crtB and the RBS pJL1 to crtI, further the synthetic construct was transferred to the V2TcR expression system to evaluate lycopene production (FIG. 9D). B-carotene production was evaluated with the incorporation of the crtY gene with the J23119 RBS to V2TcR-crtEBI yielding V2TcR-crtEBIY (FIG. 9E). The dual control of the crtEBI operon together with the crtY gene to produce B-carotene was achieved by adapting the crtEBI to the V2TcR expression system and the crtY gene to V2(3)LacI expression system present in the dual vector (FIG. 9F). All measurements were performed using LB as growth medium as the scope of this work is not to optimize the production of terpenoids. After transforming the plasmids listed in Table 3 in the three bacterial species no difference in the wild type strains was observed, thereby confirming the stability of the dual expression systems.

(134) TABLE-US-00008 TABLE 3 List of Plasmids. Expression Name system 1 ORF1 system 2 ORF2
 pσ_{sup.70} V2TcR-mCardinal V2TcR mCardinal pσ_{sup.70} V2lacI-mCardinal V2lacI mCardinal pσ_{sup.70} V3LacI-mCardinal V3LacI mCardinal pσ_{sup.70} V2TcR-sfGFP V2TcR sfGFP pσ_{sup.70} V2lacI-sfGFP V2lacI sfGFP pσ_{sup.70} V3LacI-sfGFP V3LacI sfGFP pJH0204F pσ_{sup.70} V2TcR-mCardinal/ V2TcR mCardinal V2LacI mCardinal V2LacI-mCardinal pσ_{sup.70} V2TcR-mCardinal/ V2TcR mCardinal V3LacI mCardinal V3LacI-mCardinal pσ_{sup.70} V2TcR-sfGFP/ V2TcR mCardinal V2LacI sfGFP V2LacI-mCardinal pσ_{sup.70} V2TcR-sfGFP/ V2TcR mCardinal V3LacI sfGFP V3LacI-mCardinal pUC19-crtEBI — crtEBI pσ_{sup.70} V2TcR-crtEBI V2TcR crtEBI pσ_{sup.70} V2TcR-crtEBIY V2TcR crtEBIY pσ_{sup.70} V2TcR-crtEBI/ V2TcR crtEBI V2LacI crtY V2LacI-crtY pσ_{sup.70} V2TcR-crtEBI/ V2TcR crtEBI V3LacI crtY V3LacI-crtY Note: SEQ ID NOS: are listed in the SEQUENCES section of this paper.

Example 13: Synergy of the Dual Expression System V2TcR/V2(3)LacI

(135) Increasing the copy number in *E. coli* efficiently boosted production of mCardinal by the pσ₇₀V2TcR expression system but had a negative impact in the pσ₇₀ V3LacI expression system due to the toxicity of the transcriptional regulator LacI. Therefore, the synergistic effect of the duet expression system pσ₇₀V2TcR/V3LacI both controlling mCardinal as different studies demonstrated that combined promoters enhance production of the target protein. The *E. coli* strain containing the duet expression system V2TcR/V3LacI produced 2-fold and 4-fold less red fluorescence than the strains containing the V2TcR and V3LacI when induced with aTc and IPTG respectively (FIG. 10A). Synergy was observed by the addition of IPTG and aTc to the V2TcR/V3LacI strain improving mCardinal production by 1.5-fold against the induction obtained with aTc alone (FIG. 10A). However, the synergy in mCardinal production in the duet system reached equal amounts of mCardinal as the V2TcR expression system (FIG. 10A). These results indicate that the tandem version of the synthetic promoters V2TcR/V3LacI offer an alternative to co-express different genes at high levels using different inducers.

(136) In *P. putida* the V2TcR and V2lacI expression systems showed the best performance activating transcription of heterologous proteins. Consequently, we integrated both promoters controlling expression of mCardinal into the chromosome of this host. Single induction of the dual system by aTc resulted in induction of mCardinal equivalent to V2TcR system; single induction of the dual system with IPTG resulted in a 3-fold decrease relative to V2lacI alone (FIG. 10B). However, the addition of both inducers (aTc+IPTG) increased mCardinal by 1.1-fold over the V2TcR and V2lacI expression system, thus showing a minimal synergistic effect of the dual promoters in *P. putida* (FIG. 10B).

(137) The duet expression system V2TcR/V2LacI decreased substantially the efficiency of the synthetic promoters in *V. natriegens* compared to the single expression system counterparts by 12 and 2-fold for V2LacI and V2TcR after induction with IPTG and aTc, respectively (FIG. 10C). Co-induction with IPTG+aTc did not have a synergistic effect in mCardinal production, but reached similar yields as obtained by induction with aTc alone (FIG. 10C). Together, this information demonstrates that synergy cannot be achieved with the dual expression system V2TcR/V2(3)LacI in medium copy plasmids as is the case for *E. coli* and *V. natriegens*. On the contrary, the duet expression system diminishes the activity of each promoter probably due to the constitutive expression of the transcriptional regulators TetR and LacI, and competition of the σ_{sup.70} for both promoters. In *P. putida* the duet expression system was directly integrated into the chromosome and reduced the quantities of the transcriptional regulators TetR and LacI. Surprisingly, in this host the V2TcR expression system gained strength in the dual version, the V2lacI expression system lost efficiency, and synergy was observed, thus indicating that duet expression system gain performance as single copies in the chromosome of the host bacterium.

Example 14: Expression of Different Recombinant Proteins with the Dual Expression System V2TcR/V2(3)LacI (138) Prior to the expression systems provided herein there was not a universal dual expression system that allows differential expression of distinct set of genes via two independent inducible promoters that is portable among different Gram-negative species. The V2TcR/V2(3)LacI dual expression system demonstrated that it can control the co-expression of mCardinal, although synergy was only observed in *P. putida*. Two different reporter proteins were expressed to evaluate their production in *E. coli*, *P. putida* and *V. natriegens* using the V2TcR/V2(3)LacI expression system. The transcriptional unit V2TcR controlling the expression of sfGFP while the V2(3)LacI unit controlling mCardinal production.

(139) The three bacterial species under study containing V2TcR-sfGFP/V2(3)LacI-mCardinal expression system showed no green fluorescence in the uninduced state, thus confirming the tight regulation of the V2TcR expression system, however, the V2(3)LacI-mCardinal is prone to leak and red fluorescence was observed in the absence of IPTG (FIGS. 11A-11C). Addition of aTc induced the production of sfGFP, and similar result was observed with the addition of IPTG which triggered production of mCardinal. Interestingly, when the cultures were co-induced with aTc+IPTG the green and red fluorescence reached similar levels compared with the cultures induced with only one inducer, thus indicating that both promoters (V2TcR/V2(3)LacI) are unaffected by the activation of each other reaching their maximum activity either in their sole activation or their independent combined activation (FIGS. 11A-11C).

Example 15: Production of Lycopene and B-Carotene in *E. coli*, *P. putida* And *V. natriegens* with the Expression System V2TcR

(140) To decipher the capability of the dual-expression system V2TcR/V2(3)LacI to produce natural products we first had to test the ability of the V2TcR expression system to control the expression of multiple genes, we therefore decided to evaluate the production of lycopene and B-carotene in *E. coli*, *P. putida* and *V. natriegens*. Lycopene production has already been demonstrated in *E. coli* and *P. putida* via heterologous expression of the lycopene-producing operon (LYC) containing the crtEBI genes from *Pantoea ananatis*. Lycopene is the biosynthetic precursor of B-carotene, thus expression of the LYC operon together with crtY yields B-carotene production, which also has been demonstrated in *E. coli* and *V. natriegens*. The LYC and the crtEBIY operons were accommodated in the V2TcR expression system and lycopene and B-carotene production were measured by UHPLC after 4 hours of induction with aTc. In *E. coli* production of both, lycopene and B-carotene, was achieved but the induced culture produced only 1.7 and 1.2-fold more lycopene and B-carotene respectively than the uninduced culture, thus demonstrating that the tight control of the V2TcR can't totally repress the expression of the crtEBI and crtEBIY genes (FIGS. 12A-12B). Lycopene biosynthesis in *E. coli* using the pET expression system also demonstrated abundant leakage and production of 170 mg/L lycopene after 40 hours of induction, 1.1-fold more than the uninduced culture. Despite the V2TcR expression system produced 94 mg/L, this was achieved after 4 hours of induction, against the 40 hours reported with the pET expression system. This result confirms the advantage of the V2TcR expression system over the pET expression system as described in Examples 2-10. In *P. putida* and *V. natriegens* containing the V2TcR-crtEBI (SEQ ID NO: 70) and V2TcR-crtEBIY (SEQ ID NO: 71) transcriptional units no lycopene or B-carotene production was observed in the uninduced state, and activation of by aTc yielded production of the terpenoids (FIGS. 12A-12B). *P. putida* was reported to produce 1.22 ng/mL of lycopene after 24 hours of growth with the pSEVA421 vector, with the V2TcR expression system lycopene production reached 17 mg/L after 4 hours of cultivation, which is a substantial increment (13.000×), and B-carotene production by *P. putida* reached 840 mg/L, the highest among the three species under study (FIG. 12). *V. natriegens* was reported to produce 2.93 mg/L of B-carotene, while the V2TcR expression system achieved 597 mg/L of B-carotene and 31 mg/L of lycopene. These results demonstrate that the V2TcR can be efficiently programmed for production of biosynthetic gene clusters in Gram-negative bacteria.

Example 16: Coordinated Expression of the Lycopene and B-Carotene Biosynthetic Gene Clusters (BGC) with the Dual Expression System V2TcR/V2(3)LacI

(141) Natural products are encoded by BGC and ability to activate each gene at a different time point with a different inducer represent an advantage when the genes are toxic, or the metabolites produce by the BGC affect the cell growth of the heterologous host. As provided herein, the V2TcR expression system can activate the production of lycopene and B-carotene. In order to control the production of B-carotene with two different inducers, aTc and IPTG were used. The LYC operon was incorporated to the V2TcR expression system and the crtY gene was controlled by the V2(3)LacI expression system. The plasmid pø.sup.70 V2TcR-crtEBI/V3LacI-crtY (SEQ ID NO: 73) was transformed into *E. coli* and the plasmid pø.sup.70 V2TcR-crtEBI/V2lacI-crtY (SEQ ID NO: 72) was transformed into *P. putida* and *V. natriegens*. In the three bacteria species no lycopene production was observed despite the V2TcR expression system was controlling the crtEBI genes and activation by aTc should trigger production of lycopene (FIG. 13A). This result is explained in *E. coli* by the leakage of the crtEBIY genes under the control of the synthetic promoters V2TcR and V3LacI, demonstrating the high processivity of the B-carotene BGC to metabolize the endogenous IPP isopentenyl diphosphate directly into B-carotene. However, the V2TcR-crtEBI showed no leakage in *P. putida* and *V. natriegens*, but the V2lacI expression system was reported to leak, consequently, the lycopene produced after addition of aTc is directly transformed into B-carotene by the CrtY present in the cells due its minimal expression and high processivity.

(142) B-carotene production in *E. coli* was observed no matter the absence or presence of the inducers aTc and IPTG (FIG. 13B), thus demonstrating that the V2TcR/V3LacI expression system failed to keep downregulated the B-carotene BGC in this host. In *P. putida* and *V. natriegens* B-carotene production was only achieved when the cultures were

induced by aTc or with aTc+IPTG (FIG. 13A). IPTG induction alone did not conduce to B-carotene production because the crtEBI genes were not transcribed by the V2TcR expression system. Addition of aTc resulted in B-carotene levels similar to induction with aTc+IPTG (FIG. 13A), thus indicating that the minimal leakage of the V2lacI-crtY is sufficient to metabolize the lycopene produced by the V2TcR-crtEBI into B-carotene.

(143) Importantly, the yields of B-carotene reached by *E. coli* were always lower than the production achieved by *P. putida* and *V. natriegens*. Thus, *P. putida* can be used to produce complex biosynthetic gene clusters. Therefore, a dual expression system developed in this study expands the genetic tool kit of *P. putida* to produce complex proteins. Also, *V. natriegens* outperformed *E. coli*, and the dual expression system allowed higher production of B-carotene compared with the V2TcR by 1.2-fold. Together these results demonstrate the potential of the dual expression system presented here to fully capitalize the potential of heterologous host in industrial applications.

(144) TABLE-US-00009 SEQUENCES FIG. 2A Sequences: Lac (SEQ ID NO: 3)

AGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACA
GCTATG lacUV5 (SEQ ID NO: 4)

AGGCTTTACACTTTATGCTTCCGGCTCGTATAATGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACA
GCTATG tacI (SEQ ID NO: 5)

GAGCTGTTGACAATTAATCATCGGCTCGTATAATGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACA
GAATCATATG Vilac (SEQ ID NO: 6)

AGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAATTGTGAGCGGATAACAACGCAGTAAGAGAGGAA
TGTACATATG V2lac (SEQ ID NO: 7)

GAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGTAAGAGA
GGAATGTACATATG V3lac (SEQ ID NO: 8)

GAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGAATTGTGA
GCGGATAACAATTTACACAGGAAACAGAATCATATG V4lac (SEQ ID NO: 9)

CTTTATGCTTCCGGCTCGTTGACAGTGTGGAATTGTGAGCGGATAACAATATAATGTGTGGAATTGTGAGCGGAT
AACAATTTACACAGGAAACAGAATCATATG FIG. 2B Sequences: Tet (SEQ ID NO: 10)

GGATCCTTGACACTCTATCATTGATAGAGTTATTTTACCACTCCCTATCAGTGATAGAGAAAAGTGAAATG
V1Tc (SEQ ID NO: 11)

GGATCCTTGACACTCTATCATTGATAGAGTTATTTTACCACTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGA
ATGTACATATG V2Tc (SEQ ID NO: 12)

GAGCTGTTGACAACCTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGAAT
GTACATATG V3Tc (SEQ ID NO: 13)

GAGCTGTTGACAACCTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGAGTGAATTGTGAGCGGA
TAACAATTTACACAGGAAACAGAATCATATG V4Tc (SEQ ID NO: 14)

TTGACACTCTATCATTGATAGAGTTTGACATCCCTATCAGTGATAGAGATATAATGTGTGGAATTGTGAGCGGAT
AACAATTTACACAGGAAACAGAATCATATG SEQ ID NOS: 15-38 are provided in Table 2. SEQ

ID NOS: 39-43 are provided in Example 2. SEQ ID NOS: 44-53 are provided in the
Detailed Description. FIG. 14 Full-length Sequence: V2Tc/tetR-V2lac/lacI (SEQ ID NO: 54)

GAGCTGTTGACAACCTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGAAT
GTACATATGAAGCTTCTCGGTACCAAATTCAGAAAAGAGGGCCTCCCGAAAGGGGGGCTTTTTTTCGTTTTGGTC
CGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTACACATGTCCCGCC
TGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTA
AATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATG
CCCTCGCTATTGAAATGTTGGATCGCCACCATACCCATTTCTGTCCACTGGAGGGCGAGTCTTGGCAGGACTTTC
TCCGCAACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGCACTTGGGCACCC
GTCCCACCGAAAAACAATACGAAACCTTGGAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTTAGCTTGGAGA
ATGCTCTCTATGCGCTCTCGGCTGTGCGGCACCTTACGTTGGGGTGCCTGTTGGAGGACCAGGAGCATCAAGTCG
CAAAAGAGGAGCGTGAAACCCCAACCACGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATC
ATCAGGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTTGAAAAACAACCTCAAGTGTG
AAAGCGGGTCCTAACTGCAGTCACTGCCCGCTTTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCG
GCCAACGCGCGGGGAGAGGCGGTTTGGCTATTGGGCGCCAGGGTGGTTTTTCTTTTACCAGTGAGACGGGCAAC
AGCTGATTGCCCTTACCGCCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGGCCCAGCAGGCGA
AAATCCTGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTTCGTATCCCACTACCGAG
ATATCCGCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTTGCGCCAGCGCCATCTGATCGTTGGCAACC
AGCATCGCAGTGGGAACGATGCCCTCATTACGCAATTTGCATGGTTTTGTTGAAAACCGGACATGGCACTCCAGTCG
CCTTCCCGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGACGCAGACGCGCC
GAGACAGAACTTAATGGGCCCCGCTAACAGCGCGATTGTGCTGGTGACCCAATGCGACCAGATGCTCCACGCCCAGT
CGCGTACCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGAAATAACGCCGGA
ACATTAGTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATCAGCCCAGTACG
CGTTGCGCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCGCGCTTCGTTCTACCATCGACACCACCAG
CTGGCACCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTTGCGACGGCGCGTGCAGGGCCAGACTGGAG
GTGGCAACGCCAATCAGCAACGACTGTTTGGCCGCCAGTTGTTGTGCCACGCGGTTGGGAATGTAATTCAGCTCC

CCATATCCGCGGCTTCCTCCCGCTTTTTCGCGGAGAAACAGTGGCTGGCTTCACCAACCGGGAACCGGTC
TGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTCACATTCACCACCCTGAATTGACTC
TCTTCCGGGCGCTATCATGCCATAACCGCGAAAGGTTTTGCGCCATTTCGATGGTGTCCGGGATCTCGACGCTCTCC
CTTATGCGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGGCCGTTGAGCACCGCCGCCGCAAGGAATGG
TGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagtc aaaagcctccgggtcgagggttttgact
TCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGT
AAGAGAGGAATGTACCCATGGCCATGGCTCGAGgacgaacaataaggcctccctaacggggggcctttttattg ataacaaa
Additional sequences: V2Tc/tetR-V3lac/lacI (SEQ ID NO: 55)
GAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGAAT
GTACATATGAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCCCGAAAGGGGGGCTTTTTTCGTTTTGGTC
CGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTACACATGTCCCGCC
TGGATAAATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTA
AATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATG
CCCTCGCTATTGAAATGTTGGATCGCCACCATAACCATTTCTGTCCACTGGAGGGCGAGTCCTGGCAGGACTTTC
TCCGCAACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGCACTTGGGCACCC
GTCCACCGAAAAACAATACGAAACCTTGGAATAATCAATTGGCGTTTTTTGTGCCAGCAAGGGTTTAGCTTGGAGA
ATGCTCTCTATGCGCTCTCGGCTGTGCGGCACTTTACGTTGGGGTGCGTGTGGAGGACCAGGAGCATCAAGTCG
CAAAAGAGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATC
ATCAGGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTTGAAAAACAACCTCAAGTGTG
AAAGCGGGTCCTAACTGCAGTCACTGCCCGCTTTCCAGTCGGGAAACCTGTCTGTCAGCTGCATTAATGAATCG
GCCAACGCGCGGGGAGAGGCGGTTTTCGCTATTGGGCGCCAGGGTGGTTTTCTTTTACCAGTGAGACGGGCAAC
AGCTGATTGCCCTTCACCGCCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGCCCCAGCAGGCGA
AAATCCTGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTCGTATCCCACTACCGAG
ATATCCGCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTGCGCCAGCGCCATCTGATCGTTGGCAACC
AGCATCGCAGTGGGAACGATGCCCTCATTACGCATTTGCATGGTTTGTGAAAACCGGACATGGCACTCCAGTCG
CCTTCCCGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGACGCAGACGCGCC
GAGACAGAACTTAATGGGCCCGCTAACAGCGCGATTGTCTGGTGACCCAATGCGACCAGATGCTCCACGCCCAGT
CGCGTACCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGAAATAACGCCGGA
ACATTAGTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATCAGCCCACTGACG
CGTTGCGCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCGCGCTTCGTTCTACCATCGACACCACCAG
CTGGCACCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTTGCGACGGCGCGTGCAGGGCCAGACTGGAG
GTGGCAACGCCAATCAGCAACGACTGTTTGCCCGCCAGTTGTTGTGCCACGCGGTTGGGAATGTAATTCAGCTCC
GCCATCGCCGCTTCCACTTTTTTCCCGCGTTTTTCGCGAGAAACGTGGCTGGCCTGGTTCACCACGCGGGAAACGGTC
TGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTCACATTCACCACCCTGAATTGACTC
TCTTCCGGGCGCTATCATGCCATAACCGCGAAAGGTTTTGCGCCATTTCGATGGTGTCCGGGATCTCGACGCTCTCC
CTTATGCGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGGCCGTTGAGCACCGCCGCCGCAAGGAATGG
TGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagtc aaaagcctccgggtcgagggttttgact
TCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGA
TTGTGAGCGGATAACAATTTACACAGGAAACAGAATCCCATGGCTCGAGgacgaacaataaggcctccctaacg
gggggcctttttattgataacaaa

(145) TABLE-US-00010 TABLE 3 List of Plasmids pσ.sup.70 V2TcR-mCardinal (SEQ ID NO: 56)
ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGGTGAGTAAGGGTGAGGAGCTCATTAAAGGAGAACATGCACATGAAGCTGTATATGGAGGGC
ACCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCCTACGAGGGGACGCAGACCCAACGC
ATCAAGGTCGTGGAGGGCGGCGCCGCTGCCTTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCTCGAAG
ACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGCTTCACCTGGGAGCGC
GTCACCACGTATGAAGACGGTGGGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGATTAC
AACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGCCAGTGATGCAGAAAAAGACGCTGGGTTGGGAGGCC
ACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATATGGCCCTGAAATTGGTCGGCGGC
GGTCATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTGGTGT
TATTTTGTGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCC
GTGGCTCGTTATTGCGATCTGCCGTGGAAGCTGGGTACAAAACCTGAACGGCATGGATGAGCTGTACAAAGATTAT
AAGGATGATGACGACAAGTAAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCCCGAAAGGGGGGCTTTT
TTCGTTTTTGGTCCGAATTTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTAC
ACATGTCCCGCCTGGATAAATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGC
TGACGACCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGG
CATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATAACCATTTCTGTCCACTGGAGGGCGAGTCCT
GGCAGGACTTTCTCCGCAACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGC
ACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGGAATAATCAATTGGCGTTTTTTGTGCCAGCAAGGGT
TTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTGCGGCACTTTACGTTGGGGTGCGTGTGGAGGACCAGG
AGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCG

AACTCTTCGATCATCAGGCGCGGAGCCAGGACCTTCTCTTTGGCTGGAGCTGATTCTGCGGTTGGGAAAAAC
AACTCAAGTGTGAAAGCGGGTCCTAACTGCAGTCTAGACCATGGctcgag V2lacI-mCardinal (SEQ ID
NO: 57)
GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGT
AAGAGAGGAATGTACATATGGTGAGTAAGGGTGAGGAGCTCATTAAAGGAGAACATGCACATGAAGCTGTATATGG
AGGGCACCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCCTACGAGGGGACGCAGACCC
AACGCATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCT
CGAAGACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGCTTCACCTGGG
AGCGCGTCACCACGTATGAAGACGGTGGGGTGCTCACCCTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGA
TTTACAACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGGCCAGTGATGCAGAAAAAGACGCTGGGTTGGG
AGGCCACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATATGGCCCTGAAATTGGTCTG
GCGGCGGTCAATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTG
GTGTTTATTTTGTGCGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAG
TGGCCGTGGCTCGTTATTGCGATCTGCCGTCAAGCTGGGTACAAACTGAACGGCATGGATGAGCTGTACAAAG
ATTATAAGGATGATGACGACAAGTAAAAGCTTCTCGGTACCAAATTCCAGAAAAGAGGCCTCCCGAAAGGGGGGC
CTTTTTTCGTTTTTGGTCCGAATTCCCCCGTGGCCGGGGGACTGTTGGGCGCCATCTCCTTGCATGCACCATTCCT
TGCGGCGGCGGTGCTCAACGGCCTCAACCTACTACTGGGCTGCTTCTTAATGCAGGAGTCGCATAAGGGAGAGCG
TCGAGATCCCGGACACCATCGAATGGCGCAAAACCTTTCGCGGTATGGCATGATAGCGCCCGGAAGAGAGTCAAT
TCAGGGTGGTGAATGTGAAACCAGTAACGTTATACGATGTCGCAGAGTATGCCGGTGTCTCTTATCAGACCGTTT
CCCGCGTGGTGAACCAGGCCAGCCACGTTTCTGCGAAAACGCGGGAAAAAGTGGAAGCGGCGATGGCGGAGCTGA
ATTACATTCCCAACCGCGTGGCACAACAACCTGGCGGGCAACAGTCGTTGCTGATTGGCGTTGCCACCTCCAGTC
TGGCCCTGCACGCGCCGTGCGCAAATTGTCGCGGCGATTAAATCTCGCGCCGATCAACTGGGTGCCAGCGTGGTGG
TGTCGATGGTAGAACGAAGCGGCGTCAAGCCTGTAAAGCGGCGGTGCACAATCTTCTCGCGCAACGCGTCAGTG
GGCTGATCATTAACCTATCCGCTGGATGACCAGGATGCCATTGCTGTGGAAGCTGCCTGCACTAATGTTCCGGCGT
TATTTCTTGATGTCTCTGACCAGACACCCATCAACAGTATTATTTTCTCCCATGAAGACGGTACGCGACTGGGCG
TGGAGCATCTGGTCGCATTGGGTCAACCAGCAAATCGCGCTGTTAGCGGGCCCATTAAGTTCTGTCTCGGCGCGTC
TGCGTCTGGCTGGCTGGCATAAATATCTCACTCGCAATCAAATTCAGCCGATAGCGGAACGGGAAGGCGACTGGA
GTGCCATGTCCGGTTTTTCAACAAACCATGCAAATGCTGAATGAGGGCATCGTTCCTCACTGCGATGCTGGTTGCCA
ACGATCAGATGGCGCTGGGCGCAATGCGCGCCATTACCGAGTCCGGGCTGCGCGTTGGTGCGGATATCTCGGTAG
TGGGATACGACGATACCGAAGACAGCTCATGTTATATCCCGCCGTTAACCACCATCAAACAGGATTTTCGCCTGC
TGGGGCAAACCAGCGTGGACCGCTTGCTGCAACTCTCTCAGGGCCAGGCGGTGAAGGGCAATCAGCTGTTGCCCG
TCTCACTGGTGAAAAGAAAAACCACCCTGGCGCCCAATACGCAAACCGCCTCTCCCCGCGCGTTGGCCGATTAT
TAATGCAGCTGGCACGACAGGTTTCCCGACTGGAAGCGGGCAGTGACTGCAGCTCGAG pσ⁷⁰

V3LacI-mCardinal (SEQ ID NO: 60)

GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGA
TTGTGAGCGGATAACAATTTACACAGGAAACAGAATCATATGGTGAGTAAGGGTGAGGAGCTCATTAAAGGAGAA
CATGCACATGAAGCTGTATATGGAGGGCACCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAA
ACCCTACGAGGGGACGCAGACCCAACGCATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCT
GGCGACCTGTTTTATGTACGGCTCGAAGACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAG
CTTCCCTGAGGGCTTCACCTGGGAGCGCGTCACCACGTATGAAGACGGTGGGGTGCTCACCCTGACCCAGGACAC
GAGCTTGCAGGATGGCTGCTTGATTACAACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGGCCAGTGAT
GCAGAAAAAGACGCTGGGTTGGGAGGCCACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGT
CGATATGGCCCTGAAATTGGTTCGGCGGCGGTCAATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACC
CGCCAAAAACCTGAAGATGCCTGGTGTTTATTTTGTGCGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGA
GACGTACGTGGAACAGCACGAAGTGCCGTGGCTCGTTATTGCGATCTGCCGTCAAGCTGGGTACAAACTGAA
CGGCATGGATGAGCTGTACAAAGATTATAAGGATGATGACGACAAGTAAAAGCTTCTCGGTACCAAATTCCAGAA
AAGAGGCCTCCCGAAAGGGGGGCTTTTTTTCGTTTTTGGTCCGAATTCCCCCGTGGCCGGGGGACTGTTGGGCGCC
ATCTCCTTGCATGCACCATTCCTTGCGGCGGCGGTGCTCAACGGCCTCAACCTACTACTGGGCTGCTTCTTAATG
CAGGAGTCGCATAAGGGAGAGCGTCGAGATCCCGGACACCATCGAATGGCGCAAAACCTTTCGCGGTATGGCATG
ATAGCGCCCGGAAGAGAGTCAATTCAGGGTGGTGAATGTGAAACCAGTAACGTTATACGATGTGCGAGAGTATGC
CGGTGTCTCTTATCAGACCGTTTCCCGCGTGGTGAACCAGGCCAGCCACGTTTCTGCGAAAACGCGGGAAAAAGT
GGAAGCGGCGATGGCGGAGCTGAATTACATTCCCAACCGCGTGGCACAACAACCTGGcGGGCAAACAGTCGTTGCT
GATTGGCGTTGCCACCTCCAGTCTGGCCCTGCACGCGCCGTGCGCAAATTGTGCGGCGGATTAAATCTCGCGCCGA
TCAACTGGGTGCCAGCGTGGTGGTGTGATGGTAGAACGAAGCGGCGTCAAGCCTGTAAAGCGGCGGTGCACAA
TCTTCTCGCGCAACGCGTCAGTGGGCTGATCATTAACTATCCGCTGGATGACCAGGATGCCATTGCTGTGGAAGC
TGCCTGCACTAATGTTCCGGCGTTATTTCTTGATGTCTCTGACCAGACACCCATCAACAGTATTATTTTCTCCCA
TGAAGACGGTACGCGACTGGGCGTGGAGCATCTGGTTCGATTGGGTCAACCAGCAAATCGCGCTGTTAGCGGGCCC
ATTAAGTTCTGTCTCGGCGCGTCTGCGTCTGGCTGGCTGGCATAAATATCTCACTCGCAATCAAATTCAGCCGAT
AGCGGAACGGGAAGGCGACTGGAGTGCCATGTCCGGTTTTTCAACAAACCATGCAAATGCTGAATGAGGGCATCGT
TCCCACTGCGATGCTGGTTGCCAACGATCAGATGGCGCTGGGCGCAATGCGCGCCATTACCGAGTCCGGGCTGCG
CGTTGGTGCAGATATCTCGGTAGTGGGATACGACGATACCGAAGACAGCTCATGTTATATCCCGCCGTTAACCAC

CATCAAACAGGATTTTCGCTGCTGGGGCAAACACGAGCTGGAGCCGCTGCTGCAACTCTCTCAGGGGCCAGGGCGGT
GAAGGGCAATCAGCTGTTGCCCGTCTCACTGGTGAAAAGAAAAACCACCCTGGCGCCCAATACGCAAACCGCCTC
TCCCCGCGCGTTGGCCGATTCATTAATGCAGCTGGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGACTGCA
GCTCGAG pσ.sup.70 V2TcR-sfGFP (SEQ ID NO: 61)

GGATCCGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCATTCTGGTGGAGCTGGATGGCGAC
GTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGCCACGAACGGTAAGCTGACGCTGAAATTC
ATTTGCACCACCGGCAAATTGCCTGTACCCTGGCCCACCCTGGTGACCACGCTCACCTACGGCGTACAGTGCTTC
AGCCGTTACCCGGACCACATGAAGCGTCACGACTTCTTCAAAGCGCCATGCCGGAGGGTTACGTGCAGGAGCGT
ACGATTAGTTTCAAGGACGACGGCACCTATAAGACCCGTGCCGAAGTGAAGTTCGAAGGCGATACGTTGGTGAAC
CGTATCGAGTTGAAGGGTATCGACTTTAAGGAAGACGGCAACATCCTGGGCCATAAGCTGGAGTACAATTTCAAC
AGCCATAACGTTTACATCACCGCCGATAAACAGAAGAACGGCATTAAAGGCCAACTTTAAGATCCGCCACAACGTC
GAAGACGGCTCGGTGCAGCTGGCCGACCATTATCAGCAAACACCCCCATCGGTGATGGGCCCGTGCTGCTGCCG
GATAACCATTATCTGAGCACGCAGTCGGTGCTCAGCAAGGACCCTAACGAAAAGCGCGATCACATGGTGCTGCTG
GAGTTCGTACGGCGGCGGGGATCACCCATGGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATGACAAG
TAAAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCCCGAAAGGGGGGCCTTTTTTCGTTTTTGGTCCGAATT
CTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTACACATGTCCCGCCTGGATA
AATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTAAATTGG
CACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATGCCCTCG
CTATTGAAATGTTGGATCGCCACCATAACCCATTTCTGTCCACTGGAGGGCGAGTCCTGGCAGGACTTTCTCCGCA
ACAACGCGAAATCCTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGCACTTGGGCACCCGTCCCA
CCGAAAAACAATACGAAACCTTGGAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTC
TCTATGCGCTCTCGGCTGTGCGGCACCTTACGTTGGGGTGCGTGTTGGAGGACCAGGAGCATCAAGTCGCAAAAG
AGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATCATCAGG
GCGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAACAACCTCAAGTGTGAAAGCG
GGTCCTAACTGCAGCTCGAG pσ.sup.70 V2lacI-sfGFP (SEQ ID NO: 62)

GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGT
AAGAGAGGAATGTACATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCATTCTGGTGGAGCTGGATG
GCGACGTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGCCACGAACGGTAAGCTGACGCTGA
AATTCATTTGCACCACCGGCAAATTGCCTGTACCCTGGCCCACCCTGGTGACCACGCTCACCTACGGCGTACAGT
GCTTCAGCCGTTACCCGGACCACATGAAGCGTCACGACTTCTTCAAAGCGCCATGCCGGAGGGTTACGTGCAGG
AGCGTACGATTAGTTTCAAGGACGACGGCACCTATAAGACCCGTGCCGAAGTGAAGTTCGAAGGCGATACGTTGG
TGAACCGTATCGAGTTGAAGGGTATCGACTTTAAGGAAGACGGCAACATCCTGGGCCATAAGCTGGAGTACAATT
TCAACAGCCATAACGTTTACATCACCGCCGATAAACAGAAGAACGGCATTAAAGGCCAACTTTAAGATCCGCCACA
ACGTGCAAGACGGCTCGGTGCAGCTGGCCGACCATTATCAGCAAACACCCCCATCGGTGATGGGCCCGTGCTGC
TGCCGGATAACCATTATCTGAGCACGCAGTCGGTGCTCAGCAAGGACCCTAACGAAAAGCGCGATCACATGGTGC
TGCTGGAGTTCGTACGGCGGCGGGGATCACCCATGGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATG
ACAAGTAAAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCCCGAAAGGGGGGCCTTTTTTCGTTTTTGGTCC
GAATTCCTCCCGTGGCCGGGGGACTGTTGGGCGCCATCTCCTTGCATGCACCATTCCTTGCGGCGGCGGTGCTCAA
CGGCCTCAACCTACTACTGGGCTGCTTCCTAATGCAGGAGTCGCATAAGGGAGAGCGTCGAGATCCCGGACACCA
TCGAATGGCGCAAAACCTTTCGCGGTATGGCATGATAGCGCCCGGAAGAGAGTCAATTCAGGGTGGTGAATGTGA
AACCAGTAACGTTATACGATGTCGCAGAGTATGCCGGTGTCTCTTATCAGACCGTTTCCCGCGTGGTGAACCAGG
CCAGCCACGTTTCTGCGAAAACGCGGGAAAAAGTGGAAGCGGCGATGGCGGAGCTGAATTACATTCCCAACCGCG
TGGCACAACAACCTGGCGGGCAAACAGTCGTTGCTGATTGGCGTTGCCACCTCCAGTCTGGCCCTGCACGCGCCGT
CGCAAATTGTCGCGGCGATTAAATCTCGCGCCGATCAACTGGGTGCCAGCGTGGTGGTGTGATGGTAGAACGAA
GCGGCGTCGAAGCCTGTAAAGCGGCGGTGCACAATCTTCTCGCGCAACGCGTCAGTGGGCTGATCATTAACATC
CGCTGGATGACCAGGATGCCATTGCTGTGGAAGCTGCCTGCACTAATGTTCCGGCGTTATTTCTTGATGTCTCTG
ACCAGACACCCATCAACAGTATTATTTTCTCCCATGAAGACGGTACGCGACTGGGCGTGGAGCATCTGGTTCGAT
TGGGTCACCAGCAAATCGCGCTGTTAGCGGGCCCATTAAGTTCTGTCTCGGCGCGTCTGCGTCTGGCTGGCTGGC
ATAAATATCTCACTCGCAATCAAATTCAGCCGATAGCGGAACGGGAAGGCGACTGGAGTGCCATGTCCGGTTTTTC
AACAAACCATGCAAATGCTGAATGAGGGCATCGTTCCCACTGCGATGCTGGTTGCCAACGATCAGATGGCGCTGG
GCGCAATGCGCGCCATTACCGAGTCCGGGCTGCGCGTTGGTGCGGATATCTCGGTAGTGGGATACGACGATACCG
AAGACAGCTCATGTTATATCCCGCCGTTAACACCACATCAAACAGGATTTTCGCTGCTGGGGCAAACCAGCGTGG
ACCGCTTGCTGCAACTCTCTCAGGGCCAGGCGGTGAAGGGCAATCAGCTGTTGCCCGTCTCACTGGTGAAGAA
AAACCACCCTGGCGCCCAATACGCAAACCGCCTCTCCCCGCGCGTTGGCCGATTCATTAATGCAGCTGGCACGAC
AGGTTTCCCGACTGGAAAGCGGGCAGTGACTGCAGCTCGAG pσ.sup.70 V3LacI-sfGFP (SEQ ID NO:

63)
GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGA
TTGTGAGCGGATAACAATTTACACAGGAAACAGAATCATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGT
GCCCATTTCTGGTGGAGCTGGATGGCGACGTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGC
CACGAACGGTAAAGCTGACGCTGAAATTCATTTGCACCACCGGCAAATTGCCTGTACCCTGGCCCACCCTGGTGAC

ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAACATGCACATGAAGCTGTATATGGAGGGC
ACCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCCTACGAGGGGACGCAGACCCAACGC
ATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCTCGAAG
ACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGGCTTCACCTGGGAGCGC
GTCACCACGTATGAAGACGGTGCGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGATTTAC
AACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGCCAGTGATGCAGAAAAAGACGCTGGGTTGGGAGGCC
ACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATATGGCCCTGAAATTGGTCGGCGGC
GGTCATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTGGTGT
TATTTTGTGCGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCC
GTGGCTCGTTATTGCGATCTGCCGTGCAAGCTGGGTACAAAACCTGAACGGCATGGATGAGCTGTACAAAGATTAT
AAGGATGATGACGACAAGTAAAAGCTTCTCGGTACCAAATTCCAGAAAAGAGGCCTCCCGAAAGGGGGGCGCTTTT
TTCGTTTTGGTCCGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTAC
ACATGTCCCGCCTGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGC
TGACGACCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGG
CATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATACCCATTTCTGTCCACTGGAGGGCGAGTCCT
GGCAGGACTTTTCTCCGCAACAACGCGAAATCCTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGC
ACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGT
TTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTGCGGCACCTTTACGTTGGGGTGCGTGTGGAGGACCAGG
AGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACACGGAAGCTCGATGCCACCTCTGCTCCGCCAAGCTATCG
AACTCTTCGATCATCAGGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAAC
AACTCAAGTGTGAAAGCGGGTCCTAACTGCAGTCACTGCCCGCTTTCAGTCGGGAAACCTGTCGTGCCAGCTGC
ATTAATGAATCGGCCAACGCGCGGGGAGAGGGCGTTTGCCTATTGGGCGCCAGGGTGGTTTTTCTTTTCACCACT
GAGACGGGCAACAGCTGATTGCCCTTCACCGCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGC
CCCAGCAGGGCGAAAATCCTGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTTCGTAT
CCCCTACCGAGATATCCGCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTGCGCCCAGCGCCATCTGA
TCGTTGGCAACCAGCATCGCAGTGGGAACGATGCCCTCATTACGATTTGCATGGTTTGTGGAAAACCGGACATG
GCACTCCAGTCGCCTTCCCGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGA
CGCAGACGCGCCGAGACAGAACTTAATGGGCCCGCTAACAGCGCGATTGCTGGTGACCCAATGCGACCAGATGC
TCCACGCCCAGTCGCGTACCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGA
AATAACGCCGGAACATTAGTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATC
AGCCCACTGACGCGTTGCGCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCGCTTCGTTCTACCATC
GACACCACACGCTGGCACCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTTGCGACGGCGCGTGCAGG
GCCAGACTGGAGGTGGCAACGCCAATCAGCAACGACTGTTTGCCCGCCAGTTGTTGTGCCACGCGGTTGGGAATG
TAATTCAGCTCCGCCATCGCCGCTTCCACTTTTTCCCGCGTTTTTCGCAGAAACGTGGCTGGCCTGGTTCACCACG
CGGGAAACGGTCTGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTCACATTCACCACC
CTGAATTGACTCTCTTCCGGGCGCTATCATGCCATAACCGCGAAAGGTTTTGCGCCATTTCGATGGTGTCCGGGATC
TCGACGCTCTCCCTTATGCGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGGCCGTTGAGCACCGCCGC
CGCAAGGAATGGTGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagtgcaaaagcctccgggtcgg
aggcttttgactTCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGA
TAACAACGCAGTAAGAGAGGAATGTACCCATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAACATGCACATGA
AGCTGTATATGGAGGGCACCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCCTACGAGG
GGACGCAGACCCAACGCATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGACCTGTT
TTATGTACGGCTCGAAGACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGG
GCTTCACCTGGGAGCGCGTCACCACGTATGAAGACGGTGGGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGG
ATGGCTGCTTGATTTACAACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGCCAGTGATGCAGAAAAAGA
CGCTGGGTTGGGAGGCCACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATATGGCCC
TGAAATTGGTCGGCGGCGGTCAATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAAAAACC
TGAAGATGCCTGGTGTATTATTTGTGCGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGG
AACAGCACGAAGTGGCCGTGGCTCGTTATTGCGATCTGCCGTGCAAGCTGGGTACAAAACCTGAACGGCATGGATG
AGCTGTACAAAGATTATAAGGATGATGACGACAAGTAACCTCGAG pσ.sup.70 V2TcR-mCardinal/V3LacI-

mCardinal (SEQ ID NO: 66)
ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAACATGCACATGAAGCTGTATATGGAGGGC
ACCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCCTACGAGGGGACGCAGACCCAACGC
ATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCTCGAAG
ACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGGCTTCACCTGGGAGCGC
GTCACCACGTATGAAGACGGTGCGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGATTTAC
AACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGCCAGTGATGCAGAAAAAGACGCTGGGTTGGGAGGCC
ACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATATGGCCCTGAAATTGGTCGGCGGC
GGTCATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTGGTGT

TATTTCGTCGACCGCGCGCTGGAGCGGCATCAAGGAAGGAGGACGAGCTGGAAACGACGAAGTGGCC
GTGGCTCGTTATTGCGATCTGCCGTGCAAGCTGGGTCACAACTGAACGGCATGGATGAGCTGTACAAAGATTAT
AAGGATGATGACGACAAGTAAAGCTTCTCGGTACCAAATTCCAGAAAAGAGGCCTCCCGAAAGGGGGGCGCTTTT
TTCGTTTTGGTCCGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTAC
ACATGTCCCGCCTGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGC
TGACGACCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGG
CATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATACCCATTTCTGTCCACTGGAGGGCGAGTCCT
GGCAGGACTTTCTCCGCAACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGC
ACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGAAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGT
TTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTGCGGCACTTTACGTTGGGGTGCCTGTTGGAGGACCAGG
AGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCG
AACTCTTCGATCATCAGGGCGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAAC
AACTCAAGTGTGAAAGCGGGTCCTAACTGCAGTCACTGCCCGCTTTCAGTCGGGAAACCTGTCGTGCCAGCTGC
ATTAATGAATCGGCCAACGCGCGGGGAGAGGGCGTTTTCGCTATTGGGCGCCAGGGTGGTTTTTCTTTTACCAGT
GAGACGGGCAACAGCTGATTGCCCTTACCGCCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGC
CCCAGCAGGGCGAAAATCCTGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTCGTAT
CCCACTACCGAGATATCCGCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTGCGCCCAGCGCCATCTGA
TCGTTGGCAACCAGCATCGCAGTGGGAACGATGCCCTCATTACGATTTGCGATGGTTTGTGAAAACCGGACATG
GCACTCCAGTCGCCTTCCCGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGA
CGCAGACGCGCCGAGACAGAACTTAATGGGCCCGCTAACAGCGCGATTGCTGGTGACCCAATGCGACCAGATGC
TCCACGCCCAGTCGCGTACCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGA
AATAACGCCGGAACATTAGTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATC
AGCCCACTGACGCGTTGCGCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCGCTTCGTTCTACCATC
GACACCACACGCTGGCACCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTTGCGACGGCGCGTGCAGG
GCCAGACTGGAGGTGGCAACGCCAATCAGCAACGACTGTTTGGCCGCCAGTTGTTGTGCCACGCGGTTGGGAATG
TAATTCAGCTCCGCCATCGCCGCTTCCACTTTTTTCCCGCGTTTTTCGCAGAAACGTGGCTGGCCTGGTTCACCACG
CGGGAAACGGTCTGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTCACATTCACCACC
CTGAATTGACTCTCTTCCGGGCGCTATCATGCCATACCGCGAAAGGTTTTGCGCCATTTCGATGGTGTCCGGGATC
TCGACGCTCTCCCTTATGCGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGGCCGTTGAGCACCGCCGC
CGCAAGGAATGGTGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagtcacaaagcctccggctcg
aggcttttgactTCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGA
TAACAAGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACAGAATCCCATGGTGAGTAAGGGTGAGGAGCT
CATTAAAGGAGAACATGCACATGAAGCTGTATATGGAGGGCACCGTAAACAACCACCACTTCAAGTGTAACCACGA
GGGTGAAGGTAAACCCTACGAGGGGACGCAGACCCAACGCATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCCG
ATTCGACATTCTGGCGACCTGTTTTATGTACGGCTCGAAGACCTTCATCAACCACACCCAAGGCATCCCGGACTT
CTTCAAGCAGAGCTTCCCTGAGGGCTTACCTGGGAGCGCGTCACCACGTATGAAGACGGTGGGGTGCTCACCGT
GACCCAGGACACGAGCTTGCAGGATGGCTGCTTGATTACAACGTCAAGCTGCGCGGGGTGAACCTCCCTAGCAA
CGGGCCAGTGATGCAGAAAAAGACGCTGGGTGGGAGGCCACCACCGAGACCCTGTACCCGGCCGACGGGGGGCT
GGAAGGGCGGTGCGATATGGCCCTGAAATTGGTTCGGCGGCGGTCAATTTGCACTGCAATCTCAAGACCACGTACCG
CTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTGGTGTATTATTTGTGACCGGCGCCTGGAGCGCATCAAGGA
AGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGCCCGTGGCTCGTTATTGCGATCTGCCGTGCAAGCTGGG
TCACAACTGAACGGCATGGATGAGCTGTACAAAGATTATAAGGATGATGACGACAAGTAACTCGAG

pσ.sup.70 V2TcR-sfGFP/V2LacI-mCardinal (SEQ ID NO: 67)

ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCCATTTCTGGTGGAGCTGGATGGCGAC
GTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGCCACGAACGGTAAGCTGACGCTGAAATTC
ATTTGCACCACCGGCAAATTGCCTGTACCCTGGCCACCCCTGGTGACCACGCTCACCTACGGCGTACAGTGCTTC
AGCCGTTACCCGGACCACATGAAGCGTCACGACTTCTTCAAAGCGCCATGCCGGAGGGTTACGTGCAGGAGCGT
ACGATTAGTTTCAAGGACGACGGCACCTATAAGACCCGTGCCGAAGTGAAGTTCGAAGGCGATACGTTGGTGAAC
CGTATCGAGTTGAAGGTATCGACTTTAAGGAAGACGGCAACATCCTGGGCCATAAGCTGGAGTACAATTTCAAC
AGCCATAACGTTTACATCACCGCCGATAAACAGAAGAACGGCATTAAAGCCAACTTTAAAGATCCGCCACAACGTC
GAAGACGGCTCGGTGCAGCTGGCCGACCATATCAGCAAAACACCCCATCGGTGATGGGCCCGTGCTGCTGCCG
GATAACCATTATCTGAGCACGCAGTCGGTGCTCAGCAAGGACCCTAACGAAAAGCGCGATCACATGGTGCTGCTG
GAGTTTCGTCACGGCGGCGGGGATACCCATGGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATGACAAG
TAAAAGCTTCTCGGTACCAAATTCCAGAAAAGAGGCCTCCCGAAAGGGGGGCGCTTTTTTTCGTTTTTGGTCCGAATT
CTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTACACATGTCCCGCCTGGATA
AATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTAAATTGG
CACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATGCCCTCG
CTATTGAAATGTTGGATCGCCACCATACCCATTTCTGTCCACTGGAGGGCGAGTCCTGGCAGGACTTTCTCCGCA
ACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGCACTTGGGCACCCGTCCCA
CCGAAAAACAATACGAAACCTTGAAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTC

TCTATGCGCTCTCGGCTGTCTGGGCACTTTACGTTTGGGTGCGTGTGTTGGAGGACCAGGAGCATCAAGTCGCAAAAG
 AGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATCATCAGG
 GCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAACAACCTCAAGTGTGAAAGCG
 GGTCTTAAGTGCAGTCACTGCCCCGCTTTCCAGTCGGGAAACCTGTCTGTGCCAGCTGCATTAATGAATCGGCCAAC
 GCGCGGGGAGAGGCGGTTTTCGCTATTGGGCGCCAGGGTGGTTTTTCTTTTACCAGTGAGACGGGCAACAGCTGA
 TTGCCCTTACCAGCCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGGCCCAGCAGGCGAAAATCC
 TGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTCTGATCCCACTACCGAGATATCC
 GCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTGCGCCCAGCGCCATCTGATCGTTGGCAACCAGCATC
 GCAGTGGGAACGATGCCCTCATTAGCATTTGCATGGTTTGTGAAAACCGGACATGGCACTCCAGTCGCCTTCC
 CGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGACGCAGACGCGCCGAGACA
 GAACTTAATGGGCCCCGCTAACAGCGCGATTTGCTGGTGACCCAATGCGACCAGATGCTCCACGCCAGTCGCGTA
 CCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGAAATAACGCCGGAACATTA
 GTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATCAGCCCACTGACGCGTTGC
 GCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCCGCTTCGTTCTACCATCGACACCACCACGCTGGCA
 CCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTTGCGACGGCGCGTGCAGGGCCAGACTGGAGGTGGCA
 ACGCCAATCAGCAACGACTGTTTGGCCGCCAGTTGTTGTGCCACGCGGTTGGGAATGTAATTCAGCTCCGCCATC
 GCGCTTCCACTTTTTCCCGCGTTTTTCGCAGAAACGTGGCTGGCCTGGTTACCACGCGGGAAACGGTCTGATAA
 GAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTTACATTACCACCCTGAATTGACTCTCTTCC
 GGGCGCTATCATGCCATACCGCGAAAGGTTTTGCGCCATTCGATGGTGTCGGGATCTCGACGCTCTCCCTTATG
 CGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGGCCGTTGAGCACCGCCGCCGCAAGGAATGGTGCATG
 CAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGAgatcaaaagcctccggtcgaggcttttgactTCTAGA
 GAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGTAAGAGA
 GGAATGTACCCATGGTGAGTAAGGGTGAGGAGCTCATTAAAGGAGAACATGCACATGAAGCTGTATATGGAGGGCA
 CCGTAAACAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCCTACGAGGGGACGCAGACCCAACGCA
 TCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCTCGAAGA
 CCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGCTTCACCTGGGAGCGCG
 TCACCACGTATGAAGACGGTGGGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGATTTACA
 ACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGGCCAGTGATGCAGAAAAAGACGCTGGGTGGGAGGCCA
 CCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATATGGCCCTGAAATTGGTTCGGCGGCG
 GTCATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAAAAACCTGAAGATGCCTGGTGTTT
 ATTTTGTGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCCG
 TGGCTCGTTATTGCGATCTGCCGTCGAAGCTGGGTCACAACTGAACGGCATGGATGAGCTGTACAAAGATTATA
 AGGATGATGACGACAAGTAACTCGAG p_o.sup.70 V2TeR-sfGFP/V3LacI-mCardinal (SEQ ID NO: 68)
 ggatccGAGCTGTTGACAACCTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
 AGGAATGTACATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCATTCTGGTGGAGCTGGATGGCGAC
 GTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGCCACGAACGGTAAGCTGACGCTGAAATTC
 ATTTGCACCACCGGCAAATTGCCTGTACCCTGGCCACCCTGGTGACCACGCTCACCTACGGCGTACAGTGCTTC
 AGCCGTTACCCGGACCACATGAAGCGTCACGACTTCTTCAAAGCGCCATGCCGGAGGGTTACGTGCAGGAGCGT
 ACGATTAGTTTCAAGGACGACGGCACCTATAAGACCCGTGCCGAAGTGAAGTTTGAAGGCGATACGTTGGTGAAC
 CGTATCGAGTTGAAGGGTATCGACTTTAAGGAAGACGGCAACATCCTGGGCCATAAGCTGGAGTACAATTTCAAC
 AGCCATAACGTTTACATCACCGCCGATAAACAGAAGAACGGCATTAAGGCCAACTTTAAGATCCGCCACAACGTC
 GAAGACGGCTCGGTGCAGCTGGCCGACCATTATCAGCAAAACACCCCATCGGTGATGGGCCCGTGCTGCTGCCG
 GATAACCATATCTGAGCACGCAGTCGGTGCTCAGCAAGGACCCTAACGAAAAGCGCGATCACATGGTGCTGCTG
 GAGTTCGTACGGCGGCGGGGATCACCCATGGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATGACAAG
 TAAAAGCTTCTCGGTACCAAATTCCAGAAAAGAGGCCTCCCGAAAGGGGGGCTTTTTTTCGTTTTGGTCCGAATT
 CTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTACACATGTCCCGCCTGGATA
 AATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTAAATTGG
 CACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATGCCCTCG
 CTATTGAAATGTTGGATCGCCACCATAACCAATTTCTGTCCACTGGAGGGCGAGTCCTGGCAGGACTTTCTCCGCA
 ACAACGCGAAATCCTTTCGCTGTGCACTCTTGTCATCGGGACGGTGCTAAGGTGCACTTGGGCACCCGTCCCA
 CCGAAAAACAATACGAAACCTTGGAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTC
 TCTATGCGCTCTCGGCTGTGCGGCACTTTACGTTGGGGTGCGTGTGTTGGAGGACCAGGAGCATCAAGTCGCAAAAG
 AGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATCATCAGG
 GCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAACAACCTCAAGTGTGAAAGCG
 GGTCTTAAGTGCAGTCACTGCCCCGCTTTCCAGTCGGGAAACCTGTCTGTGCCAGCTGCATTAATGAATCGGCCAAC
 GCGCGGGGAGAGGCGGTTTTCGCTATTGGGCGCCAGGGTGGTTTTTCTTTTACCAGTGAGACGGGCAACAGCTGA
 TTGCCCTTACCAGCCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGGCCCAGCAGGCGAAAATCC
 TGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTCTGATCCCACTACCGAGATATCC
 GCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTGCGCCCAGCGCCATCTGATCGTTGGCAACCAGCATC
 GCAGTGGGAACGATGCCCTCATTAGCATTTGCATGGTTTGTGAAAACCGGACATGGCACTCCAGTCGCCTTCC
 CGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGACGCAGACGCGCCGAGACA

GAACCTTAATGGGCCCCGCTAACAGCGCGATTGCTGGTGACCCCAATGCGACAGATGCTCCACGCCCAGTCGCGTA
CCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGAAATAACGCCGGAACATTA
GTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATCAGCCCACTGACGCGTTGC
GCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCGGCTTCGTTCTACCATCGACACCACCACGCTGGCA
CCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTTGCGACGGCGCGTGACAGGGCCAGACTGGAGGTGGCA
ACGCCAATCAGCAACGACTGTTTGCCCGCCAGTTGTTGTGCCACGCGGTTGGGAATGTAATTCAGCTCCGCCATC
GCCGCTTCCACTTTTTCCCGCGTTTTTCGCAGAAACGTGGCTGGCCTGGTTTACCACGCGGGAAACGGTCTGATAA
GAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTCACATTACACCACCCTGAATTGACTCTCTTCC
GGGCGCTATCATGCCATACCGCGAAAGGTTTTGCGCCATTTCGATGGTGTCCGGGATCTCGACGCTCTCCCTTATG
CGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGGCCGTTGAGCACCGCCGCGCAAGGAATGGTGCATG
CAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagtcaaaagcctccggtcggaggcttttgactTCTAGA
GAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGAATTGTGA
GCGGATAACAATTTACACAGGAAACAGAATCCCATGGTGAGTAAGGGTGAGGAGCTCATTAAAGGAGAACATGCA
CATGAAGCTGTATATGGAGGGCACCGTAAACAACCACCACTTCAAGTGTAACACCGAGGGTGAAGGTAAACCCTA
CGAGGGGACGCAGACCCAACGCATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGAC
CTGTTTTATGTACGGCTCGAAGACCTTCATCAACCACACCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCC
TGAGGGCTTCACCTGGGAGCGCGTACCCACGTATGAAGACGGTGGGGTGCTACCGTGACCCAGGACACGAGCTT
GCAGGATGGCTGCTTGATTTACAACGTCAAGCTGCGCGGGGTGAACTTCCCTAGCAACGGGGCCAGTGATGCAGAA
AAAGACGCTGGGTTGGGAGGCCACCACCGAGACCCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATAT
GGCCCTGAAATTGGTCGGCGGCGGTCAATTTGCACTGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAA
AAACCTGAAGATGCCTGGTGTATTATTTGTGCGACCGGCGCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTA
CGTGGAACAGCACGAAGTGCCGTGGCTCGTTATTGCGATCTGCCGTGCAAGCTGGGTACAAACTGAACGGCAT
GGATGAGCTGTACAAAGATTATAAGGATGATGACGACAAGTAACTCGAG pUC19-crtEBI (SEQ ID NO:

69)
CATATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAAACAGTTCATCTCACTCGCGATGCT
GCGGAGCAGTTACTGGCTGATATTGATCGACGCCTTGATCAGTTATTGCCCGTGAGGGGAGAACGGGATGTTGTG
GGTGCCGCGATGCGTGAAGGTGCGCTGGCACCGGGAAACGTATTGCCCCCATGTTGCTGTTGCTGACCGCCCGC
GATCTGGGTTGCGCTGTCAGCCATGACGGATTACTGGATTTGGCCTGTGCGGTGGAAATGGTCCACGCGGCTTCG
CTGATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTCAATTCTCATTAC
GGAGAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGATGCAGATGGC
CTCACGCCGCTGGCAAAAAATCGGGCGGTTTTCTGAACTGTCAAACGCCATCGGCATGCAAGGATTGGTTCAGGGT
CAGTTCAAGGATCTGTCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCACTTTAAACC
AGCACGCTGTTTTGTGCCTCCATGCAGATGGCCTCGATTGTTGCGAATGCCTCCAGCGAAGCGCGTGATTGCCTG
CATCGTTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTTGACCGATGGCATGACCGACACCGGT
AAGGATAGCAATCAGGACGCCGGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGGCGGTTGAAGAACGTCTG
AGACAACATCTTCAGCTTGCCAGTGAGCATCTCTCTGCGGCCTGCCAACACGGGCACGCCACTCAACATTTTATT
CAGGCCTGGTTTGACAAAAAACTCGCTGCCGTCAAGTAAACGCAGTAAGAGAGGAATGTAGATATGAATAATCCGT
CGTTACTCAATCATGCGGTGCAAAACGATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCAAAGTTATTTGATG
CAAAAACCCGGCGCAGCGTACTGATGCTCTACGCCTGGTGCCGCCATTGTGACGATGTTATTGACGATCAGACGC
TGGGCTTTCAGGCCCGGCAGCCTGCCTTACAAACGCCCGAACAACGTCTGATGCAACTTGAGATGAAAACGCGCC
AGGCCTATGCAGGATCGCAGATGCACGAACCGGCGTTTTGCGGCTTTTCAGGAAGTGGCTATGGCTCATGATATCG
CCCCGGCTTACGCGTTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATACAGCCAACCTGGATG
ATACGCTGCGCTATTGCTATCACGTTGCAGGCGTTGTGCGCTTGATGATGGCGCAAATCATGGGCGTGCGGGATA
ACGCCACGCTGGACCGCGCCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGCGATATTGTGGACG
ATGCGCATGCGGGCCGCTGTTATCTGCCGGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGAATTATGCGG
CACCTGAAAACCGTCAGGCGCTGAGCCGTATCGCCCGTCGTTTGGTGACGGAAGCAGAACCTTACTATTTGTCTG
CCACAGCCGGCCTGGCAGGGTTGCCCTGCGTTCCGCCTGGGCAATCGCTACGGCGAAGCAGGTTTACCGGAAAA
TAGGTGTCAAAGTTGAACAGGCCGGTCAGCAAGCCTGGGATCAGCGGCAGTCAACGACCACGCCCGAAAAATTAA
CGCTGCTGCTGGCCGCCTCTGGTCAGGCCCTTACTTCCCGGATGCGGGCTCATCCTCCCCGCCCTGCGCATCTCT
GGCAGCGCCCGCTCTGAAATAATTTGTTTAACTTTAAGAAGGAGATATAATGAAACCAACTACGGTAATTGGTG
CAGGCTTCGGTGGCCTGGCAATTCGTCTACAAGCTGCGGGGATTCCCGTCTTACTGCTTGAACAACGTG
ATAAACCCGGCGGTGCGGCTTATGTCTACGAGGATCAGGGGTTTACCTTTGATGCAGGCCCGACGGTTATACCG
ATCCCAGTGCCATTGAAGAACTGTTTGAAGCTGGCAGGAAAACAGTTAAAAGAGTATGTCGAACTGCTGCCGGTTA
CGCCGTTTTACCGCCTGTGTTGGGAGTCAGGGAAGGTCTTTAATTACGATAACGATCAAACCCGGCTCGAAGCGC
AGATTACAGCAGTTTAATCCCCGCGATGTCGAAGGTTATCGTCAGTTTCTGGACTATTCACGCGCGGTGTTTAAAG
AAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTTATCGTTCAGAGACATGCTTCGCGCCGCACCTCAACTGGCGA
AACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAAGATGAACATCTGCGCCAGGCGTTTT
CTTCCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATTTATACGTTGATACACGCGCTGGAGC
GTGAGTGGGGCGTCTGTTTTCCGCGTGCGCGCACCGGCGCATTAGTTCAGGGGATGATAAAGCTGTTTCAGGATC
TGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAAACGACAGGAAACAAGATTGAAGCCGTGCATT
TAGAGGACGGTCGCAGGTTCCCTGACGCAAGCCGTGCGGTCAAATGCAGATGTGGTTCATACCTATCGCGACCTGT

TAAGCCACGACCTTACCGGCTTAAGCAAGCTTACGAGACTGACAGCTAGTAAGTAACCTCTCTGTTTGTGC
TCTATTTTGGTTTGAATCACCATCATGATCAGCTCGCGCATCACACGGTTTGTTCGGCCCCGCGTTACCGCGAGC
TGATTGACGAAATTTTAAATCATGATGGCCTCGCAGAGGACTTCTCACTTTATCTGCACGCGCCCTGTGTCAACGG
ATTCGTCACCTGGCGCCTGAAGGTTGCGGCAGTTACTATGTGTTGGCGCCGGTGCCGCATTTAGGCACCGCGAACC
TCGACTGGACGGTTGAGGGGGCCAAACTACGCGACCGTATTTTTGCGTACCTTGAGCAGCATTACATGCCTGGCT
TACGGAGTCAGCTGGTCACGCACCGGATGTTTACGCCGTTTGATTTTCGCGACCAGCTTAATGCCTATCATGGCT
CAGCCTTTTCTGTGGAGCCCGTTCTTACCCAGAGCGCCTGGTTTCGGCCGCATAACCGCGGATAAAACCATTACTA
ATCTCTACCTGGTCGGCGCAGGCACGCATCCCGGCGCAGGCATTCTGGCGTCATCGGCTCGGCAAAAGCGACAG
CAGGTTTGATGCTGGAGGATCTGATTTGA pσ.sup.70 V2TeR-crtEBI (SEQ ID NO: 70)
GGATCCGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTATCCGTTTATAAAGGACAGCCCGAATGACGGTCTGCGCAAAAAAACAGTTTCATCTCACT
CGCGATGCTGCGGAGCAGTTACTGGCTGATATTGATCGACGCCTTGATCAGTTATTGCCCGTGAGAGGAGAACGG
GATGTTGTGGGTGCCGCGATGCGTGAAGGTGCGCTGGCACC GGGAACGTATTCGCCCCATGTTGCTGTTGCTG
ACCGCCCGCGATCTGGGTTGCGCTGTCAGCCATGACGGATTACTGGATTGCGCTGTGCGGTGGAATGGTCCAC
GCGGCTTCGCTGATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTTCAT
TCTCATTACGGAGAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGAT
GCAGATGGCCTCACGCCGCTGGCAAAAAATCGGGCGGTTTCTGAACTGTCAAACGCCATCGGCATGCAAGGATTG
GTTCAGGGTCAGTTCAAGGATCTGTCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCAC
TTTAAAACCAGCACGCTGTTTTGTGCCTCCATGCAGATGGCCTCGATTGTTGCGAATGCCTCCAGCGAAGCGCGT
GATTGCCTGCATCGTTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTGACCGATGGCATGACC
GACACCGGTAAGGATAGCAATCAGGACGCCGGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGGCGGTTGAA
GAACGTCTGAGACAACATCTTCAGCTTGCCAGTGAGCATCTCTGCGGCCTGCCAACACGGGCACGCCACTCAA
CATTTTATTACGGCCTGTTTTGACAAAAAACTCGCTGCCGTCAGTTAACGCAGTAAGAGAGGAATGTAGATATGA
ATAATCCGTCGTTACTCAATCATGCGGTGCAAACGATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCAAAGT
TATTTGATGCAAAAACCCGGCGCAGCGTACTGATGCTCTACGCCTGGTGCCGCCATTGTGACGATGTTATTGACG
ATCAGACGCTGGGCTTTCAGGCCCCGCGAGCCTGCCTTACAAACGCCCGAACAACGTCTGATGCAACTTGAGATGA
AAACGCGCCAGGCCTATGCAGGATCGCAGATGCACGAACCGGCGTTTGCGGCTTTCAGGAAGTGGCTATGGCTC
ATGATATCGCCCCGGCTTACGCGTTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATACAGCC
AACTGGATGATACGCTGCGCTATTGCTATCACGTTGCAGGCGTTGTGCGCTTGATGATGGCGCAAATCATGGGCG
TGCGGGATAACGCCACGCTGGACCGCGCCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGCGATA
TTGTGGACGATGCGCATGCGGGCCGCTGTTATCTGCCGGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGA
ATTATGCGGCACCTGAAAACCGTCAGGCGCTGAGCCGTATCGCCCGTCGTTTGGTGCGAGGAAGCAGAACCTTACT
ATTTGTCTGCCACAGCCGGCCTGGCAGGGTTGCCCTGCGTTCCGCTGGGCAATCGCTACGGCGAAGCAGGTTT
ACCGGAAAATAGGTGTCAAAGTTGAACAGGCCGGTCAGCAAGCCTGGGATCAGCGGCAGTCAACGACCACGCCCG
AAAAATTAACGCTGCTGCTGGCCGCCTCTGGTCAGGCCCTTACTTCCCGGATGCGGGCTCATCCTCCCCGCCCTG
CGCATCTCTGGCAGCGCCCGCTCTGAAATAATTTGTTTAACTTTAAGAAGGAGATATAATGAAACCAACTACGG
TAATTGGTGCAAGGCTTCGGTGCCCTGGCACTGGCAATTTCGTCTACAAGCTGCGGGGATTCCCGTCTTACTGCTTG
AACAACGTGATAAACCCGGCGGTTCGGGCTTATGTCTACGAGGATCAGGGGTTTACCTTTGATGCAGGCCCGACGG
TTATCACCGATCCCAGTGCCATTGAAGAACTGTTTGCATGGCAGGAAAACAGTTAAAAGAGTATGTCGAACTGC
TGCCGGTTACGCCGTTTTACCGCCTGTGTTGGGAGTCAGGGAAGGTCTTTAATTACGATAACGATCAAACCCGGC
TCGAAGCGCAGATTACAGCAGTTTAATCCCCGCGATGTCGAAGGTTATCGTCAGTTTCTGGACTATTACGCGCGG
TGTTTAAAGAAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTTATCGTTTCAGAGACATGCTTCGCGCCGACCTC
AACTGGCGAAACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAAGATGAACATCTGCGCC
AGGCGTTTTCTTTCCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATTATACGTTGATACAG
CGCTGGAGCGTGAGTGGGGCGTCTGGTTTCCGCGTGGCGGCACCGGCGCATTAGTTCAGGGGATGATAAAGCTGT
TTCAGGATCTGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAAACGACAGGAAACAAGATTGAAG
CCGTGCATTTAGAGGACGGTCGCAGGTTCTGACGCAAGCCGTGCGGTCAAATGCAGATGTGGTTCATACCTATC
GCGACCTGTTAAGCCAGCACCTTGCCGCGGTTAAGCAGTCCAACAAACTGCAGACTAAGCGCATGAGTAACTCTC
TGTTTGTGCTCTATTTTGGTTTGAATCACCATCATGATCAGCTCGCGCATCACACGGTTTGTTCGGCCCCGCGTT
ACCGCGAGCTGATTGACGAAATTTTAAATCATGATGGCCTCGCAGAGGACTTCTCACTTTATCTGCACGCGCCCT
GTGTCACGGATTCGTCACTGGCGCCTGAAGGTTGCGGCAGTTACTATGTGTTGGCGCCGGTGCCGCATTTAGGCA
CCGCGAACCTCGACTGGACGGTTGAGGGGGCCAAACTACGCGACCGTATTTTTGCGTACCTTGAGCAGCATTACA
TGCCTGGCTTACGGAGTCAGCTGGTCACGCACCGGATGTTTACGCCGTTTGATTTTCGCGACCAGCTTAATGCCT
ATCATGGCTCAGCCTTTTCTGTGGAGCCCGTTCTTACCCAGAGCGCCTGGTTTCGGCCGCATAACCGCGATAAAA
CCATTACTAATCTCTACCTGGTCGGCGCAGGCACGCATCCCGGCGCAGGCATTCTGGCGTCATCGGCTCGGCAA
AAGCGACAGCAGGTTTGATGCTGGAGGATCTGATTTGAAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCC
CGAAAGGGGGGCGCTTTTTTTCGTTTTTGGTCCGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCA
GTAAGAGAGGAATGTACACATGTCCCGCCTGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGA
AGTCGGTATCGAGGGGCTGACGACCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCA
CGTCAAAAATAAGCGGGCATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATAACCATTTCTGTCC
ACTGGAGGGCGAGTCCTGGCAGGACTTTCTCCGCAACAACGCGAAATCCTTTCGCTGTGCACTCTTGTTCCCATCG

GGACGTTGGCTGCTAAGCTGCTTGGCCACCGTCCACCGGCAAAACCAATCAAGCAATGCTGCTT
TTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTCGGGCACCTTTACGTTGGGGTG
CGTGTGGAGGACCAGGAGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCT
GCTCCGCCAAGCTATCGAACTCTTCGATCATCAGGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTAT
CTGCGGTTTGGAAAAACAACCTCAAGTGTGAAAGCGGGTCCTAACTGCAGCTCGAG p0.sup.70 V2TcR-
crtEBIY (SEQ ID NO: 71)
GGATCCGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAAACAGTTTCATCTCACT
CGCGATGCTGCGGAGCAGTTACTGGCTGATATTGATCGACGCCTTGATCAGTTATTGCCCGTGGAGGGAGAACGG
GATGTTGTGGGTGCCGCGATGCGTGAAGGTGCGCTGGCACC GGGAACGTAATTCGCCCCATGTTGCTGTTGCTG
ACCGCCCGCGATCTGGGTTGCGCTGTCAGCCATGACGGATTACTGGATTGGCCTGTGCGGTGGAAATGGTCCAC
GCGGCTTCGCTGATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTCA
TCTCATTACGGAGAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGAT
GCAGATGGCCTCACGCCGCTGGCAAAAAATCGGGCGGTTTCTGAACTGTCAAACGCCATCGGCATGCAAGGATTG
GTTACAGGGTCAGTTCAAGGATCTGTCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCAC
TTTAAAACCAGCACGCTGTTTTGTGCCTCCATGCAGATGGCCTCGATTGTTGCGAATGCCTCCAGCGAAGCGCGT
GATTGCCTGCATCGTTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTGACCGATGGCATGACC
GACACCGGTAAGGATAGCAATCAGGACGCCGGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGGCGGTTGAA
GAACGTCTGAGACAACATCTTCAGCTTGCCAGTGAGCATCTCTGCGGCCTGCCAACACGGGCACGCCACTCAA
CATTTTATTCAGGCCTGGTTTGACAAAAAACTCGCTGCCGTCAGTTAACGCAGTAAGAGAGGAATGTAGATATGA
ATAATCCGTCGTTACTCAATCATGCGGTCGAAACGATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCAAAGT
TATTTGATGCAAAAACCCGGCGCAGCGTACTGATGCTCTACGCCTGGTGCCGCCATTGTGACGATGTTATTGACG
ATCAGACGCTGGGCTTTCAGGCCCGGCAGCCTGCCTTACAAACGCCCGAACAACGTCTGATGCAACTTGAGATGA
AAACGCGCCAGGCCTATGCAGGATCGCAGATGCACGAACCGGCGTTTGCGGCTTTTCAGGAAGTGGCTATGGCTC
ATGATATCGCCCCGGCTTACGCGTTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATACAGCC
AACTGGATGATACGCTGCGCTATTGCTATCACGTTGCAGGCGTTGTCGGCTTGATGATGGCGCAAATCATGGGCG
TGCGGGATAACGCCACGCTGGACCGCGCCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGCGATA
TTGTGGACGATGCGCATGCGGGCCGCTGTTATCTGCCGGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGA
ATTATGCGGCACCTGAAAACCGTCAGGCGCTGAGCCGTATCGCCCGTCGTTTGGTGCAGGAAGCAGAACCTTACT
ATTTGTCTGCCACAGCCGGCCTGGCAGGGTTGCCCTGCGTTCCGCCTGGGCAATCGCTACGGCGAAGCAGGTTT
ACCGGAAAATAGGTGTCAAAGTTGAACAGGCCGGTCAGCAAGCCTGGGATCAGCGGCAGTCAACGACCACGCCCG
AAAAATTAACGCTGCTGCTGGCCGCCTCTGGTCAGGCCCTTACTTCCCGGATGCGGGCTCATCCTCCCCGCCCTG
CGCATCTCTGGCAGCGCCCCGCTCTGAAATAATTTTGTTTAACTTTAAGAAGGAGATATAATGAAACCAACTACGG
TAATTGGTGCAGGCTTCGGTGCCCTGGCACTGGCAATTTCGTCTACAAGCTGCGGGGATTCCCGTCTTACTGCTTG
AACAACGTGATAAACCCGGCGGTTCGGGCTTATGTCTACGAGGATCAGGGGTTTACCTTTGATGCAGGCCCGACGG
TTATCACCGATCCCAGTGCCATTGAAGAACTGTTTGCCTGGCAGGAAAACAGTTAAAAGAGTATGTGCAACTGC
TGCCGGTTACGCCGTTTTACCGCCTGTGTTGGGAGTCAGGGAAGGTCTTTAATTACGATAACGATCAAACCCGGC
TCGAAGCGCAGATTACGACGTTTAAATCCCCGCGATGTGCAAGGTTATCGTCAGTTTCTGGACTATTACGCGCGG
TGTTTAAAGAAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTTATCGTTTCAGAGACATGCTTCGCGCCGCACCTC
AACTGGCGAAACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAAGATGAACATCTGCGCC
AGGCGTTTTCTTTCCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATTTATACGTTGATACACG
CGCTGGAGCGTGAGTGGGGCGTCTGGTTTCCGCGTGCGCGCACCGGCGCATTAGTTCAGGGGATGATAAAGCTGT
TTCAGGATCTGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAAACGACAGGAAACAAGATTGAAG
CCGTGCATTTAGAGGACGGTCGCAGGTTCTTGACGCAAGCCGTCGCGTCAAATGCAGATGTGGTTCATACCTATC
GCGACCTGTAAAGCCAGCACCTGCGCGGTTAAGCAGTCCAACAACTGCAGACTAAGCGCATGAGTAACTCTC
TGTTTGTGCTCTATTTTGGTTTGAATCACCATCATGATCAGCTCGCGCATCACACGTTTTGTTTCGGCCCCGCGTT
ACCGCGAGCTGATTGACGAAATTTTAAATCATGATGGCCTCGCAGAGGACTTCTCACTTTATCTGCACGCGCCCT
GTGTCACGGATTCTGCTACTGGCGCCTGAAGGTTGCGGCAGTTACTATGTGTTGGCGCCGGTGCCGCATTTAGGCA
CCGCGAACCTCGACTGGACGGTTGAGGGGCCAAAACCTACGCGACCGTATTTTTGCGTACCTTGAGCAGCATTACA
TGCCTGGCTTACGGAGTCAGCTGGTCACGCACCGGATGTTTACGCCGTTTGATTTTCGCGACCAGCTTAATGCCT
ATCATGGCTCAGCCTTTTCTGTGGAGCCCGTTCTTACCCAGAGCGCCTGGTTTCGGCCGCATAACCGCGATAAAA
CCATTACTAATCTCTACCTGGTCGGCGCAGGCACGCATCCCGGCGCAGGCATTCTGGCGTCATCGGCTCGGCAA
AAGCGACAGCAGGTTTGATGCTGGAGGATCTGATTTGACGCAGTAAGAGAGGAATGTAGATATGGGAGCGGCTAT
GCAACCGCATTATGATCTGATTCTCGTGGGGGCTGGACTCGCGAATGGCCTTATCGCCCTGCGTCTCCAGCAGCA
GCAACCTGATATGCGTATTTTGCTTATCGACGCCGCACCCAGGCGGGCGGGAATCATACGTGGTCATTTACCA
CGATGATTTGACTGAGAGCCAACATCGTTGGATAGCTCCGCTGGTGGTTCATCACTGGCCCGACTATCAGGTACG
CTTTCCACACGCCGTCGTAAGCTGAACAGCGGCTACTTTTGTATTACTTCTCAGCGTTTCGCTGAGGTTTTACA
GCGACAGTTTGGCCCGCACTTGTGGATGGATACCGCGGTTCGCAGAGGTTAATGCGGAATCTGTTTCGGTTGAAAA
GGGTCAGGTTATCGGTGCCCGCGCGGTGATTGACGGGGGGGTTATGCGGCAAATTCAGCACTGAGCGTGGGCTT
CCAGGCGTTTATTGGCCAGGAATGGCGATTGAGCCACCCGCATGGTTTATCGTCTCCCATTTATCATGGATGCCAC
GGTCGATCAGCAAAATGGTTATCGCTTCGTGTACAGCCTGCCGCTCTCGCCGACCAGATTGTTAATTGAAGATAC

GCATTAATGATAATGCGAATCTAGATCTGTAATGCGCGCGGCAAAATTTGCGACTATGCCGCGCAACAGGG
TTGGCAGCTTCAGACACTGCTGCGAGAAGAACAGGGCGCCTTACCCATTACTCTGTCTGGGCAATGCCGACGCATT
CTGGCAGCAGCGCCCCCTGGCCTGTAGTGGATTACGTGCCGGTCTGTTCCATCCTACCACCGGCTATTCACTGCC
GCTGGCGGTTGCCGTGGCCGACCGCCTGAGTGCACCTTGATGTCTTTACGTCTGGCCTCAATTACCCATGCCATTAC
GCATTTTGGCCGCGAGCGCTGGCAGCAGCAGGGCTTTTTCCGCATGCTGAATCGCATGCTGTTTTTAGCCGGACC
CGCCGATTCACGCTGGCGGGTTATGCAGCGTTTTTATGGTTTACCTGAAGATTAAATTGCCCGTTTTTATGCGGG
AAAACCTCACGCTGACCGATCGGCTACGTATTCTGAGCGGCAAGCCGCCTGTTCCGGTATTAGCAGCATTGCAAGC
CATTATGACGACTCATCGTTGAAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCCCGAAAGGGGGGCGCTTT
TTTCGTTTTGGTCCGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCAGTAAGAGAGGAATGTA
CACATGTCCCGCCTGGATAAATCGAAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGG
CTGACGACCCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGG
GCATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATAACCCATTTCTGTCCACTGGAGGGCGAGTCC
TGGCAGGACTTTCTCCGCAACAACGCGAAATCCTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTG
CACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGAAAAATCAATTGGCGTTTTTGTGCCAGCAAGGG
TTTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTGGGCACCTTACGTTGGGGTGCCTGTTGGAGGACCAG
GAGCATCAAGTCGAAAAGAGGAGCGTGAAACCCCAACACGGAATCGATGCCACCTCTGCTCCGCCAAGCTATC
GAACTCTTCGATCATCAGGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGAAAAA
CAACTCAAGTGTGAAAGCGGGTCCTAACTGCAGCTCGAG pσ.sup.70 V2TcR-crtEBI/V2LacI-crtY (SEQ
ID NO: 72)

ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAAACACGTTTCATCTCACT
CGCGATGCTGCGGAGCAGTTACTGGCTGATATTGATCGACGCCTTGATCAGTTATTGCCCGTGGAGGGAGAACGG
GATGTTGTGGGTGCCGCGATGCGTGAAGGTGCGCTGGCACCGGGAAAACGTATTCGCCCCATGTTGCTGTTGCTG
ACCGCCCGCGATCTGGGTTGCGCTGTGACCCATGACGGATTACTGGATTTGGCCTGTGCGGTGGAAATGGTCCAC
GCGGCTTCGCTGATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTCAT
TCTCATTACGGAGAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGAT
GCAGATGGCCTCACGCCGCTGGCAAAAAATCGGGCGGTTTTCTGAACTGTCAAACGCCATCGGCATGCAAGGATTG
GTTACAGGGTCAGTTCAAGGATCTGTCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCAC
TTTAAAACCAGCACGCTGTTTTGTGCCTCCATGCAGATGGCCTCGATTGTTGCGAATGCCTCCAGCGAAGCGCGT
GATTGCCTGCATCGTTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTTGACCGATGGCATGACC
GACACCGGTAAGGATAGCAATCAGGACGCCGGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGGCGGTTGAA
GAACGTCTGAGACAACATCTTCAGCTTGCCAGTGAGCATCTCTCTGCGGCCTGCCAACACGGGCACGCCACTCAA
CATTTTATTTCAGGCCTGTTTTGACAAAAAACTCGCTGCCGTCAGTTAACGCAGTAAGAGAGGAATGTAGATATGA
ATAATCCGTCGTTACTCAATCATGCGGTGCAAACGATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCAAAGT
TATTTGATGCAAAAACCCGGCGCAGCGTACTGATGCTCTACGCCTGGTGCCGCCATTGTGACGATGTTATTGACG
ATCAGACGCTGGGCTTTCAGGCCCGGCAGCCTGCCTTACAAACGCCCGAACAACGTCTGATGCAACTTGAGATGA
AAACGCGCCAGGCCTATGCAGGATCGCAGATGCACGAACCGGCGTTTTGCGGCTTTTCAGGAAGTGGCTATGGCTC
ATGATATCGCCCCGGCTTACGCGTTTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATACAGCC
AACTGGATGATACGCTGCGCTATTGCTATCACGTTGCAGGCGTTGTGCGGCTTGATGATGGCGCAAATCATGGGCG
TGCGGGATAACGCCACGCTGGACCGCGCCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGCGATA
TTGTGGACGATGCGCATGCGGGCCGCTGTTATCTGCCGGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGA
ATTATGCGGCACCTGAAAACCGTCAGGCGCTGAGCCGTATCGCCCGTCGTTTGGTGCAGGAAGCAGAACCTTACT
ATTTGTCTGCCACAGCCGGCCTGGCAGGGTTGCCCTGCGTTCCGCTGGGCAATCGCTACGGCGAAGCAGGTTT
ACCGGAAAATAGGTGTCAAAGTTGAACAGGCCGGTCAGCAAGCCTGGGATCAGCGGCAGTCAACGACCACGCCCC
AAAAATTAACGCTGCTGCTGGCCGCCTCTGGTCAGGCCCTTACTTCCCGGATGCGGGCTCATCCTCCCCGCCCTG
CGCATCTCTGGCAGCGCCCGCTCTGAAATAATTTTGTTTAACTTTAAGAAGGAGATATAATGAAACCAACTACGG
TAATTGGTGCAGGCTTCGGTGGCCTGGCACTGGCAATTTCGTCTACAAGCTGCGGGGATTCCCGTCTTACTGCTTG
AACAACGTGATAAACCCGGCGGCTCGGGCTTATGTCTACGAGGATCAGGGGTTTACCTTTGATGCAGGCCCGACGG
TTATCACCGATCCCAGTGCCATTGAAGAACTGTTTGCACTGGCAGGAAAACAGTTAAAAGAGTATGTCGAACTGC
TGCCGGTTACGCCGTTTTACCGCCTGTGTTGGGAGTCAGGGAAGGTCTTTAATTACGATAACGATCAAACCCGGC
TCGAAGCGCAGATTACGACGTTTAAATCCCCGCGATGTCGAAGGTTATCGTCAGTTTCTGGACTATTACGCGCGG
TGTTTAAAGAAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTTATCGTTTACAGAGACATGCTTCGCGCCGCACCTC
AACTGGCGAAACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAAGATGAACATCTGCGCC
AGGCGTTTTCTTTCCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATTTATACGTTGATACACG
CGCTGGAGCGTGAGTGGGGCGTCTGGTTTTCCGCGTGGCGGCACCGGCGCATTAGTTCAGGGGATGATAAAGCTGT
TTCAGGATCTGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAAACGACAGGAAACAAGATTGAAG
CCGTGCATTTAGAGGACGGTCGCAGGTTCTTGACGCAAGCCGTGCGCTCAAATGCAGATGTGGTTCATACCTATC
GCGACCTGTAAAGCCAGCACCTGCGCGGTTAAGCAGTCCAACAACTGCAGACTAAGCGCATGAGTAACTCTC
TGTTTGTGCTCTATTTTGGTTTTGAATCACCATCATGATCAGCTCGCGCATCACACGGTTTGTTTCGGCCCCGCGTT
ACCGCGAGCTGATTGACGAAATTTTAAATCATGATGGCCTCGCAGAGGACTTCTCACTTTATCTGCACGCGCCCT
GTGTCACGGATTCGTCACTGGCGCCTGAAGGTTGCGGCAGTTACTATGTGTTGGCGCCGGTGCCGCATTTAGGCA

CCGCGAACCTGACCTGAGCTTGAGGGGACCAAAATACGCGACCGGATTTTTCGCGTACCTTGAGCAGCATTACA
TGCCTGGCTTACGGAGTCAGCTGGTCACGCACCGGATGTTTACGCCGTTTGATTTTCGCGACCAGCTTAATGCCT
ATCATGGCTCAGCCTTTTCTGTGGAGCCCGTTCTTACCCAGAGCGCCTGGTTTTCGGCCGCATAACCGCGATAAAA
CCATTACTAATCTCTACCTGGTCGGCGCAGGCACGCATCCCGGCGCAGGCATTCTGGCGTCATCGGGCTCGGCAA
AAGCGACAGCAGGTTTGATGCTGGAGGATCTGATTTGAAAGCTTCTCGGTACCAAATTCAGAAAAAGAGGCCTCC
CGAAAGGGGGGCTTTTTTTCGTTTTTGGTCCGAATTCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCA
GTAAGAGAGGAATGTACACATGTCCCGCCTGGATAAATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGA
AGTCGGTATCGAGGGGCTGACGACCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCA
CGTCAAAAATAAGCGGGCATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATAACCCATTTCTGTCC
ACTGGAGGGCGAGTCCTGGCAGGACTTTCTCCGCAACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCG
GGACGGTGCTAAGGTGCACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGAAAAATCAATTGGCGTT
TTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTGCGGCACTTTACGTTGGGGTG
CGTGTTGGAGGACCAGGAGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACCACGGACTCGATGCCACCTCT
GCTCCGCCAAGCTATCGAACTCTTCGATCATCAGGGCGCGGAGCCAGCCTTCCTCTTTGGGCTGGAGCTGATTAT
CTGCGGTTTGGAAAAACAACCTCAAGTGTGAAAGCGGGTCCTAACTGCAGTCACTGCCCGCTTTCAGTCGGGAAA
CCTGTCTGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCCTATTGGGCGCCAGGGTGG
TTTTTCTTTTACCAGTGAGACGGGCAACAGCTGATTGCCCTTACCAGCCTGGCCCTGAGAGAGTTGCAGCAAGC
GGTCCACGCTGGTTTGGCCCAGCAGGCGAAAATCCTGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGT
CTTCGGTATCGTCGTATCCCACTACCGAGATATCCGCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTG
CGCCAGCGCCATCTGATCGTTGGCAACCAGCATCGCAGTGGGAACGATGCCCTCATTACGATTTGCATGGTTT
GTTGAAAACCGGACATGGCACTCCAGTCGCCTTCCCGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATT
TATGCCAGCCAGCCAGACGCAGACGCGCCGAGACAGAACTTAATGGGCCCCGCTAACAGCGCGATTTGCTGGTGAC
CCAATGCGACCAGATGCTCCACGCCCAGTCGCGTACCGTCTTCATGGGAGAAAAATAACTGTTGATGGGTGTCT
GGTCAGAGACATCAAGAAATAACGCCGGAACATTAGTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCA
GCGGATAGTTAATGATCAGCCCACTGACGCGTTGCGCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGC
CGTTTCGTTCTACCATCGACACCACCACGCTGGCACCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTT
GCGACGGCGCGTGCAGGGCCAGACTGGAGGTGGCAACGCCAATCAGCAACGACTGTTTGGCCGCCAGTTGTTGTG
CCACGCGGTTGGGAATGTAATTCAGCTCCGCCATCGCCGCTTCCACTTTTTTCCCGCGTTTTTCGCAGAAACGTGGC
TGGCCTGGTTCACCACGCGGGAAACGGTCTGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTG
GTTTCACATTACCAACCTGAATTGACTCTCTTCCGGGCGCTATCATGCCATACCGCGAAAGGTTTTTGCGCCATT
CGATGGTGTCCGGGATCTCGACGCTCTCCCTTATGCGACTCCTGCATTAGGAAGCAGCCCAGTAGTAGGTTGAGG
CCGTTGAGCACCGCCGCCGCAAGGAATGGTGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagtg
caaaagcctccggtcgaggcttttgactTCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGT
GTGGAATTGTGAGCGGATAACAACGCAGTAAGAGAGGAATGTACCCATGGAGCGGCTATGCAACCGCATTATGAT
CTGATTCTCGTGCGGGGCTGGACTCGCGAATGGCCTTATCGCCCTGCGTCTCCAGCAGCAGCAACCTGATATGCGT
ATTTTGCTTATCGACGCCGCACCCAGGCGGGCGGGAATCATACGTGGTCATTTACCAACGATGATTTGACTGAG
AGCCAACATCGTTGGATAGCTCCGCTGGTGGTTCATCACTGGCCCGACTATCAGGTACGCTTTCCACACGCCGT
CGTAAGCTGAACAGCGGCTACTTTTGTATTACTTCTCAGCGTTTCGCTGAGGTTTTACAGCGACAGTTTGGCCCG
CACTTGTGGATGGATAACCGCGGTTCGCAGAGGTTAATGCGGAATCTGTTTCGGTTGAAAAAGGGTCAGGTTATCGGT
GCCCCGCGCGGTGATTGACGGGCGGGGTTATGCGGCAAATTCAGCACTGAGCGTGCGGCTTCCAGGCGTTTATTGGC
CAGGAATGGCGATTGAGCCACCCGCATGGTTTATCGTCTCCCATTTATCATGGATGCCACGGTCGATCAGCAAAAT
GGTTATCGCTTCGTGTACAGCCTGCCGCTCTCGCCGACCAGATTGTTAATTGAAGATACGCACTATATTGATAAT
GCGACATTAGATCCTGAATGCGCGCGGCAAAATATTTGCGACTATGCCGCGCAACAGGGTTGGCAGCTTCAGACA
CTGCTGCGAGAAGAACAGGGCGCCTTACCCATTACTCTGTGCGGCAATGCCGACGCATTCTGGCAGCAGCGCCCC
CTGGCCTGTAGTGGATTACGTGCCGGTCTGTTCCATCCTACCACCGGCTATTAAGTCCGCTGGCGGTTGCCGTG
GCCGACCGCCTGAGTGCATTTGATGTCTTTACGTCGGCCTCAATTCACCATGCCATTACGCATTTTGGCCGCGAG
CGCTGGCAGCAGCAGGGCTTTTTTCCGCATGCTGAATCGCATGCTGTTTTTACCGGACCCGCCGATTACGCTGG
CGGGTTATGCAGCGTTTTTATGGTTTACCTGAAGATTTAATTGCCCGTTTTTATGCGGGAAAACTCACGCTGACC
GATCGGCTACGTATTCTGAGCGGCAAGCCGCTGTTCCGGTATTAGCAGCATTGCAAGCCATTATGACGACTCAT
CGTTGACTCGAG pσ.sup.70 V2TcR-crtEBI/V3LacI-crtY (SEQ ID NO: 73)
ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAAACAGTTTCATCTCACT
CGCGATGCTGCGGAGCAGTTACTGGCTGATATTGATCGACGCCTTGATCAGTTATTGCCCGTGGAGGGGAGAACGG
GATGTTGTGGGTGCCGCGATGCGTGAAGGTGCGCTGGCACCGGGAAAAACGTATTCGCCCCATGTTGCTGTTGCTG
ACCGCCCCGCGATCTGGGTTGCGCTGTACGCCATGACGGATTACTGGATTGCGCCTGTGCGGTGGAAATGGTCCAC
GCGGCTTCGCTGATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTTCAT
TCTCATTACGGAGAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGAT
GCAGATGGCCTCACGCCGCTGGCAAAAAATCGGGCGGTTTTCTGAACTGTCAAACGCCATCGGCATGCAAGGATTG
GTTCAGGGTCAGTTCAAGGATCTGTCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCAC
TTTAAAACCAGCACGCTGTTTTGTGCCTCCATGCAGATGGCCTCGATTGTTGCGAATGCCTCCAGCGAAGCGCGT
GATTGCCTGCATCGTTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTGACCGATGGCATGACC

GACACCGGTAAAGCAATAGCAATACGAGACGCGCGGTAATCGATCGGTAATCTGTTAGGCGCCGAGGGCGGTTGAA
GAACGTCTGAGACAACATCTTCAGCTTGCCAGTGAGCATCTCTCTGCGGCCTGCCAACACGGGCACGCCACTCAA
CATTTTATTTCAGGCCTGTTTTGACAAAAAACTCGCTGCCGTCAGTTAACGCAGTAAGAGAGGAATGTAGATATGA
ATAATCCGTCGTTACTCAATCATGCGGTCGAAACGATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCAAAGT
TATTTGATGCAAAAACCCGGCGCAGCGTACTGATGCTCTACGCCTGGTGCCGCCATTGTGACGATGTTATTGACG
ATCAGACGCTGGGCTTTTCAGGCCCGGCAGCCTGCCTTACAAACGCCCGAACAACGTCTGATGCAACTTGAGATGA
AAACGCGCCAGGCCTATGCAGGATCGCAGATGCACGAACCGGCGTTTTGCGGCTTTTCAGGAAGTGGCTATGGCTC
ATGATATCGCCCCGGCTTACGCGTTTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATACAGCC
AACTGGATGATACGCTGCGCTATTGCTATCACGTTGCAGGCGTTGTGCGGCTTGATGATGGCGCAAATCATGGGCG
TGCGGGATAACGCCACGCTGGACCGCGCCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGCGATA
TTGTGGACGATGCGCATGCGGGCCGCTGTTATCTGCCGGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGA
ATTATGCGGCACCTGAAAACCGTCAGGCGCTGAGCCGTATCGCCCGTCGTTTTGGTGACAGGAAGCAGAACCTTACT
ATTTGTCTGCCACAGCCGGCCTGGCAGGGTTGCCCTGCGTTCCGCTGGGCAATCGCTACGGCGAAGCAGGTTT
ACCGGAAAATAGGTGTCAAAGTTGAACAGGCCGGTCAGCAAGCCTGGGATCAGCGGCAGTCAACGACCACGCCCC
AAAAATTAACGCTGCTGCTGGCCGCCTCTGGTCAGGCCCTTACTTCCGGATGCGGGCTCATCCTCCCCGCCCTG
CGCATCTCTGGCAGCGCCCGCTCTGAAATAATTTTGTTTAACTTTAAGAAGGAGATATAATGAAACCAACTACGG
TAATTGGTGACGGCTTCGGTGGCCTGGCAATTTCGCTCTACAAGCTGCGGGGATTCCCGTCTTACTGCTTG
AACAACGTGATAAACCCGGCGGTCGGGCTTATGTCTACGAGGATCAGGGGTTTACCTTTGATGCAGGCCCGACGG
TTATCACCGATCCCAGTGCCATTGAAGAACTGTTTGCCTGGCAGGAAAACAGTTAAAAGAGTATGTGAACTGC
TGCCGGTTACGCCGTTTTTACCGCCTGTGTTGGGAGTCAGGGAAGGTCTTTAATTACGATAACGATCAAACCCGGC
TCGAAGCGCAGATTCAGCAGTTTAATCCCCGCGATGTCGAAGGTTATCGTCAGTTTCTGGACTATTACGCGCGG
TGTTTAAAGAAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTTATCGTTTCAGAGACATGCTTCGCGCCGCACCTC
AACTGGCGAAACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAAGATGAACATCTGCGCC
AGGCGTTTTCTTTCCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATTTATACGTTGATACACG
CGCTGGAGCGTGAGTGGGGCGTCTGGTTTTCCGCGTGGCGGCACCGGCGCATTAGTTCAGGGGATGATAAAGCTGT
TTCAGGATCTGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAAACGACAGGAAACAAGATTGAAG
CCGTGCATTTAGAGGACGGTCGCAGGTTCCCTGACGCAAGCCGTGCGCTCAAATGCAGATGTGGTTCATACCTATC
GCGACCTGTAAAGCCAGCACCTGCGCGGTTAAGCAGTCCAACAACTGCAGACTAAGCGCATGAGTAACTCTC
TGTTTGTGCTCTATTTTGGTTTTGAATCACCATCATGATCAGCTCGCGCATCACACGTTTTGTTTCGGCCCCGCGTT
ACCGCGAGCTGATTGACGAAATTTTTAATCATGATGGCCTCGCAGAGGACTTCTCACTTTATCTGCACGCGCCCT
GTGTCACGGATTCGTCACTGGCGCCTGAAGGTTGCGGCAGTTACTATGTGTTGGCGCCGGTGCCGCATTTAGGCA
CCGCGAACCTCGACTGGACGGTTGAGGGGCCAAAACCTACGCGACCGTATTTTTTGCGTACCTTGAGCAGCATTACA
TGCCTGGCTTACGGAGTCAGCTGGTCACGCACCGGATGTTTACGCCGTTTGATTTTTCGCGACCAGCTTAATGCCT
ATCATGGCTCAGCCTTTTTCTGTGGAGCCCGTTCTTACCCAGAGCGCCTGGTTTTCGGCCGCATAACCGCGATAAAA
CCATTACTAATCTCTACCTGGTCGGCGCAGGCACGCATCCCGGCGCAGGCATTCCTGGCGTCATCGGCTCGGCAA
AAGCGACAGCAGGTTTGATGCTGGAGGATCTGATTTGAAAGCTTCTCGGTACCAAATTCAGAAAAGAGGCCTCC
CGAAAGGGGGGCTTTTTTTCGTTTTTGGTCCGAATTCCTTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGCA
GTAAGAGAGGAATGTACACATGTCCCGCCTGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGA
AGTCGGTATCGAGGGGCTGACGACCCGTAAATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCA
CGTCAAAAATAAGCGGGCATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATAACCCATTTCTGTCC
ACTGGAGGGCGAGTCCTGGCAGGACTTTTCTCCGCAACAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCCATCG
GGACGGTGCTAAGGTGCACTTGGGCACCCGTCCCACCGAAAAACAATACGAAACCTTGAAAAATCAATTGGCGTT
TTTGTGCCAGCAAGGGTTTAGCTTGGAGAATGCTCTCTATGCGCTCTCGGCTGTGCGGCACCTTACGTTGGGGTG
CGTGTGAGGAGCAGGAGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACACGGACTCGATGCCACCTCT
GCTCCGCCAAGCTATCGAACTCTTCGATCATCAGGGCGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTAT
CTGCGGTTTTGAAAAACAACCTCAAGTGTGAAAGCGGGTCCTAACTGCAGTCACTGCCCGCTTTCCAGTCGGGAAA
CCTGTGCTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTTGCGTATTGGGCGCCAGGGTGG
TTTTTCTTTTACCAGTGAGACGGGCAACAGCTGATTGCCCTTACCCGCTGGCCCTGAGAGAGTTGCAGCAAGC
GGTCCACGCTGGTTTTGCCCCAGCAGGCGAAAATCCTGTTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGT
CTTCGGTATCGTCGTATCCCACTACCGAGATATCCGCACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTG
CGCCAGCGCCATCTGATCGTTGGCAACCAGCATCGCAGTGGGAACGATGCCCTCATTCAGCATTGTCATGGTTT
GTTGAAAACCGGACATGGCACTCCAGTCGCCCTTCCCGTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATT
TATGCCAGCCAGCCAGACGCAGACGCGCCGAGACAGAACTTAATGGGCCCGCTAACAGCGCGATTTGCTGGTGAC
CCAATGCGACCAGATGCTCCACGCCCAGTCGCGTACCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGTCT
GGTCAGAGACATCAAGAAATAACGCCGGAACATTAGTGACAGGACGCTTCCACAGCAATGGCATCCTGGTCATCCA
GCGGATAGTTAATGATCAGCCCACTGACGCGTTGCGCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGC
CGCTTCGTTCTACCATCGACACCACCGCTGGCACCCAGTTGATCGGCGCGAGATTTAATCGCCGCGACAATTT
GCGACGGCGCGTGACAGGGCCAGACTGGAGGTGGCAACGCCAATCAGCAACGACTGTTTGCCCGCCAGTTGTTGTG
CCACGCGGTTGGGAATGTAATTCAGCTCCGCCATCGCCGCTTCCACTTTTTTCCCGCGTTTTTCGCAGAAACGTGGC
TGGCCTGGTTTACCACGCGGGAAACGGTCTGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTG
GTTTCACATTCACCACCCTGAATTGACTCTCTTCCGGGCGCTATCATGCCATACCGCGAAAGGTTTTTGCGCCATT

CGATGTTGCTCCGCTGCTCGAGTCTCGACCTCTCCCTTATGCGAGCTCTGCTGATTAGGAAGACGAGCCAGTAGTAGGTTGAGG
CCGTTGAGCACCCGCCGCCGCAAGGAATGGTGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGGagt
caaaagcctccggctcgaggcttttactTCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGT
GTGGAATTGTGAGCGGATAACAAGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACAGAATCCCATGGAG
CGGCTATGCAACCGCATTATGATCTGATTCTCGTGGGGGCTGGACTCGCGAATGGCCTTATCGCCCTGCGTCTCC
AGCAGCAGCAACCTGATATGCGTATTTTGCTTATCGACGCCGCACCCAGGCGGGCGGGAATCATACGTGGTCAT
TTCACCACGATGATTTGACTGAGAGCCAACATCGTTGGATAGCTCCGCTGGTGGTTCATCACTGGCCCGACTATC
AGGTACGCTTTCCACACGCCGTCGTAAGCTGAACAGCGGCTACTTTTGTATTACTTCTCAGCGTTTCGCTGAGG
TTTTACAGCGACAGTTTGCCCCGCACTTGTGGATGGATACCGCGGTTCGCAGAGGTTAATGCGGAATCTGTTCCGT
TGAAAAAGGGTCAGGTTATCGGTGCCCCGCGCGGTGATTGACGGGGGGGGTTATGCGGCAAATTCAGCACTGAGCG
TGGGCTTCCAGGCGTTTATTGGCCAGGAATGGCGATTGAGCCACCCGCATGGTTTATCGTCTCCCATTATCATGG
ATGCCACGGTCGATCAGCAAAATGGTTATCGCTTCGTGTACAGCCTGCCGCTCTCGCCGACCAGATTGTTAATTG
AAGATACGCACTATATTGATAATGCGACATTAGATCCTGAATGCGCGCGGCAAAATATTTGCGACTATGCCGCGC
AACAGGGTTGGCAGCTTCAGACACTGCTGCGAGAAGAACAGGGCGCCTTACCCATTACTCTGTCGGGCAATGCCG
ACGCATTCTGGCAGCAGCGCCCCCTGGCCTGTAGTGGATTACGTGCCGGTCTGTTCCATCCTACCACCGGCTATT
CACTGCCGCTGGCGGTTGCCGTGGCCGACCGCCTGAGTGCATTGATGTCTTTACGTGGGCCTCAATTCACCATG
CCATTACGCATTTTGCCCGCGAGCGCTGGCAGCAGCAGGGCTTTTTCCGCATGCTGAATCGCATGCTGTTTTTAG
CCGACCCCGCGATTACGCTGGCGGGTTATGCAGCGTTTTTATGGTTTACCTGAAGATTTAATTGCCCGTTTTT
ATGCGGGGAAAACTCACGCTGACCGATCGGCTACGTATTCTGAGCGGCAAGCCGCCTGTTCCGGTATTAGCAGCAT
TGCAAGCCATTATGACGACTCATCGTTGACTCGAG cocE DNA sequence (SEQ ID NO: 74)

ATGGTGGACGGTAATTATTCGGTAGCGTCCAACGTTATGGTGCCGATGCGCGACGGGGTGCGCTTGGCTGTAGAT
CTGTACCGCCCCGGACGCAGATGGCCCTGTACCGGTCCTGCTGGTCCGCAACCCCTACGACAAATTCGACGTGTT
GCTTGAGTACGCAGAGCACGAAGTGGCTGGAATTTGTGCGCGATGGGTACGCCGTCGTCATCCAAGACACCCGG
GGCCTCTTTGCATCCGAAGGTGAGTTCGTTCCACATGTTGATGACGAGGCGGATGCGGAAGACACGCTGAGCTGG
ATCTTGGAACAAGCATGGTGGCAGCGCAATGTGGGTATGTTCCGGTGTAAAGCTACCTGGGCGTTACGCAGTGGCAA
GCTGCTGTTAGCGGTGTGGGTGGTTTGAAGGCAATCGCCCCGAGCATGGCGAGCGCGGATCTGTACCGTGCCCC
TGGTACGGTCCTGGCGGCGCCCTGAGCGTGGAAGCACTCCTGGGCTGGAGCGCATTGATCGGTACGGGCCTGATT
ACCAGCCGTAGCGATGCCCCGCCGGAAGACGCAGCCGACTTCGTACAGCTGGCAGCCATCCTGAACGATGTGGCC
GGTGCCGCAAGCGTGACCCCTCTGGCCGAACAGCCCTTGTGGGCCGCGCTGATCCCTTGGGTGATCGACCAGGTG
GTGGACCATCCAGACAACGACGAGTCGTGGCAGAGCATCTCGCTCTTTGAACGTTTGGGTGGGCTCGCTACCCCG
GCCTTGATTACCGCCGGGTGGTACGATGGCTTCGTGGGCGAGAGCCTCCGTACCTTCGTAGCTGTGAAGGACAAC
GCGGATGCGCGTCTGGTGGTGGGGCCGTGGAGCCACAGCAATCTGACCGGCCGTAATGCCGACCGTAAGTTTGGG
ATCGCCGCGACCTACCCCATCCAGGAGGCGACGACCATGCACAAGGCTTTTTTCGACCGGCACCTCCGTGGCGAG
ACCGATGCCCTGGCAGGGGTGCCCAAGGTGCGCCTCTTCGTAATGGGTATCGATGAGTGGCGCGACGAGACCGAC
TGGCCATTGCCAGATACCGCTTACACGCCTTTTTACCTCGGGGGCTCCGGTGCGGCCAACACGAGCACGGGTGGT
GGGACCCTGTGACCTCGATCAGCGGCACGGAGTCGGCGGACACCTACCTGTATGATCCTGCCGACCCCGTGCCA
AGTCTGGGCGGCACCCCTCCTCTTCCATAATGGGGACAACGGTCCAGCTGACCAGCGCCCGATTACGATCGCGAC
GACGTGCTGTGCTACTCCACCGAGGTGTTGACCGACCCCGTGGAAGTAACGGGGACGGTTTCGGCTCGCCTGTT
GTGTCTCGTCGGCCGTGGATACCGATTTTACCGCCAAGTTGGTCGACGTGTTCCCGATGGTCGGGCAATCGCT
CTCTGCGACGGCATCGTGCGTATGCGCTACCGGGAGACCTTGGTAAATCCTACGCTCATTGAGGCCGGTGAGATT
TACGAGGTGGCTATTGATATGCTGGCCACCAGCAACGTGTTTTTGGCGGGCCACCGCATCATGGTGCAAGTTAGC
AGCTCGAACTTCCCGAAGTACGACCGCAACTCCAACACCGGCGGCGTCATCGCTCGCGAGCAACTGGAGGAAATG
TGCACCGCCGTAAACCGCATTACCGCGGGCCCCGAACACCCGTCCCATATCGTGCTGCCGATCATTAAAGCGCGAC
TATAAGGACGACGACGATAAGTGA cocE Amino acid sequence (SEQ ID NO: 75)

MVDGNYSVASNVMVPMRDGVR LAVDLYRPDADGPVPVLLVRNPYDKEDVFAWSTQSTNWLEFVRDGYAVVIQDTR
GLFASEGEFVPHVDDEADAEDTL SWILEQAWCDGNVGMFGVSYLGVTQWQA AVSGVGGLKAIAPSMASADLYRAP
WYGPGGALSVEALLGWSALIGTGLITSRSDARPEDAADFVQLAAILNDVAGAASVTPLAEQPLLGR LIPWVIDQV
VDHPDNDES WQSI SLFERL GGLATPALITAGWYDGFVGESLRTFVAVKDNADARLVVGPWSH SNLTGRNADRKFG
IAATYPIQEATTMHKAFFDRHLRGETDALAGVPKVRLFVMGIDEWRDET DWPLPDTAYTPFYLGGSGAANTSTGG
GTLSTSISGTESADTYLYDPADPVPSLGGTLLFHNGDNGPADQRPIHDRDDVLCYSTEVLTD PVEVTGTVSARLF
VSSAVD TDFTAKLVDVFPDGRAIALCDGIVRMRYRETLVNPTLIEAGEIYEVAIDMLATS NVELPGRHIMVQVS
SSNFPKYDRNSNTGGV IAREQLEEMCTAVNRIHRGPEHPSHIVLPIIKRDYKDDDDK- crtE (SEQ ID NO:

76)
ATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAAACACGTTTCATCTCACTCGCGATGCTGCG
GAGCAGTTACTGGCTGATATTGATCGACGCCTTGATCAGTTATTGCCCGTGGAGGGAGAACGGGATGTTGTGGGT
GCCGCGATGCGTGAAGGTGCGCTGGCACCGGGAACCGTATTGCCCCATGTTGCTGTTGCTGACCGCCCCGCGAT
CTGGGTTGCGCTGTCAGCCATGACGGATTACTGGATTTGGCCTGTGCGGTGGAAATGGTCCACGCGGCTTCGCTG
ATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTCAATTCTCATTACGGA
GAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGATGCAGATGGCCTC
ACGCCGCTGGCAAAAAATCGGGCGGTTTCTGAACTGTCAAACGCCATCGGCATGCAAGGATTGGTTCAGGGTCAG
TTCAAGGATCTGTCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCACTTTAAACCAGC

ACGCTGTTTTGTGCTCCATGCGAGATGGCCTCGATTGTTGCGAATGCCTCCAGCGAAGCGCGTGATTGCCTGCAT
CGTTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTTGACCGATGGCATGACCGACACCGGTAAG
GATAGCAATCAGGACGCCGGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGGCGGTTGAAGAACGTCTGAGA
CAACATCTTCAGCTTGCCAGTGAGCATCTCTCTGCGGCCTGCCAACACGGGCACGCCACTCAACATTTTATTTCAG
GCCTGTTTTGACAAAAAACTCGCTGCCGTCAGTTAA crtB (SEQ ID NO: 77)
ATGAATAATCCGTTCGTTACTCAATCATGCGGTCGAAACGATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCA
AAGTTATTTGATGCAAAAACCCGGCGCAGCGTACTGATGCTCTACGCCTGGTGGCGCCATTGTGACGATGTTATT
GACGATCAGACGCTGGGCTTTCAGGCCCCGGCAGCCTGCCTTACAAACGCCCGAACAACGTCTGATGCAACTTGAG
ATGAAAACGCGCCAGGCCTATGCAGGATCGCAGATGCACGAACCGGCGTTTTCAGGAAGTGGCTATG
GCTCATGATATCGCCCCGGCTTACGCGTTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATAC
AGCCAACTGGATGATACGCTGCGCTATTGCTATCACGTTGCAGGCGTTGTCGGCTTGATGATGGCGCAAATCATG
GGCGTGCGGGATAACGCCACGCTGGACCGCGCCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGC
GATATTGTGGACGATGCGCATGCGGGCCGCTGTTATCTGCCGGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAA
GAGAATTATGCGGCACCTGAAAACCGTCAGGCGCTGAGCCGTATCGCCCGTCGTTTGGTGCAGGAAGCAGAACCT
TACTATTTGTCTGCCACAGCCGGCCTGGCAGGGTTGCCCTGCGTTCCGCCTGGGCAATCGCTACGGCGAAGCAG
GTTTACCGGAAAATAGGTGTCAAAGTTGAACAGGCCGGTCAGCAAGCCTGGGATCAGCGGCAGTCAACGACCACG
CCCGAAAAATTAACGCTGCTGCTGGCCGCCTCTGGTCAGGCCCTTACTTCCCGGATGCGGGGCTCATCTCCCCGC
CCTGCGCATCTCTGGCAGCGCCCGCTCTAG crtI (SEQ ID NO: 78)
ATGAAACCAACTACGGTAATTGGTGCAGGCTTCGGTGGCCTGGCACTGGCAATTCGTCTACAAGCTGCGGGGATT
CCCGTCTTACTGCTTGAACAACGTGATAAACCCGGCGGTCGGGCTTATGTCTACGAGGATCAGGGGTTTACCTTT
GATGCAGGCCCGACGGTTATCACCGATCCCAGTGCCATTGAAGAACTGTTTGCACTGGCAGGAAAACAGTTAAAA
GAGTATGTGCAACTGCTGCCGGTTACGCCGTTTTACCGCCTGTGTTGGGAGTCAGGGAAGGTCTTTAATTACGAT
AACGATCAAACCCGGCTCGAAGCGCAGATTCAGCAGTTTAATCCCCGCGATGTGCAAGGTTATCGTCAGTTTCTG
GACTATTCACGCGCGGTGTTTAAAGAAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTTATCGTTTCAGAGACATG
CTTCGCGCCGCACCTCAACTGGCGAAACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAA
GATGAACATCTGCGCCAGGCGTTTTCTTTCCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATT
TATACGTTGATACACGCGCTGGAGCGTGAGTGGGGCGTCTGGTTTCCGCGTGGCGGCACCGGCGCATTAGTTCAG
GGGATGATAAAGCTGTTTCAGGATCTGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAAACGACA
GGAAACAAGATTGAAGCCGTGCATTTAGAGGACGGTCGCAGGTTTCTGACGCAAGCCGTGCGCTCAAATGCAGAT
GTGGTTCATACCTATCGCGACCTGTTAAGCCAGCACCTGCCGCGGTTAAGCAGTCCAACAAACTGCAGACTAAG
CGCATGAGTAACTCTCTGTTTGTGCTCTATTTTGGTTTGAATCACCATCATGATCAGCTCGCGCATCACACGGTT
TGTTTTCGGCCCCGCTTACCGCGAGCTGATTGACGAAATTTTTAATCATGATGGCCTCGCAGAGGACTTCTCACTT
TATCTGCACGCGCCCTGTGTACGGATTCTGTCACCTGGCGCCTGAAGGTTGCGGCAGTTACTATGTGTTGGCGCCG
GTGCCGCATTTAGGCACCGCGAACCCTCGACTGGACGGTTGAGGGGGCCAAACTACGCGACCGTATTTTTTGCCTAC
CTTGAGCAGCATTACATGCCTGGCTTACGGAGTCAGCTGGTCACGCACCGGATGTTTACGCCGTTTGATTTTCGC
GACCAGCTTAATGCCTATCATGGCTCAGCCTTTTCTGTGGAGCCCGTTCTTACCCAGAGCGCCTGGTTTCGGCCG
CATAACCGCGATAAAACCATTACTAATCTCTACCTGGTCGGCGCAGGCACGCATCCCGGCGCAGGCATTCTGGC
GTCATCGGCTCGGCAAAAGCGACAGCAGGTTTGATGCTGGAGGATCTGATTTGA crtY (SEQ ID NO: 79)
ATGGGAGCGGCTATGCAACCGCATATGATCTGATTCTCGTGGGGGCTGGACTCGCGAATGGCCTTATCGCCCTG
CGTCTCCAGCAGCAGCAACCTGATATGCGTATTTTGCTTATCGACGCCGCACCCAGGCGGGCGGGAATCATACG
TGGTCATTTACCACGATGATTTGACTGAGAGCCAACATCGTTGGATAGCTCCGCTGGTGGTTCATCACTGGCCC
GACTATCAGGTACGCTTTCCACACGCCGTCGTAAGCTGAACAGCGGCTACTTTTGTATTACTTCTCAGCGTTTC
GCTGAGGTTTTACAGCGACAGTTTGGCCCGCACTTGTTGGATGGATAACCGCGGTGCGAGAGGTTAATGCGGAATCT
GTTTCGGTTGAAAAAGGGTCAGGTTATCGGTGCCCGCGCGGTGATTGACGGGCGGGGTTATGCGGCAAATTCAGCA
CTGAGCGTGGGCTTCCAGGCGTTTATTGGCCAGGAATGGCGATTGAGCCACCCGCATGGTTTATCGTCTCCCAT
ATCATGGATGCCACGGTCGATCAGCAAAATGGTTATCGCTTCGTGTACAGCCTGCCGCTCTCGCCGACCAGATTG
TTAATTGAAGATACGCACTATAATTGATAATGCGACATTAGATCCTGAATGCGCGCGGCAAAATATTTGCGCACTAT
GCCGCGCAACAGGGTTGGCAGCTTCAGACACTGCTGCGAGAAGAACAGGGCGCCTTACCCATTACTCTGTGCGGC
AATGCCGACGCATTCTGGCAGCAGCGCCCCCTGGCCTGTAGTGGATTACGTGCCGGTCTGTTCCATCCTACCACC
GGCTATTCATGCGCTGGCGGTTGCCGTGGCCGACCGCCTGAGTGCATTGATGTCTTTACGTGCGCCTCAATT
CACCATGCCATTACGCATTTTGGCCGCGAGCGCTGGCAGCAGCAGGGCTTTTTCCGCATGCTGAATCGCATGCTG
TTTTTAGCCGGACCCGCCGATTACGCTGGCGGGTTATGCAGCGTTTTTATGGTTTACCTGAAGATTTAATTGCC
CGTTTTTATGCGGGAAAACCTCACGCTGACCGATCGGCTACGTATTCTGAGCGGCAAGCCGCCTGTTCCGGTATTA
GCAGCATTGCAAGCCATTATGACGACTCATCGTTGA

(146) While exemplary embodiments have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will occur to those skilled in the art. It should be understood that various alternatives to the embodiments described herein may be employed. It is intended that the following claims define the scope of the disclosure and that methods and structures within the scope of these claims and their equivalents be covered thereby.

Claims

1. A nucleic acid construct comprising a modified inducible promoter, wherein the modified inducible promoter comprises: a. a TATAATGT sequence at position –10, relative to a transcriptional start site of the promoter; b. a TTGACA sequence at position –35, relative to the transcriptional start site of the promoter; and c. a nucleic acid sequence that encodes a bacterial ribosome binding sequence.
 2. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a transgene.
 3. The nucleic acid construct of claim 2, wherein the transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBIYZW gene, an ABA1 gene, an ABA2 gene, and a CocE gene.
 4. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a sequence encoding a reporter gene.
 5. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a regulatory element.
 6. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a terminator sequence.
 7. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises: a transgene; and a second modified inducible promoter sequence, wherein the second modified inducible promoter sequence comprises: (i) a TATAATGT sequence at position –10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position –35, relative to the transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a second bacterial ribosome binding sequence.
 8. The nucleic acid construct of claim 1, wherein promoter nucleic acid sequence integrity downstream of the TATAATGT sequence at position –10 is maintained.
 9. The nucleic acid construct of claim 7, wherein promoter nucleic acid sequence integrity downstream of the TATAATGT sequence at position –10 is maintained.
 10. A composition comprising the nucleic acid construct of claim 1; and an inducer.
 11. A kit comprising the nucleic acid construct of claim 1, packaging, buffers, and, optionally, instructions for use.
 12. A method of expressing a protein encoded by a transgene in a cell, the method comprising: a. transforming a cell with a nucleic acid construct, wherein the nucleic acid construct comprises a modified inducible promoter that comprises: i. a TATAATGT sequence at position –10, relative to a transcriptional start site of the promoter; ii. a TTGACA sequence at –35 position, relative to the transcriptional start site of the promoter; iii. a nucleic acid that encodes a bacterial ribosome binding sequence; and iv. a transgene; and b. contacting the cell with an inducer, thereby expressing the protein encoded by the transgene.
 13. The method of claim 12, wherein the cell is a prokaryotic cell.
 14. The method of claim 13, wherein the prokaryotic cell is a bacterium.
 15. The method of claim 12, wherein the cell does not comprise a T7 promoter or a T7 polymerase.
 16. The method of claim 12, wherein, when contacting the cell with the inducer, the protein expression is greater than the protein expression in a comparable cell that comprises the transgene operably linked to a T7 promoter.
-